

Estimating Nonlinear Heterogeneous Agent Models with Neural Networks *

Hanno Kase

ECB

Leonardo Melosi

EUI & CEPR

Matthias Rottner

BIS &
Deutsche Bundesbank

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Abstract

We leverage recent advances in machine learning to develop an integrated framework that globally solves and estimates models featuring agent heterogeneity and nonlinear constraints, while accounting for aggregate risk. The framework enables the solution and estimation of a frontier quantitative Heterogeneous Agent New Keynesian (HANK) model in its nonlinear form, while allowing for the estimation of parameters governing the steady state. We apply it to U.S. data to estimate a HANK model with a zero lower bound (ZLB) constraint. In the estimated model, the interaction between nonlinearities, such as the ZLB, and wealth heterogeneity emerges as a quantitatively important driver of aggregate volatility.

Keywords: Neural networks, likelihood, global solution, heterogeneous agents, nonlinearity, aggregate uncertainty, HANK, zero lower bound.

JEL classification: C11, C45, D31, E32, E52.

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1 Introduction

Over the past three decades, a significant advancement in macroeconomic research has been the integration of heterogeneity into dynamic general equilibrium models. These models enable the study of distributional issues, economic fluctuations, and stabilization policies within a single micro-founded framework. The complexity of these models often necessitates introducing tractability assumptions that largely preclude scholars from considering nonlinear aggregate dynamics and their interactions with heterogeneity. Yet, accounting for these nonlinear dynamics is essential to understand recent macroeconomic events, including recurring and prolonged periods at the ZLB, deep recessions, and the recent rise in inflation. In this paper, we leverage recent advances in machine learning to develop a neural network (NN)-based method for solving and estimating models with heterogeneous agents, fully incorporating nonlinear aggregate dynamics.

Estimating structural models entails searching for parameter values that globally maximize an objective function, which could be a likelihood function, a posterior distribution, or the distance between a set of empirical moments or impulse response functions and their model counterparts. This search involves evaluating the objective function at potentially numerous parameter values. For each parameter value, two steps are typically taken. First, the model’s policy functions are characterized (*solution step*). Second, once the policy functions are properly approximated for that parameter value, the objective function is evaluated (*evaluation step*).

When estimating the parameters of an economic model, these two steps are repeated multiple times to find the global maximum of the objective function. This repetition is not problematic if the model can be solved within a fraction of a second, using a routine that can be easily automated. For instance, this is the case of linearized representative-agent models. However, repeatedly solving a highly complex nonlinear model, such as nonlinear HANK models with aggregate uncertainty, is unmanageable. Additionally, in the context of HANK models, the solution step may

involve computing the steady-state equilibrium afresh, which is a quite computationally intense task to perform.

To overcome these computational hurdles, we approximate the model’s policy functions without conditioning on specific parameter values. By treating the parameters as inputs, or pseudo-state variables, of the policy functions, we only need to solve the model once, and only before launching the estimation. Parameters are pseudo-state variables because, unlike the models’ state variables, parameters are fixed from the perspective of agents in the model, but they are allowed to vary stochastically only during NN training. While this approach expands the dimensionality of the model’s policy functions to approximate, NNs are highly effective in managing high-dimensional problems (*scalability*), and hence the increase in computational burden remains manageable.¹ Once the model’s policy functions are approximated over the parameter space, the model’s solution can be retrieved in a fraction of a second for any combination of parameters.

An additional computational challenge arises when performing likelihood estimation of nonlinear models. Evaluating the likelihood of these models requires the use of filters based on Monte Carlo (MC) methods, such as the particle filter. These filters are computationally more expensive and less accurate compared to those used for evaluating the likelihood of linearized Dynamic Stochastic General Equilibrium (DSGE) models. Therefore, minimizing the frequency of likelihood evaluations and improving the accuracy of filter-based assessments are critical objectives.

To achieve these objectives, we train another NN –*the NN particle filter*– to obtain an approximation of the likelihood function before estimation. This additional NN is trained through three steps. First, we obtain several thousand quasi-random parameter draws (training sample). Second, for each draw, we evaluate the likelihood using the NNs that approximate the model solution together with the particle filter.

¹NNs are the fundamental building blocks of deep learning, which is a subset of machine learning methods. They provide a flexible functional form that can efficiently approximate arbitrarily complex, nonlinear functions, potentially with a large set of inputs.

To perform this step we do not need to solve the model multiple times as we can utilize the NNs trained to provide a fast and reliable approximation of the model solution at each draw of the training sample. Third, the NN is trained to predict the likelihood evaluated at these draws.² The NN particle filter delivers an accurate and smooth approximation of the likelihood function by effectively ironing out the inaccuracies of the particle filter. With the approximated likelihood function at hand, there is no need to repeatedly run a time-consuming MC filter to maximize this function. Instead, the function can be evaluated almost instantaneously at any parameter value using the NN approximation.

Importantly, our methodology enables the estimation of parameters that determine the model’s steady-state equilibrium, a capability not offered by existing methods. By treating these parameters as pseudo-state variables, the steady state needs to be solved only once with a dedicated NN, eliminating a major computational bottleneck and allowing a broader set of parameters to be estimated.

We provide four proofs of concept to validate our methodology. In the first one, we show that our approach replicates the policy functions of a Representative Agent New Keynesian (RANK) model, which can be solved analytically.

In the second proof of concept, we compare our method for likelihood estimation to an alternative state-of-the-art method for nonlinear models. Since this alternative method cannot treat parameters as pseudo state variables, its scope of application is considerably limited. In light of these limitations, we are compelled to consider a stylized representative-agent model featuring only one source of aggregate nonlinearity, which is the ZLB constraint. We show that the two methods deliver remarkably similar estimation outputs, suggesting that our method can perform equally well as a state-of-the-art global method, but with the advantage of being applicable to a much broader set of models.

In the third proof of concept, we establish that our NN methods can successfully

²We minimize the mean squared errors between the likelihood values approximated by the NN and the likelihood values computed via the standard particle filter in the second step.

replicate the popular Sequence Space Jacobian (SSJ) solution to the HANK model introduced by Auclert et al. (2021). Because the SSJ solution is highly accurate for this model, our goal is to show that the NN approach can replicate its results, establishing it as a reliable tool (or at least a comparably valid method) for solving heterogeneous-agent models.

In the fourth proof of concept, we show that our method can be effectively applied to solve and estimate a frontier quantitative HANK model similar to those studied by Auclert et al. (2020); Bayer et al. (2024a). These models feature multiple aggregate shocks, making aggregate risk a key driver of macroeconomic dynamics—an aspect that our nonlinear solution method explicitly captures. Although we are not the first to solve nonlinear HANK models using neural networks, to our knowledge, the scale of the quantitative nonlinear HANK model we solve globally is unprecedented. The large number of state variables poses significant challenges for existing filtering techniques in evaluating the likelihood. Nevertheless, we show that NNs noticeably mitigate these limitations, enabling an accurate estimation of most model parameters.

Beyond demonstrating that our methods can solve and estimate frontier quantitative HANK models under full nonlinearity and aggregate risk, we also apply them to estimate a simpler nonlinear HANK model with a ZLB constraint and three aggregate shocks using U.S. data. Nonlinear models with occasionally binding constraints and meaningful aggregate risk are notoriously difficult to solve and, to our knowledge, have never been estimated.³ Yet such environments are increasingly relevant, as large shocks have become more frequent and the ZLB a persistent feature of macroeconomic stabilization. We also estimate parameters governing the deterministic steady state, which are typically not estimated in the existing literature.

The proposed methodology extends beyond HANK models, providing a flexible and scalable framework for solving and estimating a broad class of economic models whose complexity has long hindered structural estimation. This class includes nonlin-

³The current state of the art, represented by Fernández-Villaverde et al. (2025), solves a related model with a ZLB and a single aggregate shock to maintain computational tractability.

ear models with occasionally binding constraints, such as the ZLB on nominal interest rates, downward nominal wage rigidities, and other similar nonlinearities. We focus primarily on HANK models for two reasons. First, solving these models globally in the presence of aggregate shocks makes the curse of dimensionality especially severe, an issue discussed, for instance, in Fernández-Villaverde et al. (2024). We show that treating parameters as pseudo-state variables allows NNs to efficiently handle these challenges, enabling the estimation of a quantitative nonlinear HANK model. Second, existing estimation methods for HANK models typically rely on linearization, which rules out aggregate risk and occasionally binding constraints. This omission is quantitatively important since the interactions between nonlinearities –namely the ZLB constraint– and heterogeneity –namely wealth heterogeneity– is a key source of aggregate risks and macroeconomic volatility.

Literature. Since Krusell and Smith (1998) wrote their seminal paper, the literature has introduced a variety of methods to solve models with heterogeneous agents (e.g. Algan et al., 2008; Reiter, 2009; Den Haan et al., 2010; Ahn et al., 2018; Boppart et al., 2018; Bayer et al., 2019; Auclert et al., 2021; and Winberry, 2021). These methods rely on linear approximations of the aggregate dynamics of the model and are therefore not suitable for studying the interactions between nonlinearities and heterogeneity in general equilibrium. Recently, the literature has applied these linearized techniques to estimate HANK models using likelihood methods (e.g. Challe et al., 2017; Bayer et al., 2024a; Auclert et al., 2020; Lee, 2020; Auclert et al., 2021; Bilbiie et al., 2023; and Acharya et al., 2023). Unlike in our paper, estimation is very often confined to parameters that do not affect the model’s steady state and aggregate nonlinearities (e.g. the ZLB constraint) are not taken into account. Bhandari et al. (2023) and Bayer et al. (2024b) pioneered methods to solve heterogeneous agent models by approximating their aggregate dynamics to higher orders. In this paper, we introduce NN-based methods to solve these models using global methods. Our

method also allows structural estimation of models with heterogeneous agents.

Our paper is connected to the literature that develops NNs to solve complex dynamic macroeconomic models globally such as Fernández-Villaverde et al. (2020), Ebrahimi Kahou et al. (2021), Maliar et al. (2021), Azinovic et al. (2022), Duarte et al. (2024), Azinovic and Žemlička, and Valaitis and Villa (2024). Our main contribution to this literature is twofold. First, we demonstrate that NNs can effectively solve frontier HANK models with more than a thousand state variables by leveraging their scalability. Second, we show how NNs can be used to estimate nonlinear heterogeneous-agent models. Estimation – including the simulated method of moments, impulse-response-functions matching, maximum likelihood, and Bayesian estimation – adds an additional important layer of complication in that it requires solving the model at hundreds of thousands of different points in the parameter space.

Fernández-Villaverde et al. (2023) estimate a critical parameter of a model with a financial frictions and heterogeneous households with NNs using the time-series of GDP growth. We can expand the scale of the estimation exercise and estimate several parameters using multiple observable variables because (i) we combine the scalability of NNs with the intuition of treating the model parameters as pseudo-state variables when solving the model and (ii) we use NNs to facilitate the evaluation of the likelihood via the filter - which we call *NN particle filter*. Furthermore, differently from Fernández-Villaverde et al. (2023), we estimate a HANK model. To our knowledge, we are the first to estimate a HANK model in its nonlinear specification.

HANK models are attracting growing attention of scholars (e.g. Oh and Reis, 2012; McKay and Reis, 2016; McKay et al., 2016; Den Haan et al., 2017; Guerrieri and Lorenzoni, 2017; Ravn and Sterk, 2017; Kaplan and Violante, 2018; Kaplan et al., 2018; Auclert, 2019; Bilbiie, 2020 and Ravn and Sterk, 2020; Gornemann et al., 2021; Luetticke, 2021; and Auclert et al., 2024).⁴ Methods to solve these models globally by relying on NNs have been recently developed (e.g. Maliar and

⁴Scholars have also explored general equilibrium models where heterogeneity characterizes the firm side – e.g. Ottonello and Winberry (2020).

Maliar, 2020; Gorodnichenko et al., 2021; Fernández-Villaverde et al., 2025; Han et al., 2021; and Kase et al., 2025) or by introducing alternative approaches unrelated to machine learning (e.g. Schaab, 2020; and Lin and Peruffo, 2024). Compared to this strand of literature, we show how the scalability of NNs can be leveraged to enable the estimation of nonlinear HANK models. Moreover, these studies have invariably focused on models of substantially lower complexity than the quantitative nonlinear HANK models solved in this paper.

2 A neural-network framework for estimation

In this section, we describe how to leverage the properties of NNs to enable the estimation of nonlinear models with heterogeneous agents. We first provide a primer on NNs. Second, we explain how NNs can be used to efficiently characterize the policy functions of a model treating the parameters as pseudo-state variables. Third, we show how another NN can be trained to characterize the deterministic steady-state equilibrium of the model, when this equilibrium influences its policy functions. Fourth, we explain how to train the NN particle filter.

2.1 Neural Networks

NNs are function approximators well-suited for learning highly dimensional nonlinear functions.⁵ Consider a function $Y = \psi(X)$, mapping a vector of inputs X into a

⁵NNs are the fundamental building blocks of deep learning – a class of machine learning techniques. The universal approximation theorem states that a feedforward network with at least one hidden layer can approximate any continuous function on a compact subset of a finite-dimensional space to arbitrary accuracy, given sufficient width (Hornik et al., 1989.). It is important to note, however, that the universal approximation property is not unique to NNs. A key advantage of NNs is that they can mitigate the curse of dimensionality, enabling them to handle high-dimensional problems (Barron, 1993; Bach, 2017). As argued in Fernández-Villaverde et al. (2024), the “unreasonable effectiveness” of NNs is based on intuitive geometrical reasoning. They transform the model’s original state space into a geometrically more efficient representation that captures the relevant features of the underlying model.

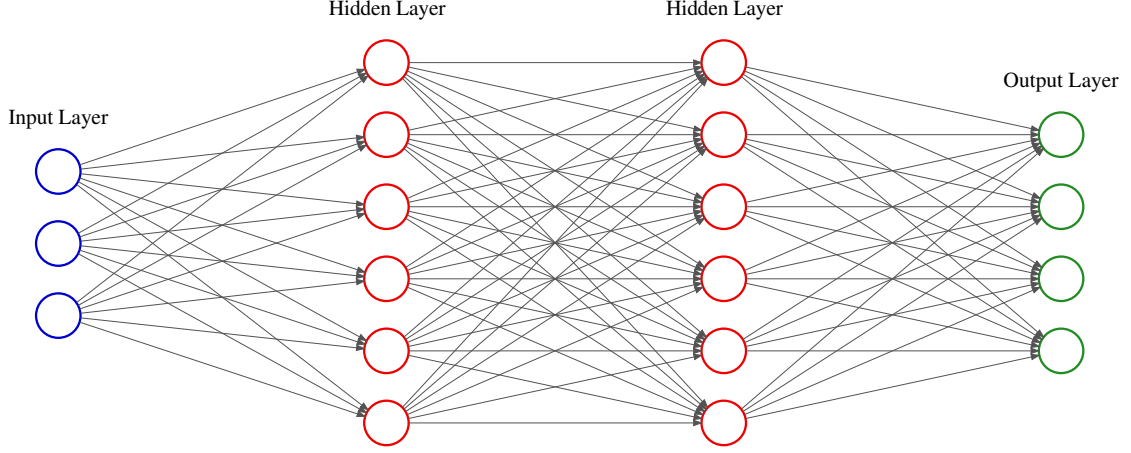


Figure 1: Illustration of a NN. An illustrative example of a NN with three inputs (blue circles), two hidden layers (the two rows of red circles) with six neurons (the red circles) each, and four outputs (the green circles).

vector of outputs Y . The function can be approximated by a NN denoted by:

$$Y = \psi_{NN}(X|W), \quad (1)$$

where W are the weights defining the NN. As illustrated in Figure 1, the NN consists of several layers, which can be divided into the input layer, hidden layer(s) in between, and the output layer. Each hidden layer consists of several neurons, which are represented as circles in the figure.

A single neuron is a composition of an affine function with inputs $x_1, x_2, \dots, x_M$, weights $w_1, w_2, \dots, w_M$, and bias w_0 and a nonlinear activation function $g(\cdot)$ that returns a single output $\tilde{y}$:⁶

$$\tilde{y} = g(w_0 + \sum_{i=1}^M w_i x_i). \quad (2)$$

The structure and weights characterize a NN. For example, the NN illustrated in Figure 1 has four layers containing in total 94 weights (including biases).⁷ In matrix

⁶The nonlinear activation function $g(\cdot)$ makes it possible for the NN to approximate nonlinear functions. Popular examples of activation functions are ReLU (rectified linear unit), CELU (continuously differentiable exponential linear units), or Tanh (hyperbolic tangent).

⁷The first hidden layer has $(3 + 1) \times 6 = 24$ weights (including biases); the second hidden layer has $(6 + 1) \times 6 = 42$; the output layer has $(6 + 1) \times 4 = 28$. The first term in the brackets is the

notation, we can write a single layer h of the NN as:

$$y = g(\mathcal{A}^h(x)), \text{ where } \mathcal{A}^h(x) \equiv W^h x, \quad (3)$$

where the inputs, outputs, and weights are now written as vectors.⁸ As implied by equation (3) and Figure 1, NNs are natural to vectorize, making them efficient to compute, especially on graphics processing units (GPUs) that excel at matrix multiplications and element-wise operations.⁹

The weights W are optimized (trained) to minimize a loss function; e.g. the mean squared error between observed and predicted values of the function to approximate. The optimization of the weights of the NNs relies on the back-propagation algorithm, which allows computing all partial derivatives of the loss with respect to the weights of the NN in approximately the same number of operations, and similar computational cost, as computing the loss itself. The partial derivatives inform the direction in which an optimization algorithm should adjust the weights of the NN to minimize the loss. In particular, we use a stochastic gradient descent based optimization algorithm called AdamW, which is currently one of the most widely used optimizers for training NNs. The accuracy of the NN approximation and the speed of the training process can be improved by adjusting the learning rate during the training, which is done using a learning rate scheduler. A common choice is to gradually decrease the learning rate as the training progresses, following, for example, a linear or exponential schedule.

weights plus the bias at each neuron, and the second term is the number of neurons in the layer.

⁸The entire NN with H hidden layers can be written as a composition of functions $\psi_{NN}(X|W) = (\mathcal{A}^{H+1} \circ g \circ \mathcal{A}^H \circ g \circ \mathcal{A}^{H-1} \circ \dots \circ g \circ \mathcal{A}^1)(X)$ where $\circ$ denotes a function composition.

⁹To further improve efficiency, the calculations are typically done for batches of inputs.

2.2 Neural-network solution method

We consider a large class of dynamic general equilibrium models, whose dynamics are governed by the following transition equations or laws of motion:

$$\mathbb{S}_t = f(\mathbb{S}_{t-1}, \nu_t | \Theta), \quad (4)$$

where $\mathbb{S}_t$ is a finite vector of S state variables of the model. The economy is subject to U exogenous shocks ν_t that affect the response of the state variables. The model's structural parameters are denoted by Θ . The function $f(\cdot)$ is possibly nonlinear and typically does not have a closed-form analytical characterization.

Solving the model amounts to finding a set of O policy functions that map the state variables, $\mathbb{S}_t$ to a set of control variables, ψ_t for given values of the model parameters, Θ : $\psi_t = \psi(\mathbb{S}_t | \Theta)$. These policy functions satisfy a set of K equilibrium conditions derived from the model equations:¹⁰

$$F(\psi(\mathbb{S}_t | \Theta)) = 0. \quad (5)$$

With the policy functions $\psi(\cdot)$ at hand, the transition equations (4) can be obtained.

Extended policy functions. A critical bottleneck for estimation is the need to solve policy functions for a potentially very large and a-priori unknown number of times at different parameter values. In the context of the models studied in this paper, this task is often unmanageable. To address this issue, we treat the parameters to estimate as pseudo-state variables when approximating the policy functions.¹¹

We define the *extended policy functions* as

$$\psi_t = \psi(\mathbb{S}_t, \tilde{\Theta} | \bar{\Theta}), \quad (6)$$

¹⁰Examples of these conditions may be the Euler equation or the resource constraint.

¹¹Examples of computational papers that use pseudo-state variables in combination with machine learning are Norets (2012), Scheidegger and Bilonis (2019), and Duarte and Fonseca (2024).

where the vector of parameters Θ is split into $\tilde{\Theta}$ – the set of parameters to estimate – and $\bar{\Theta}$ – the set of parameters that are fixed in estimation. The characterization of the extended policy functions leads to the extended transition equations:

$$\mathbb{S}_t = f\left(\mathbb{S}_{t-1}, \nu_t, \tilde{\Theta}|\bar{\Theta}\right). \quad (7)$$

Agents’ heterogeneity. In the literature studying heterogeneous agent models, it is commonly assumed that there exists a continuum of agents. However, this assumption makes the state vector infinite dimensional, which of course presents insurmountable computational challenges. Therefore, when approximating the policy functions of these models, we discretize the model population by keeping track of a finite number L of agents.

The transition equations of heterogeneous agent models can be recast in exactly the same form as in equation (4), where the state vector $\mathbb{S}_t$ comprises both the set of aggregate state variables, $\mathbb{S}_t^A$, and the set of individual state variables for each agent, $\{\mathbb{S}_t^i\}_{i=1}^L$, and the vector of shocks, ν_t , includes both aggregate shocks, ν_t^A , and idiosyncratic shock for each agent, $\{\nu_t^i\}_{i=1}^L$.

In contrast to the previous case with no heterogeneity, we now have two sets of policy functions to approximate: the aggregate policy functions, denoted as $\psi^A(\cdot)$, and the individual policy function, represented by $\psi^I(\cdot)$.¹² Following the approach introduced earlier, these functions are expressed in their extended form by treating the parameters to estimate, $\tilde{\Theta}$, as pseudo-state variables.

$$\psi_t^A = \psi^A(\mathbb{S}_t, \tilde{\Theta}|\bar{\Theta}) \quad \text{and} \quad \psi_t^i = \psi^I(\mathbb{S}_t^i, \mathbb{S}_t, \tilde{\Theta}, |\bar{\Theta}), \quad (8)$$

where $\mathbb{S}_t^i$ denotes the individual state variables. The number of individual and aggregate state variables are S^i and S_t^A , respectively, so the total number of state variables is $S = S^i \times L + S^A$. The number of exogenous shocks and policy functions is similarly

¹²Note that the individual function is the same across agents, indicating that agents with the same individual state variables $\mathbb{S}_t^i$ make the exactly same decisions, ψ_t^i .

defined as $U = U^i \times L + U^A$ and $O = O^i \times L + O^A$, respectively.¹³

NNs and training. We set up two NNs, $\psi_{NN}^I(\cdot)$ and $\psi_{NN}^A(\cdot)$, to approximate the individual and aggregate extended policy functions, $\psi^I(\cdot)$ and $\psi^A(\cdot)$, defined in equation (8).¹⁴ We train these two NNs by choosing their weights, W , so as to minimize the weighted squared residual error for the equilibrium conditions:

$$\Phi^L = \sum_{k=1}^K \alpha_k [F_k(\psi_{NN}(\mathbb{S}_t, \tilde{\Theta}|\bar{\Theta}))]^2, \quad (9)$$

where $F_k(\cdot)$ denotes the residual error associated with the k -th equilibrium condition and α_k is the weight of this condition.¹⁵ We include at least as many optimality conditions as the number of policy functions, so that $K \geq O^i \times L + O^A$.

To leverage the potential for parallelization provided by GPUs, the two NNs are trained on a batch with size B . This step amounts to having B pseudo-randomly drawn state vectors, $\mathbb{S}_{t,b}$ and parameters to estimate, $\tilde{\Theta}_b$. In each optimization step, we minimize the loss functions Φ^L averaged across batch size B :

$$\bar{\Phi}^L = \frac{1}{B} \sum_{b=1}^B \sum_{k=1}^K \alpha_k \left[F_k(\psi_{NN}(\mathbb{S}_{t,b}, \tilde{\Theta}_b|\bar{\Theta})) \right]^2. \quad (10)$$

This loss function is minimized by using a stochastic gradient descent method.

As standard, to evaluate the residual error for some of the equilibrium conditions, $F_k(\psi(\cdot))$, we are required to compute agents' expectations. We do so by using Monte Carlo (MC) integration. Specifically, we randomly draw next period's shocks to ap-

¹³In this paper, we track all the states of the agents as in Maliar et al. (2021) for instance. An alternative approach is to reduce the dimension of the individual states to track by exploiting the symmetry of the agents following Ebrahimi Kahou et al. (2021). Our method can be combined with either approach.

¹⁴Our approach, in which we treat parameters as pseudo-state variables, is consistent with many approximating functions. However, since this approach extends the state space, we naturally focus on NNs, as they alleviate the curse of dimensionality.

¹⁵The weights can vary so as to take into account that losses across different equations possibly have different magnitudes.

proximate expectations.¹⁶ To increase precision, one can conduct several optimization rounds with new sets of drawn shocks to evaluate expectations.

The training of the NNs to approximate the aggregate and idiosyncratic policy functions concludes the solution of the model. Since the parameters to be estimated are treated as pseudo-state variables of the approximated policy functions, the model can be simulated and its transition equations obtained in just a fraction of a second, as these tasks only require evaluating the previously trained NNs. This approach eliminates the need to repeatedly solve the model, which is a critical bottleneck when estimating models with a large set of state variables.

Stochastic solution domain and truncated parameter space. The grid points for the state variables and parameter values at the beginning of each optimization step are drawn randomly. We aim to train the algorithm only on the relevant domain of the state space. Following the approach of Maliar et al. (2021), we directly sample from the model to approximate the ergodic distribution of the state variables, also known as the stochastic solution domain. Specifically, we simulate each economy (batch) forward using the NNs trained at the current optimization step. The endpoint of the simulation provides an approximation of the ergodic distribution of the state variables, which we use for the next optimization step.

We truncate the space from which we draw the model parameters to estimate (the pseudo-state variables). By limiting the admissible set of parameter values, we achieve a more accurate evaluation of the ergodic distribution of the state at each optimization step. In the case of Bayesian estimation, truncating the parameter space when approximating the model’s solution can be made consistent through the use of truncated prior distributions.

Differences between state and pseudo-state variables. Consistent with rational expectations, optimizing agents form beliefs about future outcomes, knowing that

¹⁶To reduce the variance of MC integration, we use the antithetic variate method in this paper.

pseudo-states are fixed parameters while recognizing that true state variables evolve over time. To implement this distinction, pseudo-state variables and true state variables are treated differently during training. During training, pseudo-state variables are held fixed when we simulate the true state variables and compute residual errors within each iteration, but they vary across iterations, taking different values throughout the training process. In contrast, true state variables are re-simulated at the beginning of each optimization step using the neural network from the previous step and the value of fixed pseudo-state variables in every batch.

Parameters affecting the deterministic steady state (DSS). Often, solving a heterogeneous agent model requires knowing its DSS.¹⁷ Finding the DSS necessitates solving a Bewley-Huggett-Aiyagari type economy over the parameter space, which is a computationally intensive step. To address this, we train an auxiliary NN to approximate the DSS. This auxiliary NN is then used when training the other two NNs (ψ_{NN}^I and ψ_{NN}^A) that approximate the aggregate and idiosyncratic policy functions defined in equation (8).

Number of agents to track. The choice of how many agents, L , to track is delicate, as increasing this number can substantially raise computational costs when solving models. For most applications considered in this paper, we set $L = 100$. Although this may appear relatively small, we have successfully scaled the number of agents up to 1,000 in most applications, including the one-asset HANK model (Section 3.3) and the quantitative HANK model (Section 3.4). Importantly, for most of applications we do not observe any significant changes in the approximated policy functions, the stationary wealth distribution, or the impulse response functions produced by our method when the number of agents increases beyond 100 and approaches

¹⁷The DSS is typically defined as the model economy in the absence of aggregate shocks (e.g., Fernández-Villaverde et al., 2025). For example, in the context of New Keynesian models, the monetary policy rule is typically specified as a function of the model’s DSS interest rate and output.

1,000.¹⁸ For the quantitative HANK model, an appropriate number of agents is 400.

To provide some intuition for why our NN solution method does not require tracking a very large number of agents and why, in practice, around 100 agents is often sufficient, consider the training procedure. The model is simulated over a long horizon and across multiple batches, with each batch effectively representing an independent and identical economy. Within these B economies, the model is solved separately, and the NN learns from the outcomes across the parallel simulations. This setup generates a rich and diverse dataset for training, even when the number of agents in each individual economy remains moderate. This feature of NN training is crucial, as it allows heterogeneous-agent models to be solved without the need to track a very large number of agents, thereby minimizing computational costs.

Finally, a similar result arises when approximating agents' expectations, which is another delicate and potentially problematic step in solving structural models. When employing a Monte Carlo approximation, that is, using random sampling to approximate expectations, it is not necessary to generate a very large number of draws for the integration to be accurate. We mostly use 100 draws. Results, where we use up to 500 draws, shown in one of the applications presented are available upon request.

2.3 The neural network particle filter

The likelihood function of DSGE models is typically evaluated in two steps. In the *solution step*, the model's policy functions must be characterized. The NNs (NNs), trained as described in the previous section, can be utilized to perform this task quickly and at several points in the model's parameter space. With the policy functions at hand, we can construct the transition equations (7) which, combined with the measurement equations that map the observable variables to the appropriate model concepts, form the state-space representation of the model. Since the likelihood

¹⁸Results for the one-asset HANK model are shown in Appendix D.

cannot be analytically characterized, it is typically evaluated by applying a filter to the state-space representation in the *evaluation step*. Filters return the model’s one-step-ahead predictive densities over the available sample period, $p_{\tilde{\Theta}}(y_t|y_{t-1}, \dots, y_1, \tilde{\Theta})$, where y_t denotes the vector of observable variables in period t . The product of the model’s predictive densities in every period $t \in 1, \dots, T$ is the likelihood function:

$$\mathcal{L}_{\tilde{\Theta}}(\mathbb{Y}_{1:T}|\tilde{\Theta}) = \prod_{t=1}^T p_{\tilde{\Theta}}(y_t|y_{t-1}, \dots, y_1, \tilde{\Theta}). \quad (11)$$

Thus, for a given data set of T periods, $\mathbb{Y}_{1:T}$, a filter can be thought as a mapping from the set of model parameters, $\tilde{\Theta}$, to the value of the likelihood function.¹⁹ We train a new NN – the *NN particle filter* – to accurately approximate this mapping.

The likelihood function of nonlinear models requires MC filters for evaluation.²⁰ Compared to filters used with linear models, MC filters are both more time-consuming and less accurate. These challenges are particularly pronounced when estimating models with a highly dimensional state space such as nonlinear heterogeneous agent models. As we will show, our NN particle filter significantly mitigates these shortcomings of MC filters.

The NN particle filter is trained through the following three steps. First, we obtain several thousands of quasi-random draws from the parameter space. Second, we execute the (standard) particle filter to derive the likelihood value at each of these draws. It is noteworthy that for this step, we employ the trained NNs to approximate the policy functions and the DSS, which were described in Section 2.2. Consequently, this step is not particularly time-consuming. Third, the NN particle filter is trained to minimize the mean squared error between its predicted values and the likelihood

¹⁹The notation makes it clear that while the likelihood function depends on all the parameters $\Theta \in \{\tilde{\Theta}, \bar{\Theta}\}$, we maximize this function only with respect to the subset $\tilde{\Theta}$.

²⁰An example of MC filter is the particle filter (Fernández-Villaverde and Rubio-Ramírez, 2007). This filter is often used by the economic literature to estimate the distribution of the state variables or to evaluate the likelihood of nonlinear models; e.g. models with occasionally binding constraints (Gust et al., 2017; Atkinson et al., 2020; Aruoba et al., 2021) or with (endogenous) multiple equilibria (Aruoba et al., 2018; Rottner, 2023).

values evaluated in the second step.²¹

The NN particle filter undergoes training over thousands or even millions of optimization steps. Specifically, the training sample is divided into batches of size B . Each batch is fed into the NN particle filter, which generates a vector of approximated likelihoods. The loss is then calculated as the mean squared deviation with respect to the likelihood values evaluated by the standard particle filter across batches. A gradient-based optimizer adjusts the weights of the NN particle filter to minimize this loss. This process continues until all points in the training sample have been used, completing the first epoch. This procedure repeats for thousands of epochs. Over epochs, the learning rate of the gradient descent method is progressively lowered, allowing for broader exploration in earlier epochs and finer search in later epochs.

Dealing with overfitting. It is critical to prevent the NN particle filter from overfitting the training sample of drawn parameters. Overfitting occurs when NNs learn too many details from the training sample, including noise generated by computational inaccuracies in the likelihood values approximated by an MC filter. To mitigate overfitting, we evaluate the trained NN particle filter on a validation sample of quasi-randomly drawn parameter values. The network’s weights are not updated during this validation stage, so the corresponding loss provides a useful metric for determining when to stop training before overfitting occurs. A lower loss on the training sample relative to the validation sample indicates potential overfitting.

Bayesian estimation and forecasting. The NN approximation for the likelihood explained above can be used to approximate the moments of the posterior distribution for the parameters quite straightforwardly. Details on how to combine the likelihood approximated with NNs and a standard simulator of posterior draws, such as the

²¹It is often convenient to work directly with the log-likelihood. This approach entails summing the log of the one-step-ahead predictive densities across all the periods in the sample, T . It should be noted, however, that using the log-likelihood introduces a slight bias due to Jensen’s inequality (specified for the logarithm as $E(\ln(X)) \leq \ln E(X)$) because our approach relies on averaging.

Random-Walk Metropolis-Hastings (RWMH), are provided in Appendix A. Our code relies on *PyTorch* as the machine learning framework.

Real-time estimation and forecasting often entail the iterative estimation of model parameters as new data becomes available. One way to do that in the context of our methodology is to store the filtered states in the last period of the sample. As the new data become available, we run the filter only over the new periods and recompute the values of the likelihood to retrain the NN particle filter.

Another approach involves calculating the likelihood values using future data as pseudo-state variables. We can train the NN that approximates the likelihood function by randomly drawing future data that have not yet been observed. As the data are released, we can obtain the value of the likelihood in a fraction of a second from the NN particle filter trained with simulated future data. The second approach could be adjusted to accommodate data revisions for a few periods back.

2.4 Further discussion

The accuracy of estimation can be enhanced by adopting a sequential approach to the proposed NN-based estimation method. First, parameter values can be drawn from an uninformative unit hypercube to train the relevant NNs, which then provide an approximation of the likelihood. Second, we use this approximated likelihood to refine the bounds and the proposal distribution of the parameters in order to retrain the NNs approximating the policy functions as well as the NN particle filter. This iterative cycle could be repeated several times to improve accuracy.

Non-likelihood methods, such as the method of moments or impulse-response matching, can be embedded in this framework. Instead of using the NN to approximate the likelihood, NNs are trained using a loss function defined with respect to the moments to match or impulse response functions.

2.5 Applying our method: a step-by-step guide

We provide a learning code on the authors’ websites, designed to guide scholars and practitioners in applying our method to solve and estimate DSGE models. The tutorial includes Python code with detailed documentation for the linearized RANK model.

3 Validation of the method

We provide four proofs of concept aimed at showing the accuracy and robustness of our neural-network solution and estimation methods for a wide range of economic models. Specifically, we assess our method i) by comparing the policy functions and the likelihood approximated with NNs to the true solution of a linearized small-scale New Keynesian DSGE model, which admits analytical solution; ii) by comparing the policy functions approximated with NNs to those obtained with a state-of-the-art (non-NN) global method when solving a RANK model with a recurrently binding ZLB constraint; iii) by comparing the solutions —namely, the steady-state distribution, policy functions, and impulse response functions— of a one-asset HANK model with a single aggregate shock à la Auclert et al. (2021) across our NN-based method and the SSJ, which provides a highly accurate approximation of this model under the assumption of linear aggregate dynamics, iv) by showing the ability of our methods to recover the true parameter values of a quantitative HANK model of the type estimated by Auclert et al. (2020) and Bayer et al. (2024a) in its fully nonlinear specification and aggregate risk.

3.1 Proof of concept 1: a linearized small-scale RANK model

We consider an off-the-shelf three-equation RANK model with one shock (TFP shock). The model is log-linearized around its unique steady-state equilibrium and the resulting equations are shown in Appendix B.

Parameters		LB	UB	Parameters		LB	UB
β	Discount factor	0.95	0.99	θ_{Π}	Mon.pol. inflation response	1.25	2.5
σ	Relative risk aversion	1	3	θ_Y	Mon.pol. output response	0.0	0.5
η	Inverse Frisch elasticity	1	4	ρ_A	Persistence TFP shock	0.8	0.95
φ	Price duration	0.5	0.9	σ_A	Std. dev. TFP shock	0.02	0.1

Table 1: Model parameters values – first proof of concept. The table shows the parameters of the three-equation New Keynesian model. The lower bound (LB) and upper bound (UB) show how we truncate the parameter space when training the NN to approximate the model solution.

3.1.1 The extended policy functions

We use our NN approach to approximate the extended policy functions of the output gap and inflation. For reasons explained earlier, we truncate the parameter space, as shown in Table 1. We use 500,000 iterations to train the NN by minimizing the residual errors of the Euler equation and the Phillips curve. The resulting policy functions are conditioned on the state variables and parameters, giving us nine inputs in total.²² We conduct five optimization steps to approximate the policy functions. The convergence of the total loss during the training of the NN underscores a considerable degree of accuracy in approximating the two policy functions, with the mean squared residual error averaged across the two loss components reaching values around 10^{-10} . See Appendix B.

In Figure 2, we show the extended policy function of the output gap as the structural parameters of the model vary. The extended policy function is evaluated at a negative one standard deviation shock, and the unvaried parameters are fixed at the middle value of their bounds, shown in Table 1. The comparison with the analytical solution demonstrates that our NN is able to replicate the true solution extremely well as the lines almost perfectly coincide. Identical conclusions on the accuracy of the method can be drawn from a similar graph showing the policy function of inflation (Appendix B).

²²Appendix G provides the details of the NN architecture and the training process.

3.1.2 Likelihood approximation and estimation

As a next step, we estimate the likelihood of this model with our NN approach. We simulate inflation and the output gap from the linearized RANK model for 100 periods and use these simulated series as data. In the simulation, the parameters are set to the middle of their bounds.²³

We train the NN particle filter using a sample of likelihood values evaluated at 10,000 quasi-random parameter draws.²⁴ These likelihood approximations are obtained by running the standard particle filter for each draw. When running the particle filter, we deliberately choose a low number of particles (100 particles) to illustrate the NN to cut through the noise of the filter and to deliverate an accurate representation of the likelihood in this application.

We divide our sample of 10,000 likelihood approximations into a training sample (80%) and a validation sample (20%). The NN is trained using the training dataset across 20,000 epochs, wherein each epoch, the entire training dataset is processed. The training sample is divided into batches of 100. Each batch is passed simultaneously to the optimization routine. An optimization step is completed when the routine converges and returns the optimal NN weights. With 8,000 parameter draws in our training sample, this optimization step is performed 80 times within each epoch. In this application, we consider 20,000 epochs, resulting in a total of 1.6 million optimization steps across all epochs. The dynamics of total loss during the training of the NN are shown in Appendix B. The total loss converges to a value that implies fairly contained inaccuracy in approximating the likelihood function. The loss computed on the validation sample (shown in Appendix B) is very close to that obtained during training, alleviating concerns about overfitting. This evaluation is particularly important here because the NN was deliberately trained using a particle filter with only 100 particles to magnify potential overfitting in the NN particle filter.

²³We add a small measurement error (5% of the variance of each observable variable) to avoid the well-known problem of degeneracy of the likelihood function when using the particle filter.

²⁴Appendix G provides the details of the NN architecture.

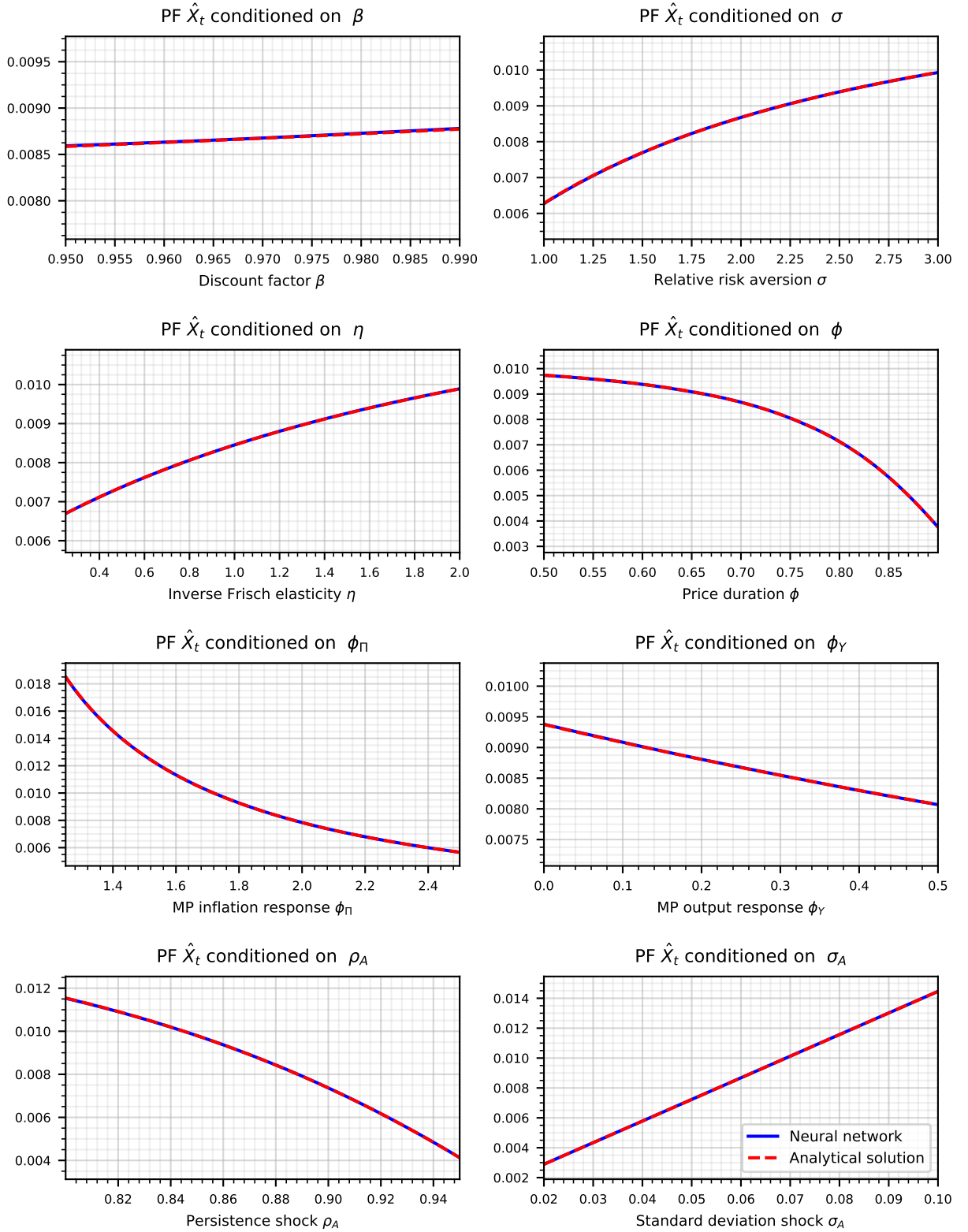


Figure 2: Accuracy of the NN solution method. Comparison between the analytical policy function for the output gap and the approximation based on NNs. The values of the fixed parameters in each graph are set to the middle of their bounds shown in Table 1.

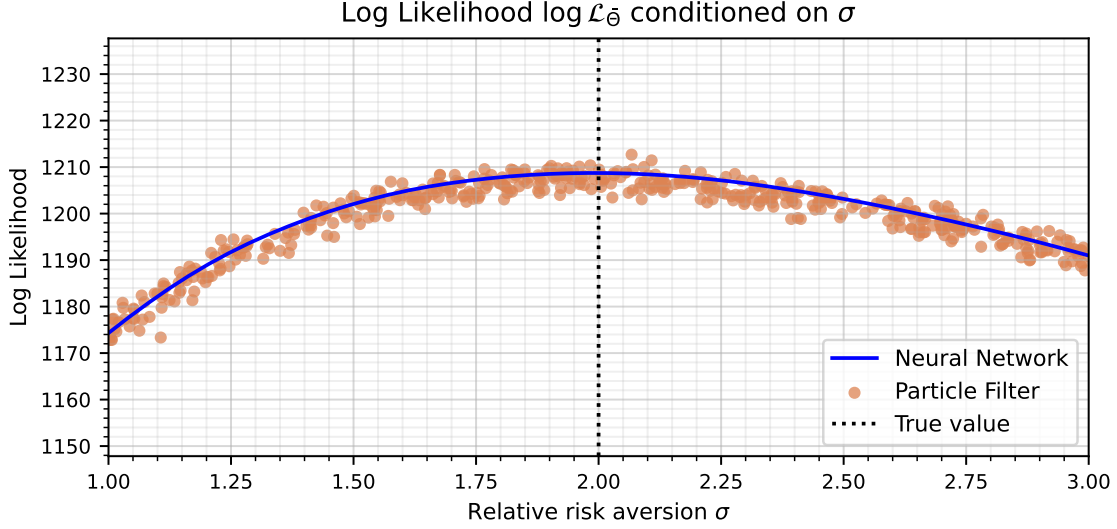


Figure 3: Accuracy in likelihood evaluations: NN particle filter vs. standard particle filter. The logarithm of the likelihood of the model as a function of the risk aversion parameter σ . The value of the fixed parameters is set to the middle of their bounds.

To more directly assess potential issues of overfitting, we compare likelihood approximations using different methods. In Figure 3, we plot how the likelihood – approximated with two alternative methods – varies with respect to the relative risk aversion, σ . The orange dots denote likelihood values evaluated with a standard particle filter using only 100 particles. As expected, this approximation is poor due to the limited number of particles used to deliberately exaggerate potential overfitting issues for instructional purposes. The solid blue line depicts the likelihood approximation using the NN particle filter, also trained with 100 particles for consistency.

Interestingly, the NN approximation cuts through the cloud of likelihood points generated by the particle filter, effectively filtering out the noise the filter introduces. Moreover, the parameter value at which the NN-approximated likelihood peaks closely aligns with the true parameter value, as indicated by the vertical dotted line. Appendix B shows how the approximated likelihood varies with respect to other parameters.

3.2 Proof of concept 2: A RANK model with the ZLB

We evaluate the effectiveness of our NN estimation strategy in recovering the true data-generating process of a stylized nonlinear RANK model augmented with a ZLB constraint, which is described in Appendix C.

First, we simulate output growth, inflation, and the nominal interest rate for 500 periods from the model. Second, we estimate the model with our NN-based estimation method and with an alternative method combining a state-of-the-art global solution method (Richter et al., 2014) and the particle filter (Herbst and Schorfheide, 2015). While this alternative method is too slow for estimating more complex nonlinear models, it serves as a useful benchmark for evaluating the effectiveness of our method.

Our method demonstrates strong performance in recovering the parameters of the true data-generating process. The posterior median closely aligns with the true value and consistently falls within the 90 percent credible interval (shown in Appendix C). Moreover, the shape of the posterior closely resembles that obtained using the alternative method, as shown in Figure 4. However, the alternative method is significantly slower than our proposed NN method.

3.3 Proof of concept 3: One-asset HANK model

In the third proof of concept, we show that our NN methods can successfully replicate the popular SSJ solution to the one-asset HANK model with one aggregate shock introduced by Auclert et al. (2021). The high accuracy of the SSJ method in solving this model, combined with the small degree of aggregate risk, makes this method a valid benchmark for assessing the reliability of our NN approach in solving heterogeneous-agent models.

Solving this HANK model globally presents challenges for our method in that it requires tracking 202 state variables.²⁵ The execution time is 4h03min on an Nvidia

²⁵We track 100 agents, and we use 100 MC draws to evaluate expectations. The model features one individual policy function, two aggregate policy functions, and one mapping for the DSS to ap-

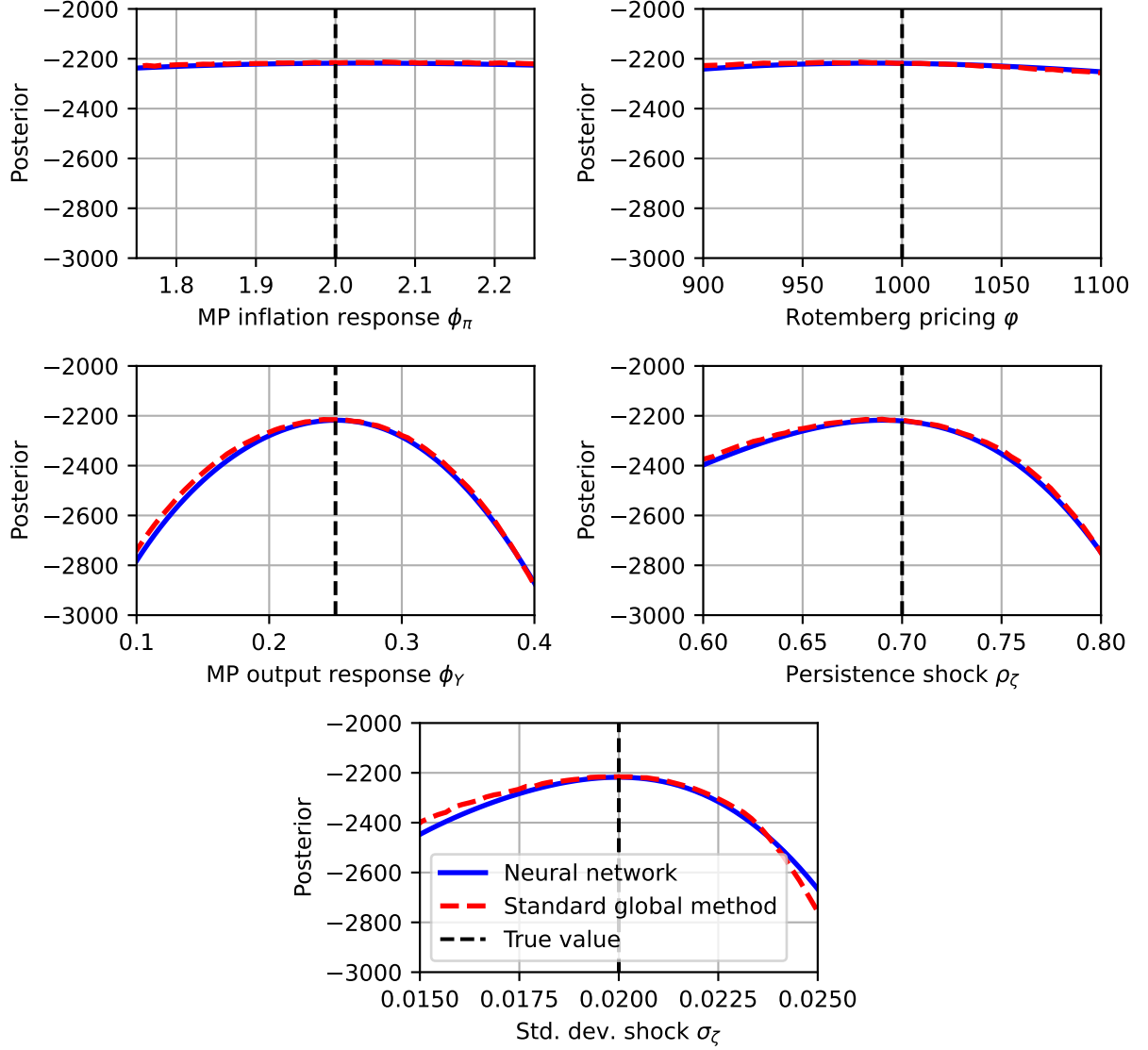


Figure 4: Posterior distribution. Comparison of the posterior distribution approximated with the NN method (solid blue), and a state-of-the-art global method (dashed red). The vertical black dash line denotes the true value of each parameter. The posterior distribution is evaluated at its median by varying one parameter at a time.

RTX 3090 GPU with 24 GB memory. These challenges make our validation test informative for assessing whether our method is a reliable tool for solving models with a highly dimensional state space due to the presence of heterogeneous agents. Appendix D reports the loss during training and the accuracy of our solution after training by presenting approximation errors for selected equilibrium conditions.

proximate. Tracking 1,000 agents does not significantly improve the accuracy of the approximation. This is shown in Appendix D.

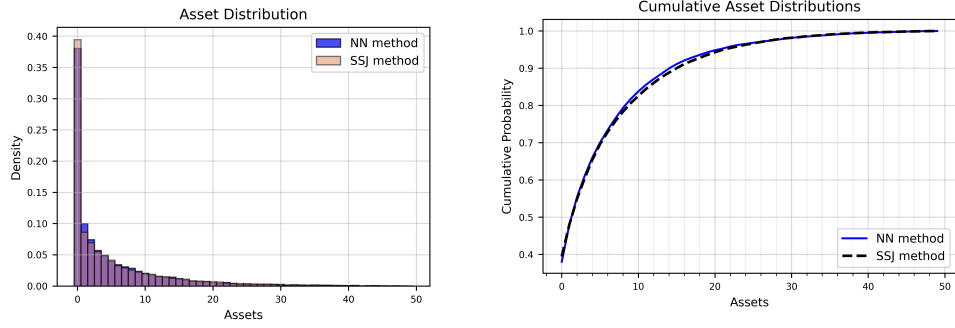


Figure 5: Asset distribution for our NN method for comparison with SSJ method. For the NN method, we evaluate the distribution at its stochastic steady state, averaging over 100 economies. For the SSJ, we use the deterministic steady state as it cannot account for aggregate uncertainty. Left panel: The histogram of asset holdings in equilibrium. Right panel: Cumulative density function for the asset holdings in equilibrium.

Figure 5 compares the asset distribution generated by our method and by the SSJ approach. For our method, we report the distribution at the stochastic steady state, which accounts for the presence of aggregate risk. In contrast, the SSJ method—by construction—assumes away aggregate uncertainty, so the distribution it produces corresponds to the deterministic steady state. Despite this key difference, the two distributions are remarkably similar. Although minor deviations are visible in the tails, the overall shape and key moments align closely, suggesting that our method provides a highly accurate approximation.

Notably, these deviations cannot be attributed solely to inaccuracies in either our method or the SSJ approach. Instead, they stem from a fundamental difference in the economic environments captured by the two methods. Incorporating aggregate risk, as our method does, leads poorer households to work more in response to increased precautionary motives, resulting in fewer households clustered at the bottom of the asset distribution, which is reflected in the shorter blue bars in the lower asset bins. At the same time, this increased labor supply and precautionary saving among the poor depresses the equilibrium real interest rate, thereby reducing the incentive for wealthier households to save. As a result, rich households accumulate less wealth in equilibrium, leading to a thinner right tail in the asset distribution. This pattern is

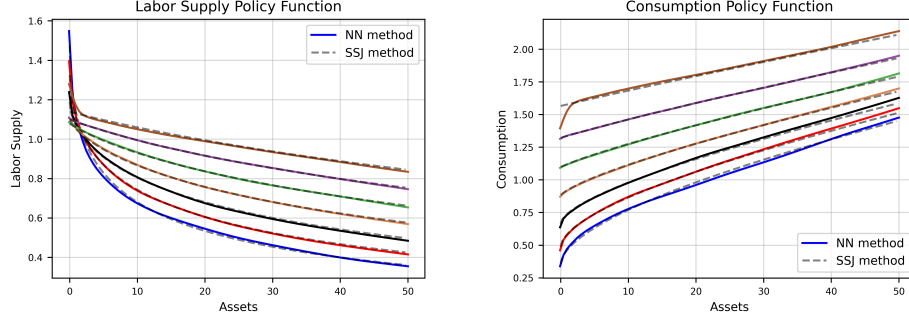


Figure 6: Individual policy functions comparison for the one-asset HANK model. The figure illustrates how the policy functions for labor and consumption vary for different asset levels and different productivity levels. We compare the NN method with the SSJ method. We use the 7 productivity levels that the discretization approach of the SSJ provides for comparison. We evaluate the policy functions at the stochastic steady state (NN) and deterministic steady state (SSJ).

clearly illustrated by the crossing of the blue solid and black dashed lines in the right panel of Figure 5, which plots the cumulative asset distribution.

The next step is to compare the two sets of policy functions that determine agents' labor supply and consumption decisions, as shown in Figure 6. The solid lines depict the policy functions approximated using our method, while the dashed lines correspond to the approximation obtained using the SSJ method from Auclert et al. (2021). Each color represents one of the seven idiosyncratic labor productivity levels considered in Auclert et al. (2021). The brown lines correspond to the most productive agents (top productivity bin), while the blue lines represent the least productive ones (bottom productivity bin.)²⁶

The comparison yields two main insights. First, our method closely replicates the policy functions obtained with the SSJ approach. Second, the only notable discrepancy concerns the behavior of highly productive agents with zero assets, a group representing an extremely small share of the population. These agents were previously low-productivity individuals constrained at the borrowing limit who experience a sudden, large productivity shock that propels them to the highest productivity level

²⁶Note that these seven productivity levels are used solely for the purpose of comparing the results from our solution method with those from SSJ. It is worth recalling that our solution method does not rely on discretizing the productivity space. Instead, we treat idiosyncratic labor productivity as the outcome of a continuous AR(1) process, as specified in the model.

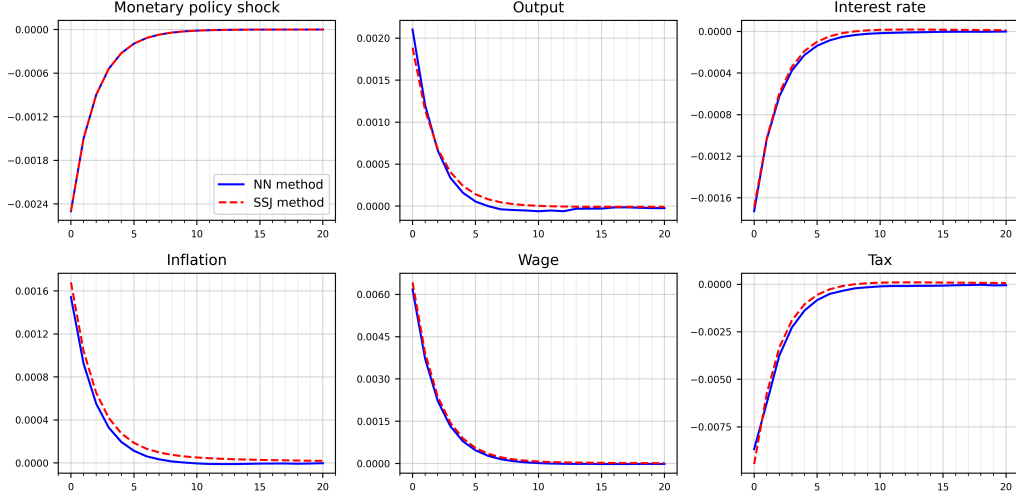


Figure 7: Impulse response functions comparison of NN method and SSJ approach. The deviations from the stochastic steady state (for the NN method) and from the deterministic steady state (for the SSJ method) are reported for a one-standard-deviation negative monetary policy shock. $L=100$ and $MC = 100$.

in a single step. Such transitions are exceedingly rare, providing the NN with little data to learn from and resulting in a less precise approximation of their behavior.

A similar argument explains why the two solutions diverge when very large asset stocks are considered in the graph. Few agents hold such high levels of wealth in equilibrium, making it difficult to train the NN accurately in that region of the state space. Nonetheless, the inaccuracies in these cases are fairly limited, highlighting the strong extrapolative properties of NNs. Nevertheless, since these “extreme” types of agents have almost zero measure in the population, the approximation error introduced by the NN in this specific region is clearly inconsequential for aggregate outcomes. Indeed, the effects of a monetary policy shock on the equilibrium dynamics of aggregate variables predicted by the two methods are quite similar, as shown in Figure 7.

3.4 Proof of concept 4: A quantitative HANK model

In this section, we show that our method can be applied to solve and estimate the likelihood of a quantitative HANK model similar to those in Auclert et al. (2020), Bayer et al. (2024a), and Hagedorn et al. (2025). The model embeds several features typically introduced to enhance the empirical fit of macroeconomic models. For instance, the model includes capital accumulation with investment subject to adjustment costs. Wage and price setting follow Rotemberg-style nominal rigidities with partial indexation, introducing backward-looking components in both Phillips curves.

The model is driven by seven aggregate shocks: a technology growth shock, price and wage markup shocks, a monetary policy shock, a government spending shock, a marginal efficiency of investment shock, and a risk-premium shock. A detailed description of the model is provided in Appendix E.

Building on NN techniques, our approach introduces two key methodological innovations for solving and estimating quantitative HANK models. First, it preserves the model’s full nonlinear aggregate dynamics while explicitly accounting for aggregate risk generated by the seven shocks. Second, it allows us to estimate the standard deviation and persistence of the idiosyncratic income process that require solving for the deterministic steady state, a step not undertaken in previous studies. Moreover, a nonlinear quantitative HANK model has not been solved before.

A quantitative HANK model. First, we address the question of whether this HANK model can truly be considered a quantitative model. To examine this critical issue for our proof of concept, we rely on the model’s predictive checks.²⁷ In practice, to compute predictive checks for our model, we draw parameter values from the support of the prior hypercube and simulate the model for 262 quarterly periods at

²⁷Prior predictive checks, widely used in Bayesian analysis, serve as a diagnostic tool to assess whether a model and its priors can plausibly generate data consistent with empirical observations. These checks are typically performed before confronting the model with the data. See Leeper et al. (2017a) for an application of prior predictive checks to macro models.

each draw. This procedure is computationally feasible because our approach does not require re-solving the model or retraining the NNs at each parameter draw, as parameters are treated as pseudo-state variables. For each simulation, we compute the standard deviations and serial correlations of the model variables, obtaining *predictive distributions* for these statistics.

We then compute the same statistics for seven observables for the US: GDP growth, consumption growth, investment growth, real wage growth, hours worked growth, inflation, and the federal funds rate, observed over the sample period 1954Q3 to 2019Q4 (262 quarters). These observable variables are standard in empirical macroeconomic research and are used in studies that estimate empirical HANK models, such as Auclert et al. (2021) and Bayer et al. (2024a). If these empirical moments lie within regions where the predictive distributions assign probability mass, the model can be regarded as capable of explaining those moments; otherwise, it is misspecified along that dimension.

The predictive checks are shown in Appendix E.3. Overall, the model’s predictive distributions align closely with the data’s standard deviations and most autocorrelations, indicating the quantitative suitability of the HANK model under consideration.

Solution accuracy for the quantitative HANK model. The second step is to demonstrate that our method can solve this quantitative HANK model, featuring more than 1,200 state variables. An approach is to evaluate the accuracy of our solution *after* the training is completed. To this end, we compute Euler equation errors following Judd (1992) and Aruoba et al. (2006). We use the same set of equilibrium conditions employed during training, namely the consumption Euler equation, the first-order conditions for capital and investment, the Phillips curves for prices and wages, and the capital aggregation residuals.²⁸

Residual errors for the six equilibrium conditions are evaluated using a 10,000-period simulation for 1,000 randomly drawn pseudo-state variables (i.e., parameters)

²⁸These conditions are shown in Appendix E.4.

from the truncated hypercube defining the support of the prior distribution that we will use in estimation. Figure 8 presents the post-training absolute residual errors. The residuals remain well-contained, even in the left tails of the distributions. The magnitude of these errors is comparable to those from the one-asset HANK model, whose solution was accurately approximated by our method as shown in Section 3.3. Each distribution is approximately bell-shaped, with modes close to the corresponding mean absolute errors, indicating that large approximation errors are rare and unlikely to distort the model’s dynamics.

Furthermore, the symmetry and tight concentration of the distributions around small values provide additional evidence for the reliability of the NN solution, indicating that the method performs consistently well across a wide range of simulated states and parameter values.

The left tail of these distributions is particularly important for assessing whether the training of the solution network is sufficiently accurate to permit reliable estimation. Since estimation requires evaluating the likelihood at many, possibly fairly dispersed, points in the parameter space, pseudo-state variables’ values that cause the trained NN to yield large residual errors would corrupt the likelihood evaluation performed by the NN particle filter. The left tails of these distributions appear well-behaved and decline rapidly in all six cases.

Estimation of the quantitative HANK model. The final step is to show that the model can be estimated using the NN particle filter and that the true parameter values used to simulate the data can be recovered. To this end, we simulate the seven observable variables used in the prior predictive checks and employed by Auclert et al. (2021) and Bayer et al. (2024a) to estimate their quantitative HANK models. The length of the simulated sample is 262 quarters.

Table 2 reports the prior and posterior moments of the estimated parameters. The estimation recovers most true parameter values accurately. Some priors are set

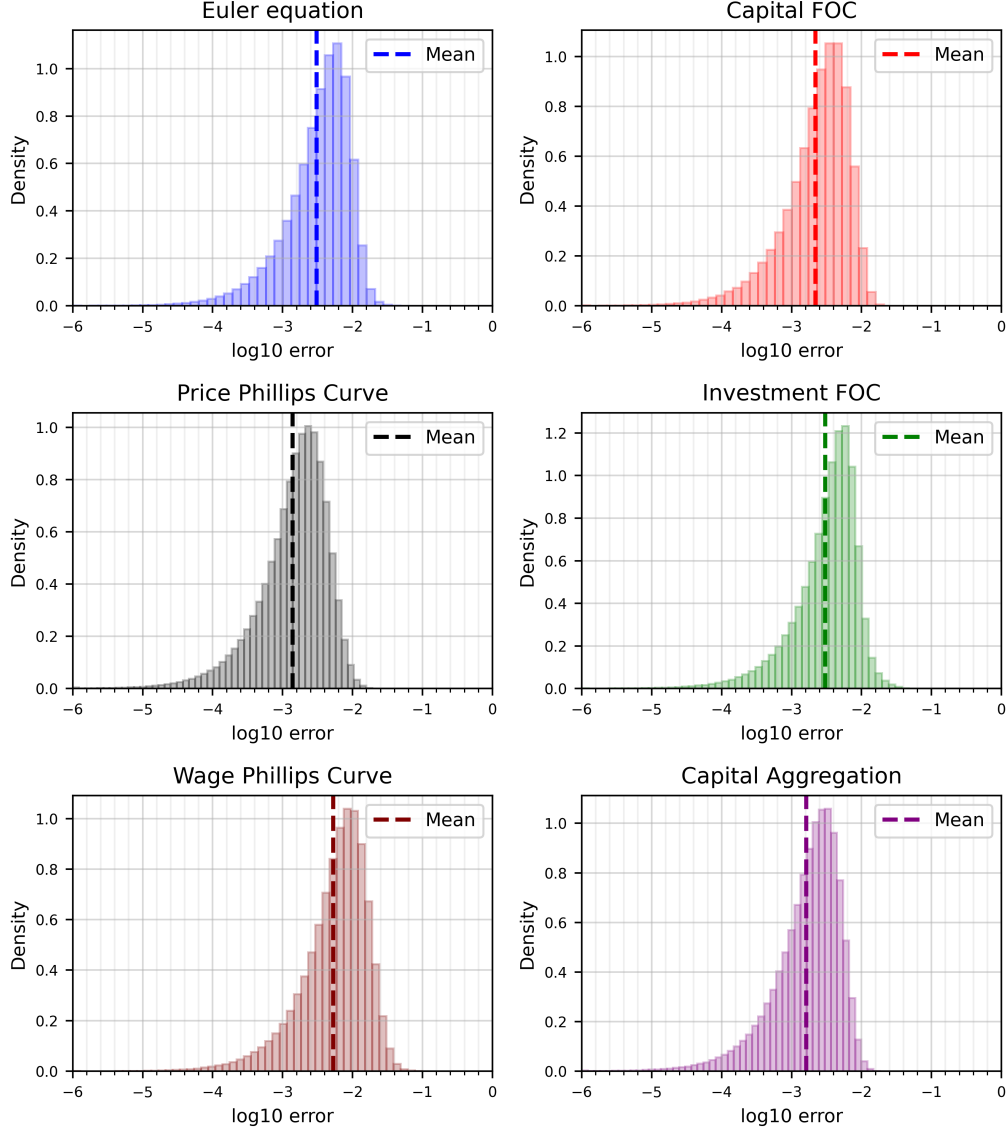


Figure 8: Distribution of post-training approximation errors. The figure shows the absolute error for the equilibrium conditions defined in Appendix E.4. Errors are evaluated after the training of the NNs is complete. The distributions are based on a simulation of 5,000 periods for 100 parameter draws. We evaluate the expectations, relevant to the modified Euler equation and the NKPC, using 1,000 Monte Carlo draws. The horizontal axis displays the $\log_{10}$ of the value.

relatively tight due to the flatness of the likelihood, partially reflecting the limited information content of the seven observed macro variables and the short sample period. The main culprit of the flatness, however, is the inaccuracy of the particle filter used to train our NN particle filter.

The issue is not specific to the type of filter used for training. Existing filters, when applied to nonlinear models of the scale of the quantitative HANK model considered here (with over 1,200 state variables), yield noisy and imprecise evaluations of the likelihood. The particle filter’s inaccuracy is evident from validation errors, shown in Appendix E.6, which are two to three orders of magnitude larger than those in the stylized RANK model, shown in Appendix B.3. This inaccuracy mainly reflects filtering error rather than solution error, as the residuals’ left tails are short and well-behaved (see Figure 8) and comparable to those from the one-asset HANK model which was solved very accurately by our methods (Section 3.3).²⁹

To generate the training sample for the NN particle filter, it takes approximately 9–10 hours for 1,000 evaluations of the likelihood on a single machine equipped with a decent GPU. This process can be parallelized across multiple machines when generating the training data to accelerate computation. The training of the NN particle filter itself takes around 10 minutes. Estimating the model from the solution stage to the Metropolis-Hastings sampling takes approximately 131 hours on an Nvidia RTX 3090 with 24GB of memory as a GPU.

Once the NN particle filter is trained, obtaining millions of posterior draws via RWMH is fast and computationally inexpensive. We obtain 10 million posterior draws. This large number of posterior draws is feasible because at each draw, the likelihood function is evaluated through our trained neural-network particle filter in a fraction of second, without any need for retraining.³⁰

²⁹The figure reporting the residual errors for the one-asset HANK model of Section 3.3 is shown in Appendix D.

³⁰Acharya and Dogra (2020) derive conditions for determinacy in HANK models with linearly approximated aggregate dynamics, and we always verify that these conditions are satisfied in our analysis. Moreover, highly persistent or volatile preference shocks can lead to the nonexistence of a stable equilibrium in DSGE models with a ZLB constraint (Bianchi et al., 2021). We truncate the priors for these parameters to minimize such issues, although this does not guarantee the existence and uniqueness of an equilibrium. Equilibrium multiplicity makes NNs difficult to train, and convergence failures often reveal it, as the stochastic nature of training amplifies instability and hinders convergence. Explosive dynamics in training, in turn, can indicate the nonexistence of a stable rational expectations equilibrium.

Estimation									
Parameters		True	Prior					Posterior	
		Value	Type	Mean	Std	Lower Bound	Upper Bound	Mean	Std
<i>Idiosyncratic risk</i>									
Persistence idiosyncratic risk	ρ_s	0.90	Trc.N	0.90	0.01	0.85	0.95	0.882	0.007
Std. dev. idiosyncratic risk	σ_s	0.20	Trc.N	0.20	0.01	0.15	0.25	0.206	0.008
<i>Other Parameters</i>									
MP inflation response	ϕ_π	2.0	Trc.N	2.00	0.5	1.50	3.00	2.317	0.084
MP output response	ϕ_y	0.25	Trc.N	0.25	0.25	0.00	0.50	0.397	0.032
Inflation indexation	ι_p	0.25	Trc.N	0.25	0.25	0.00	0.50	0.289	0.080
Wage indexation	ι_w	0.25	Trc.N	0.25	0.03	0.00	0.50	0.247	0.041
Rotemberg costs price PC	φ^p	100	Trc.N	100	2.50	75	125	103.9	2.75
Rotemberg costs wage PC	φ^w	100	Trc.N	100	2.50	75	125	98.5	2.93
Persistence risk-premium	ρ_ζ	0.90	Trc.N	0.90	0.05	0.75	0.95	0.882	0.022
Persistence wage mark-up	ρ_{μ^w}	0.90	Trc.N	0.90	0.05	0.75	0.95	0.884	0.022
Persistence price mark-up	ρ_{μ^p}	0.90	Trc.N	0.90	0.01	0.75	0.95	0.920	0.010
Persistence MP shock	ρ_m	0.90	Trc.N	0.90	0.01	0.75	0.95	0.907	0.010
Persistence investment eff.	ρ_ν	0.90	Trc.N	0.90	0.01	0.75	0.95	0.897	0.011
Persistence government	ρ_g	0.90	Trc.N	0.90	0.005	0.75	0.95	0.900	0.005
Std. dev. risk-premium	$100\sigma_\zeta$	0.2	Trc.N	0.2	0.5	0.1	0.4	0.240	0.021
Std. dev. growth rate	$100\sigma_z$	0.2	Trc.N	0.2	0.5	0.1	0.4	0.187	0.018
Std. dev. wage mark-up	$100\sigma_{\mu^w}$	5.0	Trc.N	5.0	5.0	2.0	8.0	5.40	0.258
Std. dev. price mark-up	$100\sigma_{\mu^p}$	5.0	Trc.N	5.0	5.0	2.0	8.0	5.26	0.222
Std. dev. MP shock	$100\sigma_m$	0.1	Trc.N	0.1	0.01	0.05	0.2	0.093	0.011
Std. dev. government	$100\sigma_g$	0.5	Trc.N	0.5	0.25	0.1	1.0	0.419	0.022
Std. dev. investment eff.	$100\sigma_\nu$	1.0	Trc.N	1.0	0.5	0.5	1.5	0.834	0.055

Table 2: Prior and Posterior distributions. Prior and posterior moments for the estimation of the quantitative HANK model. The prior type indicates the prior density function. Trc.N stands for truncated normal distribution.

4 Estimation of a HANK model with the ZLB

In this section, we show how to apply our methods to estimate the nonlinear HANK model with the ZLB constraint using U.S. data. We consider an off-the-shelf, stylized HANK model that features idiosyncratic and aggregate risk. A detailed description of the model is provided in Appendix F.

Estimation									
Parameters			Prior				NN		
		Type	Mean	Std	Lower Bound	Upper Bound	Posterior		
							Median	5%	95%
<i>Parameters affecting the DSS</i>									
Std. dev. idiosyncratic risk	$100\sigma_s$	Trc.N	5.00	1.000	2.50	10.0	7.04	5.67	8.10
Borrowing limit	$\underline{B}$	Trc.N	-0.50	0.010	-0.65	-0.35	-0.50	-0.54	-0.46
<i>Other parameters</i>									
Rotemberg costs NKPC	φ	Trc.N	100	5.00	70	120	101	94	107
MP inflation response	θ_Π	Trc.N	2.25	0.125	1.75	2.75	2.43	2.20	2.67
MP output response	θ_Y	Trc.N	1.00	0.025	0.75	1.25	0.96	0.92	1.00
Persistence growth rate	ρ_z	Trc.N	0.40	0.025	0.2	0.6	0.43	0.39	0.47
Persistence MP shock	ρ_m	Trc.N	0.90	0.005	0.85	0.95	0.91	0.90	0.91
Std. dev. preference	$100\sigma_\zeta$	Trc.N	1.50	0.100	1.00	2.00	1.22	1.10	1.33
Std. dev. growth rate	$100\sigma_z$	Trc.N	0.40	0.100	0.30	0.60	0.47	0.43	0.53
Std. dev. MP shock	$100\sigma_m$	Trc.N	0.06	0.010	0.05	0.20	0.15	0.14	0.16

Table 3: Prior and Posterior distributions. The table the prior and posterior moments for the estimation with real data. The prior type indicates the prior density function. Trc.N stands for truncated normal distribution.

4.1 Data, calibrated parameters, and priors

We use data on U.S. quarterly real per-capita GDP growth, U.S. quarterly GDP deflator inflation rate, and the shadow interest rate estimated by Wu and Xia (2016).³¹ The model’s counterpart of the shadow rate is the notional rate – representing the interest rate that would have been set by the central bank in the absence of the ZLB constraint.³² All observables are in percent and annualized. Our analysis covers a sample period starting from the first quarter of 1990 and extending through the fourth quarter of 2019. To avoid degeneracy problem in evaluating the likelihood, we introduce an i.i.d. measurement error for each observable.

³¹While it is in principle feasible to include cross-sectional variables directly in our estimation, we instead incorporate cross-sectional information through the prior distribution for the volatility of idiosyncratic income risk. Research combining aggregate and cross-sectional data in likelihood-based estimation remains in its early stages, and introducing such additional complexity would go beyond the scope of this study (e.g. Liu and Plagborg-Møller, 2023).

³²Since our model features a ZLB constraint, we could rely on the observed federal funds rate; however, the shadow rate provides a better measure of the policy stance at the ZLB, as it reflects the notional rate perceived by agents and the influence of unconventional tools such as forward guidance and quantitative easing.

We calibrate some parameters of the model, as presented in Appendix F. Table 3 shows the prior distributions. The prior distributions are centered at values that are broadly consistent with the parameter values of linearized RANK models estimated in the literature. Regarding the parameters affecting the DSS, the prior mean of the borrowing limit is set to approximately 2 month labor income as in Fernández-Villaverde et al. (2025), while the prior of idiosyncratic income volatility is set in line with the Gini coefficient on the US wealth inequality measured by the World Inequality Database over a period ranging from 1990 to 2019.

4.2 Estimation with neural networks

Details regarding the specifications of the NNs can be found in Appendix G, while convergence statistics are provided in F. After completing the training of these NNs, evaluating the likelihood of the nonlinear HANK model takes only seconds. Hence, we can conduct the RWMH aiming for an acceptance rate of 20% – 40%. Details on how to perform Bayesian estimation of this model are provided in Appendix A.

The posterior moments are shown in Table 3. Bayesian updating is modest for most parameters. Since the prior mean for these parameters was set to reflect typical estimates from linearized RANK models, we interpret this finding as suggesting that heterogeneity and nonlinearities do not lead to substantial revisions to the estimated values of those parameters.³³

4.3 Interactions between heterogeneity and nonlinearities

In this section, we analyze the effects of the interaction between heterogeneity and nonlinearities; namely, the ZLB constraint in a HANK model with aggregate risk. The ZLB constraint interacts significantly with idiosyncratic income risk, amplifying

³³Bayesian updating might be stronger if more observable variables (e.g., consumption or wages) are used in estimation, but this requires extending the HANK model, which is feasible but beyond the scope of this methodological paper.

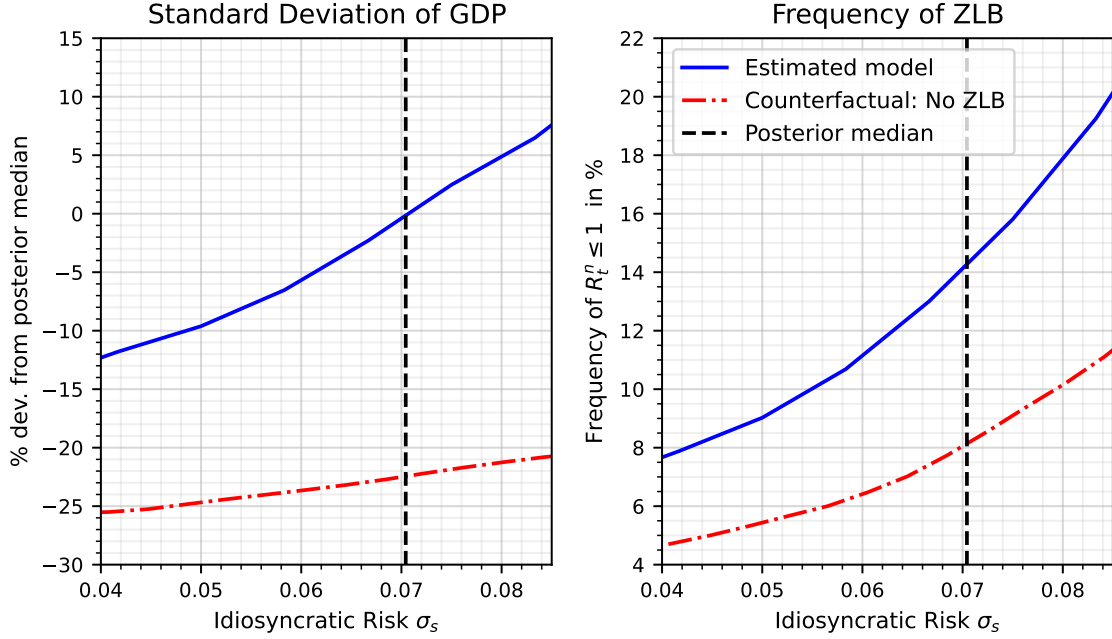


Figure 9: Volatility of output and ZLB frequency. *Left chart:* Volatility of output as a function of the idiosyncratic risk in the estimated model (blue solid line) and in the counterfactual case in which the ZLB is not enforced (red dashed-dotted line). *Right chart:* Frequency of ZLB episodes as a function of the idiosyncratic risk in the estimated model (blue solid line). Frequency of episodes when the interest rate turns negative in the counterfactual case in which the ZLB is not enforced (red dashed-dotted line). The value of the other parameters is set to their posterior median. The standard deviation of output and the frequencies are computed from simulations of 100,000 periods.

the volatility of macroeconomic variables. In Figure 9, we illustrate that a heightened risk of encountering the ZLB amplifies output volatility in our nonlinear HANK model. This amplification is represented by the vertical distance between the blue solid line and the red dashed-dotted line. The ZLB risk exacerbates output volatility because the effectiveness of monetary policy in stabilizing the economy at the ZLB is diminished. The vertical black line, representing the estimated volatility of idiosyncratic risk, shows that approximately 22 percentage points of GDP volatility are explained by the occasionally binding ZLB constraint.

Importantly, the amplification effect of the ZLB on output volatility intensifies with higher levels of idiosyncratic income risk. In Figure 9, we observe that the vertical distance between the solid blue line and the red dashed-dotted line expands as the level of idiosyncratic risk increases along the x-axis.

To delve into the rationale behind this finding, we examine how the frequency of the ZLB changes with the level of idiosyncratic risk. As shown by the blue solid line on the right chart, as idiosyncratic risk increases, the ZLB becomes binding more often. This occurs because higher idiosyncratic risk reduces the steady-state nominal interest rate, leading to a heightened frequency of encountering the ZLB.³⁴

5 Concluding remarks

We propose an NN-based method for solving and estimating nonlinear dynamic general equilibrium models with heterogeneous agents. The approach handles high-dimensional state spaces, captures nonlinear dynamics such as the zero lower bound (ZLB) and borrowing constraints, accounts for aggregate uncertainty, and enables the estimation of parameters that determine the steady state. Applying the framework to U.S. data, we estimate a nonlinear HANK model with a ZLB constraint and meaningful aggregate uncertainty, highlighting the interactions between heterogeneity and nonlinearity. The framework can be extended to large-scale applications, including multi-sector and multi-country settings, and provides a foundation for evaluating how quantitative HANK models capture episodes of large shocks and persistent ZLB conditions.

³⁴The relationship between the frequency of negative interest rates and the level of idiosyncratic risk remains positive – albeit somewhat less strong – even when the ZLB constraint is not enforced, as indicated by the red dashed-dotted line on the right chart. This is because higher idiosyncratic income risk prompts households to increase precautionary savings, thereby depressing the nominal interest rate and making it more likely for nominal interest rates to fall into the negative territory.

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Appendix

A Neural network based Bayesian estimation

The following Random Walk Metropolis-Hastings (RWMH) algorithm can be used to estimate the parameters $\tilde{\Theta}$, where we use the NN approximation of the likelihood function. To integrate our NN approach for Bayesian estimation, we adapt the RWMH algorithm of Atkinson et al. (2020).³⁵ The approach is as follows:

1. Initialize the algorithm: Obtain a first proposal density from which the parameters: $\tilde{\Theta}$ are drawn, as follows:

Draw from the prior distribution a candidate vector $\tilde{\Theta}^{New}$. Evaluate the log-likelihood of the candidate vector using the trained NN for the particle filter $\log \mathcal{L}_{\tilde{\Theta}}(\mathbb{Y}_{1:T}|\tilde{\Theta}^{New})$ and combine it with the prior to obtain the log-posterior $\log g(\mathbb{Y}_{1:T}|\tilde{\Theta}^{New})$.

Repeat these steps N^{init} times and collect all draws as $\tilde{\Theta}^{init}$. Construct an approximation for the covariance matrix by selecting draws where the likelihood is above the 90% quantile, denoted by $\hat{\Theta}$, and subtract the mean $\check{\Theta} = \hat{\Theta} - \bar{\tilde{\Theta}}$. Calculate the covariance matrix $\Sigma = (\check{\Theta}\check{\Theta}')/(0.1N^{init})$. Set $\check{\Theta}$ to equal the mode, the draw associated with the highest likelihood.

2. Run the RWMH algorithm as burn-in. Repeat the steps below N^{Burn} times:

Draw a new parameter vector $\tilde{\Theta}^{New}$ from the proposal density to evaluate the log-posterior. The proposal density is a multivariate normal distribution with mean vector $\check{\Theta}$ and covariance matrix $c\Sigma$, where c is set to achieve an acceptance rate between 20 and 40%.

³⁵See also Herbst and Schorfheide (2015) for more information on RWMH.

Evaluate the log-likelihood of the parameter vector using the trained NN for the particle filter $\log \mathcal{L}_{\tilde{\Theta}} \left(\mathbb{Y}_{1:T} | \tilde{\Theta}^{New} \right)$ and combine it with the prior to obtain the log-posterior $\log g(\mathbb{Y}_{1:T} | \tilde{\Theta}^{New})$. Accept the draw if $\exp(\log g(\mathbb{Y}_{1:T} | \tilde{\Theta}^{New}) - \log g(\mathbb{Y}_{1:T} | \tilde{\Theta}))$ is larger than the draw from a standard uniform distribution. If the draw is accepted, the mean of the proposal density is updated to $\check{\Theta} = \tilde{\Theta}^{New}$.

3. Use these N^{Burn} draws to get an updated proposal density:

Keep only the last 75% of draws, denoted as $\hat{\Theta}^b$. Calculate the deviations of each draw from the mean $\check{\Theta}^b = \hat{\Theta}^b - \bar{\tilde{\Theta}}^b$. Calculate the covariance matrix $\Sigma^b = (\check{\Theta}^b \check{\Theta}^{b'}) / (0.75 N^{Burn})$. Calculate the mode $\check{\check{\Theta}}^b$ and update the mean of the proposal density $\check{\Theta} = \check{\check{\Theta}}^b$.

4. Run the RWMH algorithm to obtain the posterior:

Use the proposal density from above and repeat the steps below N^{final} times: Draw a new parameter vector $\tilde{\Theta}^{New}$ from the proposal density to evaluate the log-posterior. The proposal density is a multivariate normal distribution with mean vector $\check{\Theta}$ and covariance matrix $c\Sigma$, where c is again tuned to have an acceptance rate that is between 20-40%. Evaluate the log-likelihood of the parameter vector using the trained NN for the particle filter $\log \mathcal{L}_{\tilde{\Theta}}^{NN} \left(\mathbb{Y}_{1:T} | \tilde{\Theta}^{New} \right)$ and combine it with the prior to obtain the log posterior $\log g(\mathbb{Y}_{1:T} | \tilde{\Theta}^{New})$. Accept the draw if $\exp(\log g(\mathbb{Y}_{1:T} | \tilde{\Theta}^{New}) - \log g(\mathbb{Y}_{1:T} | \tilde{\Theta}))$ is larger than the draw from a standard uniform distribution. If the draw is accepted, the mean of the proposal density is updated to $\check{\Theta} = \tilde{\Theta}^{New}$.

5. N^{final} draws characterize the posterior distribution.

B Linearized DSGE model

The model is a prototypical three-equation New Keynesian DSGE model with a TFP shock. The model is log-linearized around its unique steady-state equilibrium, which results in the following equations.

$$\hat{Y}_t = E_t \hat{Y}_{t+1} - \sigma^{-1} \left(\hat{R}_t - E_t \hat{\Pi}_{t+1} \right), \quad (12)$$

$$\hat{\Pi}_t = \kappa(\hat{Y}_t - \hat{Y}_t^*) + \beta E_t \hat{\Pi}_{t+1}, \quad (13)$$

$$\hat{R}_t = \phi_\Pi \hat{\Pi}_t + \phi_Y \hat{X}_t, \quad (14)$$

$$\hat{Y}_t^* = \omega \hat{A}_t, \quad (15)$$

$$\hat{A}_t = \rho_A \hat{A}_{t-1} + \sigma_A \epsilon_t^A, \quad (16)$$

where we define $\omega = (1 + \eta)/(\eta + \sigma)$ as well as $\kappa = (1 - \phi)(1 - \phi\beta)(\sigma + \eta)/\phi$. Output is defined as $\hat{Y}_t = (Y_t - Y)/Y$, inflation as $\hat{\Pi}_t = \Pi_t - \Pi$, nominal rate as $\hat{R}_t = R_t - R$, output in the flex price economy as $\hat{Y}_t^* = (Y_t^* - Y^*)/Y^*$, TFP as $\hat{A}_t = (A_t - Y)/A$, and the output gap as $\hat{X}_t = \hat{Y}_t - \hat{Y}_t^*$. The shock ϵ_t^A follows a standard normal distribution.

After some manipulations, the model equations can be written as:

$$\hat{X}_t = E_t \hat{X}_{t+1} - \sigma^{-1} \left(\phi_\Pi \hat{\Pi}_t + \phi_Y \hat{X}_t - E_t \hat{\Pi}_{t+1} - \hat{R}_t^* \right), \quad (17)$$

$$\hat{\Pi}_t = \kappa \hat{X}_t + \beta E_t \hat{\Pi}_{t+1}, \quad (18)$$

$$\hat{R}_t^* = \rho_A \hat{R}_{t-1}^* + \sigma(\rho_A - 1)\omega\sigma_A \epsilon_t^A, \quad (19)$$

where $\hat{R}_t^*$ is the natural rate of interest, which follows an exogenous process derived from the TFP process.

We set up the NN as in section B.1 and train it by minimizing the residual error constructed from equations (17) and (18). The law of motion of the exogenous state variable, described in equation (19), is used for sampling.

The analytical solution for the output gap $\hat{X}_t$ and inflation $\hat{\Pi}_t$, which can be

derived with the method of undetermined coefficients, is given as:

$$\hat{X}_t = \frac{1 - \beta\rho_A}{(\sigma(1 - \rho_A) + \theta_Y)(1 - \beta\rho_A) + \kappa(\theta_\Pi - \rho_A)} \hat{R}_t^* \quad (20)$$

$$\hat{\Pi}_t = \frac{\kappa}{(\sigma(1 - \rho_A) + \theta_Y)(1 - \beta\rho_A) + \kappa(\theta_\Pi - \rho_A)} \hat{R}_t^* \quad (21)$$

The analytical solution underscores the idea of extended policy functions, which are simultaneously conditioned on the parameters and state variables. The analytical solution for the output gap and inflation depends on both the state variable $\hat{R}_t^*$ and the parameters.

B.1 Mapping of the model to the general framework

We can map the linearized NK model in the general form of the solution method outlined in section 2.2. The state variable, the structural shock, and the control variables of the model are:

$$\mathbb{S}_t = \left\{ \hat{R}_t^* \right\}, \quad \text{and} \quad \nu_t = \left\{ \epsilon_t^A \right\}, \quad \text{and} \quad \psi_t = \left\{ \hat{X}_t, \hat{\Pi}_t \right\}. \quad (22)$$

The parameters of the model are divided into calibrated ($\bar{\theta}$), which is an empty set in this application, and estimated ones ($\tilde{\theta}$):

$$\tilde{\Theta} = \left\{ \beta, \sigma, \eta, \phi, \theta_\Pi, \theta_Y, \rho_A, \sigma_A \right\}. \quad (23)$$

The NN ψ_{NN} is trained to determine the output gap and inflation:

$$\begin{pmatrix} \hat{X}_t \\ \hat{\Pi}_t \end{pmatrix} = \psi_{NN} \left(\mathbb{S}_t, \tilde{\Theta} | \bar{\Theta} \right). \quad (24)$$

B.2 Neural network for solving the model

The NN trained to approximate the policy function achieves a very high degree of accuracy, as shown in Figure 10. Figure 11 displays the extended policy function for

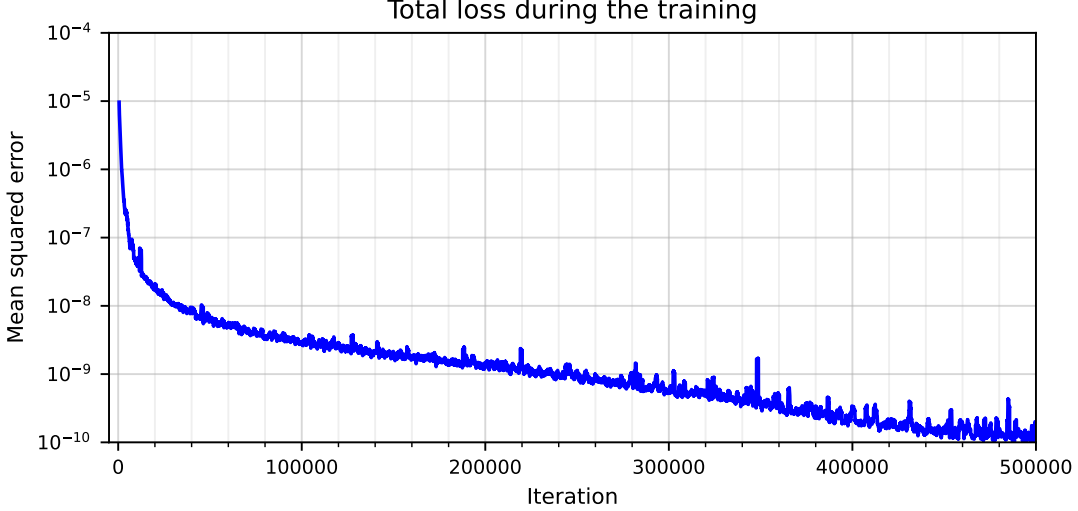


Figure 10: Convergence of the NN solution. The figure shows the dynamics of the mean squared error. The loss is calculated as a moving average over 1,000 iterations. The vertical axis has a logarithmic scale.

inflation, conditioned on the parameters and presented in the paper.

B.3 Neural network for likelihood approximation

Figure 12 shows the likelihood produced by the NN for varying different parameters (blue line). The orange dots show the mapping between the parameter and the likelihood directly coming from the particle filter. Even though the filter delivers only very noisy estimates of the likelihood (due to the choice of tracking only 100 particles in training), the NN is able to correct this inaccuracy and interpolate the likelihood function by cutting through the noise generated by the particle filter.

The dynamics of total loss during the training of the NN are depicted in Figure 13. The total loss converges to a value whose magnitude reflects the accuracy of the particle filter, primarily dependent on the number of particles used (in this case, 100 particles).

To assess potential concerns about overfitting of the NN particle filter, in the right chart of Figure 13, we report the total loss when the likelihood is evaluated at the parameters of the validation sample using the NN weights trained in every epoch (x-axis). The proximity of errors across epochs in both samples, depicted on the left and

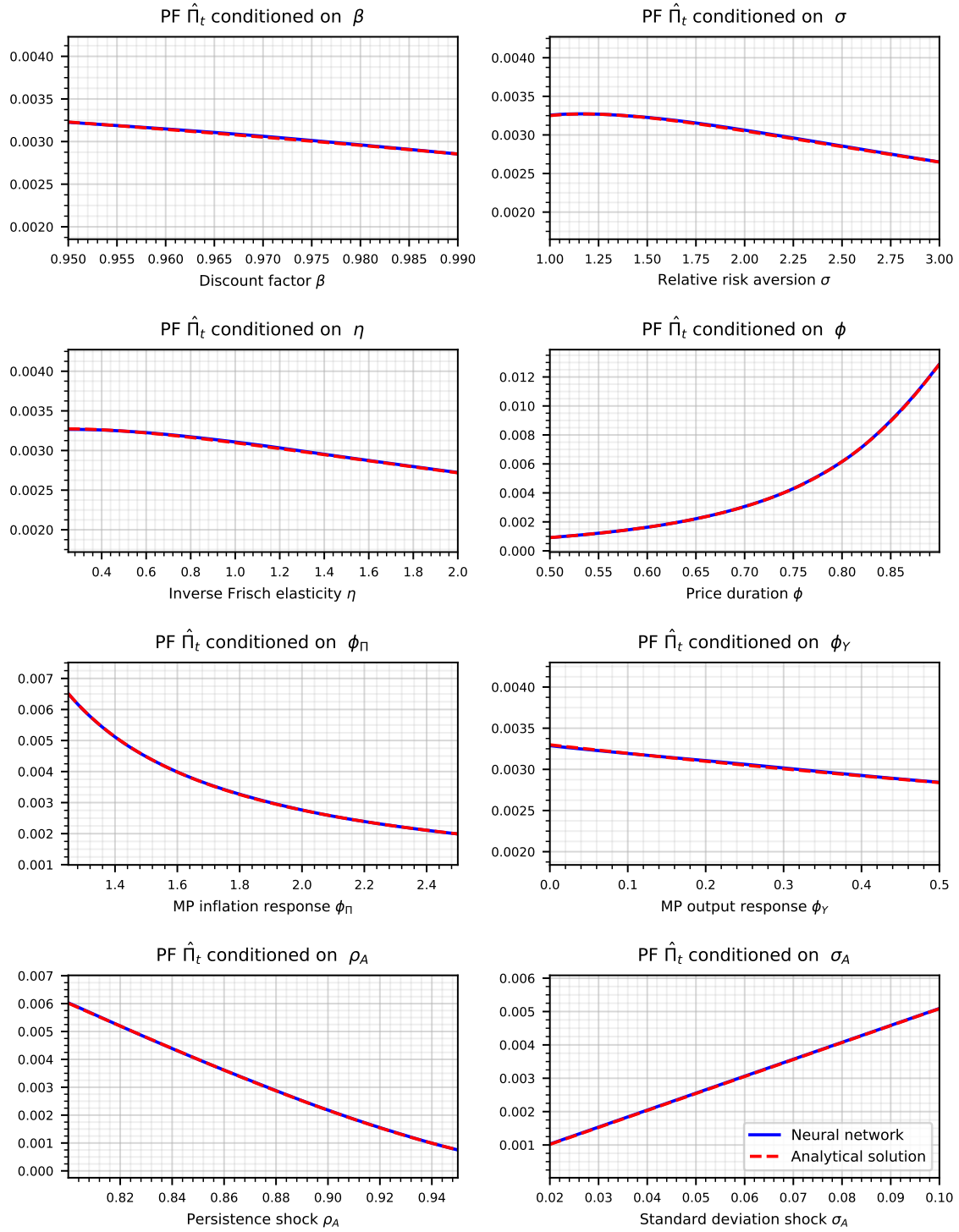


Figure 11: Accuracy of the NN solution method. Comparison between the analytical policy function for inflation and the approximation based on NNs. The values of the fixed parameters in each graph are set to the middle of their bounds.

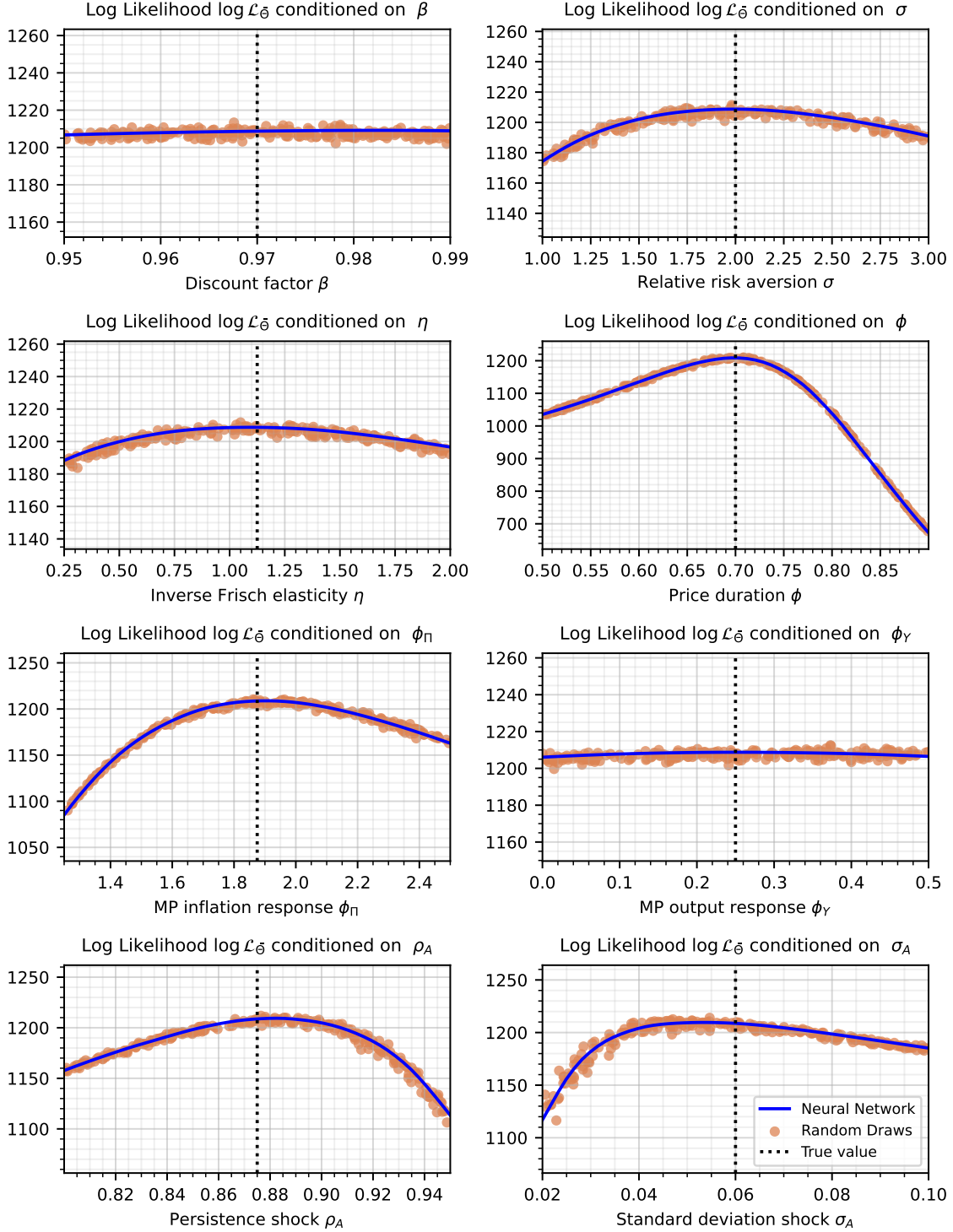


Figure 12: Accuracy in likelihood evaluations: NN particle filter vs. standard particle filter. The log-likelihood of the model as a function of different parameters. The value of the fixed parameters is set to the middle of their bounds.

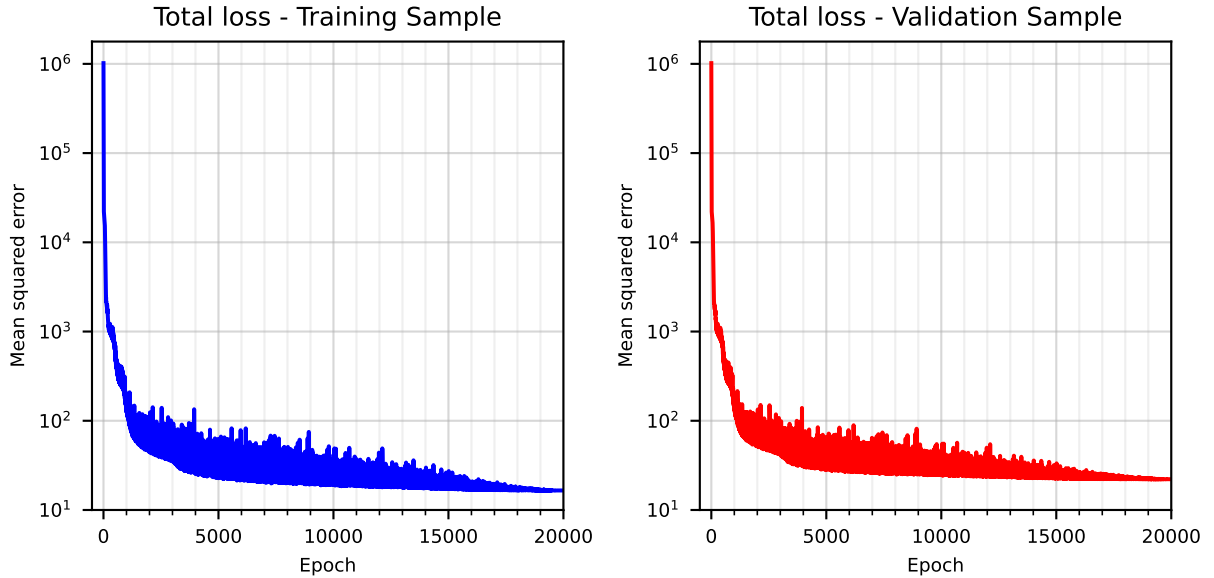


Figure 13: Training and validation convergence. The figure shows the total loss over the training sample (left) and over the validation sample (right). An epoch is completed when all the training or validation sample points are utilized. The vertical axis is expressed in a logarithmic scale.

right charts, along with the consistent decrease in error over the validation sample, suggests that the NN is not overfitting.

C Nonlinear RANK model with the ZLB constraint

We now assess whether our NN estimation method can recover the true data-generating process of a nonlinear model. Specifically, a nonlinear RANK model with a ZLB. Furthermore, we compare the results with a state-of-the-art estimation method for nonlinear macroeconomic models. This method involves solving the model using global methods and then evaluating the likelihood using a particle filter for every draw of the Metropolis-Hastings algorithm.

C.1 A nonlinear RANK model with the ZLB constraint

We use a small-scale RANK model with the ZLB as a nonlinear element.

Households. The economy consists of a representative household. The household chooses consumption C_t , labor H_t and assets B_t to maximize their utility:

$$E_0 \sum_{t=0}^{\infty} \beta^t \exp(\zeta_t) \left[\left(\frac{C_t}{1-\sigma} \right)^{1-\sigma} - \chi \left(\frac{1}{1+\eta} \right) (H_t)^{1+\eta} \right],$$

where ζ_t is an aggregate preference shock, which follows an AR(1) process $\zeta_t = \rho_\zeta \zeta_{t-1} + \epsilon_t^\zeta$. C_t is aggregate consumption. The budget constraint in real terms is:

$$C_t + B_t = W_t H_t + \frac{R_{t-1}}{\Pi_t} B_{t-1} - T_t + Div_t, \quad (25)$$

where Div_t is the real dividend, W_t is real wage, R_t is the nominal interest rate, Π_t is the inflation rate and T_t is real lump sum tax. The first-order conditions are:

$$1 = \beta R_t \mathbb{E}_t \left[\left(\frac{\exp(\zeta_{t+1})}{\exp(\zeta_t)} \right) \left(\frac{C_t}{C_{t+1}} \right)^\sigma \frac{1}{\Pi_{t+1}} \right], \quad (26)$$

$$\chi(H_t)^\eta = (C_t)^{-\sigma} W_t. \quad (27)$$

Firms. The firm sector consists of a continuum of final goods producers and intermediate goods firms. The final goods retailers buy the intermediate goods and transform them into a homogeneous final good using a CES production technology:

$$Y_t = \left(\int_0^1 (Y_t^j)^{\frac{\epsilon-1}{\epsilon}} dj \right)^{\frac{\epsilon}{\epsilon-1}}, \quad (28)$$

where Y_t^j is the output of intermediate goods firm j . The equilibrium price of the final good and the demand for the intermediate goods of firm j can be expressed as:

$$P_t = \left(\int_0^1 (P_t^j)^{1-\epsilon} dj \right)^{\frac{1}{1-\epsilon}}, \quad \text{and} \quad Y_t^j = \left(\frac{P_t^j}{P_t} \right)^{-\epsilon} Y_t. \quad (29)$$

Intermediate goods producers are monopolistically competitive. The firm j uses labor N_t^j as input to produce output Y_t^j with the following production technology:

$$Y_t^j = A N_t^j, \quad (30)$$

where A is the total factor productivity. Labor is hired in competitive markets, where the wage is determined as follows:

$$W_t = AMC_t. \quad (31)$$

The firm j sets the price of its goods to maximize its profit subject to the demand curve for intermediate goods and Rotemberg adjustment costs for changing prices:

$$\max_{P_t^j} P_t^j \left(\frac{P_t^j}{P_t} \right)^{-\epsilon} \frac{Y_t}{P_t} - MC_t \left(\frac{P_t^j}{P_t} \right)^{-\epsilon} Y_t - \frac{\varphi}{2} \left(\frac{P_t^j}{\Pi P_{t-1}^j} - 1 \right)^2 Y_t, \quad (32)$$

where Π is the inflation target of the central bank. Imposing a symmetric equilibrium and discounting future profits with the real interest rate, the New Keynesian Phillips curve can be written as:

$$\left[\varphi \left(\frac{\Pi_t}{\Pi} - 1 \right) \frac{\Pi_t}{\Pi} \right] = (1 - \epsilon) + \epsilon MC_t + \varphi \mathbb{E}_t \frac{\Pi_{t+1}}{R_t} \left[\left(\frac{\Pi_{t+1}}{\Pi} - 1 \right) \frac{\Pi_{t+1}}{\Pi} \frac{Y_{t+1}}{Y_t} \right], \quad (33)$$

where $\Pi_t = P_t/P_{t-1}$. The Rotemberg adjustment costs are given back as a lump sum. The real dividends paid out by the firm sector are $Div_t = Y_t - W_t N_t$.

Policy makers. The central bank sets the nominal interest R_t using a Taylor rule that responds to inflation and output deviations from their targets Π and Y . The ZLB restricts the nominal interest rate. The rule is given as follows:

$$R_t = \max \left[1, R \left(\frac{\Pi_t}{\Pi} \right)^{\theta_\Pi} \left(\frac{Y_t}{Y} \right)^{\theta_Y} \right]. \quad (34)$$

The fiscal authority follows a passive policy rule, where it uses lump-sum taxes T_t to keep their debts D on a constant path:

$$D = \frac{R_{t-1}}{\Pi_t} D - T_t. \quad (35)$$

C.2 Mapping of the model to the general framework

We can map the RANK model in the general form of the outlined estimation procedure. The state variable, the structural shock, and the control variables are:

$$\mathbb{S}_t = \{\zeta_t\}, \quad \text{and} \quad \nu_t = \{\epsilon_t^\zeta\}, \quad \text{and} \quad \psi_t = \{N_t, \Pi_t\}. \quad (36)$$

The parameters of the model are divided in calibrated ($\bar{\theta}$) and estimated ones ($\tilde{\theta}$):

$$\bar{\Theta} = \{\beta, \sigma, \eta, \epsilon, \chi\}, \quad (37)$$

$$\tilde{\Theta} = \{\theta_\Pi, \theta_Y, \varphi, \rho_\zeta, \sigma_\zeta\}. \quad (38)$$

The NN is trained to determine labor supply and inflation, which is sufficient to back out other variables:

$$\begin{pmatrix} N_t \\ \Pi_t \end{pmatrix} = \psi_{NN}(\mathbb{S}_t, \tilde{\Theta} | \bar{\Theta}). \quad (39)$$

C.3 Data-generating process

The calibrated model is used as a data-generating process, providing a controlled environment for the estimation. The upper panel of Table 4 summarizes the calibration of the model, with parameters chosen such that the model occasionally encounters the ZLB. We use our developed approach to estimate the nonlinear model in a Bayesian setup. We then compare the results to an estimation with a state-of-the-art (non-NN) approach. The observables are the annualized quarterly output growth rate, annualized quarterly inflation rate, and the nominal interest rate, which are generated with the calibrated model and cover a span of 500 periods.

True parameters for the data-generating process					
Parameters			Value		
β	Discount factor	0.9975	θ_{Π}	Mon. pol. inflation response	2
σ	Relative risk aversion	1	θ_Y	Mon. pol. output response	0.25
η	Inverse Frisch elasticity	1	$4 \log(\Pi)$	Inflation target (annualized)	2
ϵ	Price elasticity demand	11	Y	Output target	1
χ	Disutility labor	0.91	ρ_{ζ}	Persistence preference shock	0.7
φ	Rotemberg pricing	1000	$100\sigma^{\zeta}$	Std. dev. preference shock	2

Estimation											
Par.	Prior					NN			Alternative Approach		
	Type	Mean	Std	Lower Bound	Upper Bound	Posterior			Posterior		
						Median	5%	95%	Median	5%	95%
θ_{Π}	Trc.N	2.0	0.1	1.5	2.5	2.04	1.92	2.15	2.06	1.93	2.20
θ_Y	Trc.N	0.25	0.05	0.05	0.5	0.250	0.240	0.260	0.248	0.237	0.260
φ	Trc.N	1000	50	700	1300	985	921	1047	970	909	1033
ρ_{ζ}	Trc.N	0.7	0.05	0.5	0.9	0.69	0.671	0.707	0.688	0.670	0.707
σ^{ζ}	Trc.N	0.02	0.0025	0.01	0.025	0.020	0.019	0.021	0.020	0.019	0.021

Table 4: Parameters of the nonlinear RANK model. The upper panel shows the calibrated parameters for the nonlinear RANK model. The lower panel shows the true values used for simulating the model, the prior distributions as well as the posterior distributions. The posterior is displayed for the NN-based estimation and an alternative (non-NN) state-of-the-art approach. Trc.N stands for a truncated normal distribution.

C.4 Estimation

Priors. The prior distributions are truncated normal densities.³⁶ The prior mean corresponds to the true value, while the standard deviation σ is loose to avoid the results being driven by the prior. The truncation ensures that the drawn parameters lie inside the bounds that have been imposed while solving the NN. The lower panel of Table 4 summarizes the priors.

Measurement equation. We base the analysis on the annualized quarterly output growth rate, annualized quarterly inflation rate, and nominal interest rate. All data

³⁶The probability density function of the truncated normal is $f(x; \mu, \sigma, a, b,) = \frac{1}{\sigma} \frac{\phi((x-\mu)/\sigma)}{\Phi((b-\mu)/\sigma) - \Phi((a-\mu)/\sigma)}$. If the bounds are not symmetric, the parameter μ does not correspond to the mean of the truncated normal. For simplicity, we refer to μ as the mean independent of the bounds.

are in percent. The measurement equation is given as:

$$\begin{bmatrix} \text{Output Growth} \\ \text{Inflation} \\ \text{Interest Rate} \end{bmatrix} = \begin{bmatrix} 400 \left(\frac{Y_t}{Y_{t-1}} - 1 \right) \\ 400 (\Pi_t - 1) \\ 400 (R_t - 1) \end{bmatrix} + u_t, \quad (40)$$

where the measurement error follows a Gaussian distribution $u_t \sim \mathcal{N}(0, \Sigma_u)$. The variance of the measurement error for each time series is a fraction m_E of its own variance. We set $m_E = 0.1$.

Training of the extended policy functions. The computationally most challenging step in the estimation is the first one, where we solve for the policy functions. The NN approximates the labor and consumption policy and is conditioned on the state variables and parameters to estimate. The network is trained by minimizing the Euler equation and New Keynesian Phillips curve errors.

We use 100,000 iterations to train the NN.³⁷ The batch size is set to 100, which can be thought of as having 100 different parallel economies in each iteration. We draw new parameters after every 100 iterations for the first 40,000 iterations and after 10 iterations during the rest of the training. After drawing new parameters, we simulate the economy for 100 periods to get a new draw for the state variable for each batch. For iterations where we don't draw new parameters, we simulate for 20 periods. The expectations are evaluated using 100 draws.³⁸

The NN convergence during the training is shown in Figure 14, where the mean squared error averaged across the functions we minimize is shown. The shaded area indicates the periods 10,000 to 15,000 in which we introduce the ZLB. We slowly scale

³⁷The NN contains five hidden layers with 128 neurons each. The activation function for the four initial hidden layers is the Sigmoid Linear Unit (SiLU), while the fifth layer is Leaky Rectified Linear Unit (Leaky ReLU). The optimizer is AdamW. The learning rate follows a cosine annealing scheduler starting from an initial rate of $1e-3$ and terminating at a rate of $1e-7$.

³⁸Our algorithm follows the general setup that we have described in Section 2 in the main paper.



Figure 14: Convergence of the NN solution. The figure shows the mean squared error averaged across the two functions we minimize. The loss is calculated as a moving average over 1,000 iterations. The vertical axis has a logarithmic scale. The shaded area indicates the period in which we introduce the ZLB.

the impact of the ZLB using the following functional form:

$$R_t = \max[R_t^N, 1] + a^{ZLB} \min[R_t^N - 1, 0] \quad (41)$$

The parameter a^{ZLB} is gradually lowered from 1 to 0 to move from the economy without ZLB to the ZLB economy. The introduction of the ZLB complicates the problem, as can be seen by the increase in the loss. At the end of the training, the average mean squared error reaches a level of about 10^{-6} .

NN particle filter. We generate 15,000 likelihoods with the particle filter to train the NN-based particle filter. We use the RWMH algorithm to easily obtain 50,000 draws from the posterior distribution after a burn-in.

Comparison. We compare the results to a state-of-the-art nonlinear estimation approach that does not use machine learning techniques. The global solution method is based on time iteration with piecewise linear policy functions as in Richter et al. (2014), and the particle filter implementation follows Herbst and Schorfheide (2015).

Standard deviations			Autocorrelations			Avg. Gini coef.		
	Model	Data		Model	Data		Model	Data
GDP	0.6947	0.5831	GDP	0.1355	0.4050	Wealth	0.8793	0.8410
Inflation	1.1511	0.9045	Inflation	0.8146	0.5456			
FFR	2.561	2.7537	FFR	0.7219	0.9707			

Table 5: Moments comparison: model vs. data. The table reports the standard deviations (left panel) and the autocorrelation coefficients (middle panel) in the model and in the U.S. data for the three observable variables: real GDP growth (GDP), GDP deflator growth (Inflation), and the federal funds rate (FFR). The right panel of the table shows the average Gini coefficient in the model and in the U.S. data (World Inequality Database). The model-implied moments are obtained by simulating the estimated model for 1,000,000 periods and with parameters set at the posterior median. The empirical moments are computed over the sample period used in estimation (1990:Q1-2019:Q4).

When applying the alternative approach, we similarly use the nonlinear model as the data-generating-process.³⁹ However, this approach is much slower because we need to resolve the model and run the particle filter for each draw. These limitations justify our decision to use only 50,000 draws in the Metropolis-Hastings algorithm.

Results. The estimation results are summarized in Table 4. First, the NN approach performs very well in recovering the true data-generating process. The posterior median is close to the true value and always contained inside the 90% credible interval. Furthermore, the posterior median and credible interval are very similar to those obtained with the alternative method.

D Heterogeneous Agent New Keynesian (HANK) model

The next model is a one-asset Heterogeneous Agent New Keynesian (HANK) model, following the specification of Auclert et al. (2021). We solve the model using neural

³⁹We use the same sequence of structural shocks, measurement error shocks, and the true parameter values to generate the data. However, the solution of the model comes either from our NN or from the standard global method.

networks. In our baseline specification, we use 100 agents ($L = 100$). However, we also increase the number of agents up to 1,000, and show that in this case, 100 agents are already sufficient. Note that this setup now requires a decent GPU to solve the model.⁴⁰ The model itself is solved in two steps, as the Krusell-Smith model. Specifically, we first show how to solve the deterministic steady state. Afterward, we solve the full equilibrium of the model. Finally, we compare the results to ones obtained when using the Sequence Space Jacobian approach of Auclert et al. (2021). For ease of comparison, we follow their setup of the model.

D.1 Model

The economy consists of a continuum of households. The households choose consumption C_t^i , labor H_t^i , and assets B_t^i to maximize their utility:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\left(\frac{1}{1-\sigma} \right) (C_t^i)^{1-\sigma} - \chi \left(\frac{1}{1+\eta} \right) (H_t^i)^{1+\eta} \right].$$

The budget constraint in real terms can be written as:

$$C_t^i + B_t^i = W_t \exp(s_t^i) H_t^i + \frac{R_{t-1}}{\Pi_t} B_{t-1}^i + Div_t \exp(s_t^i) - T_t \exp(s_t^i) \quad (42)$$

where Div_t is the real dividends, W_t is the real wage, R_t is the gross nominal interest rate, T_t is the tax and Π_t is the gross inflation rate. The real taxation and real dividends are scaled with the productivity of the household. The agents individual labor productivity s_t^i is stochastic and follows an AR(1) process: $s_t^i = \rho_s s_{t-1}^i + \epsilon_t^{s,i}$. The agents face a borrowing limit $\underline{B}$, which implies:

$$B_t \geq \underline{B}. \quad (43)$$

⁴⁰To keep working with a CPU, we recommend reducing the number of households to 25.

The first-order conditions can be written as

$$1 = \beta R_t \mathbb{E}_t \left[\left(\frac{C_t^i}{C_{t+1}^i} \right)^\sigma \frac{1}{\Pi_{t+1}} \right] + \mu_t^i, \quad (44)$$

$$\chi(H_t^i)^\eta = (C_t^i)^{-\sigma} \exp(s_t^i) W_t, \quad (45)$$

where $\mu_t^i \geq 0$ is the normalized Lagrange multiplier on the individual borrowing limit in equation (180).

The New Keynesian Phillips curve, derived from Rotemberg pricing, is:

$$\varphi \log(\Pi_t) = (1 - \epsilon) + \epsilon MC_t + \varphi \mathbb{E}_t \left[\frac{\Pi_{t+1}}{R_t} \log(\Pi_{t+1}) \frac{Y_{t+1}}{Y_t} \right]. \quad (46)$$

Firms produce output using labor:

$$Y_t = N_t. \quad (47)$$

Labor is hired in competitive markets so that the wage is:

$$W_t = MC_t. \quad (48)$$

The real dividends of the firm sector can then be written as

$$Div_t = Y_t - W_t Y_t - 0.5\varphi [\log(\Pi_t)]^2 Y_t. \quad (49)$$

The central bank sets the nominal interest R_t using a Taylor rule that responds to inflation and output deviations from their targets Π and Y :

$$(R_t - 1) = (R - 1) + \theta_\Pi(\Pi_t - 1) + \theta_Y(Y_t - Y) + mp_t, \quad (50)$$

where the monetary policy shock mp_t follows an AR(1) process:

$$mp_t = \rho_m mp_{t-1} + \epsilon_t^m. \quad (51)$$

The fiscal authority sets T_t to keep their debts D_t on a constant level:

$$D = \frac{R_{t-1}}{\Pi_t} D - T_t \quad (52)$$

Market clearing for the labor market, bond market, and goods market requires

$$N_t = \int \exp(s_t^i) H_t^i di, \quad (53)$$

$$D = \int B_t^i di, \quad (54)$$

$$Y_t = \int C_t^i di + 0.5\varphi [\log(\Pi_t)]^2 Y_t, \quad (55)$$

$$s_t = \rho_s s_{t-1} + \epsilon_t^s, \quad (56)$$

$$(57)$$

Equilibrium conditions. The equilibrium conditions can be written as:

$$C_t^i + B_t^i = W_t \exp(s_t^i) H_t^i + \frac{R_{t-1}}{\Pi_t} B_{t-1}^i + Div_t \exp(s_t^i) - T_t \exp(s_t^i), \quad \forall i \quad (58)$$

$$B_t^i \geq \underline{B}, \quad \forall i, \quad (59)$$

$$1 = \beta R_t \mathbb{E}_t \left[\left(\frac{C_t^i}{C_{t+1}^i} \right)^\sigma \frac{1}{\Pi_{t+1}} \right] + \mu_t^i, \quad \forall i, \quad (60)$$

$$\chi(H_t^i)^\eta = (C_t^i)^{-\sigma} \exp(s_t^i) W_t, \quad \forall i, \quad (61)$$

$$Y_t = N_t, \quad (62)$$

$$W_t = MC_t, \quad (63)$$

$$Div_t = Y_t - W_t Y_t - 0.5\varphi [\log(\Pi_t)]^2 Y_t, \quad (64)$$

$$\varphi \log(\Pi_t) = (1 - \epsilon) + \epsilon MC_t + \varphi \mathbb{E}_t \left[\frac{\Pi_{t+1}}{R_t} \log(\Pi_{t+1}) \frac{Y_{t+1}}{Y_t} \right], \quad (65)$$

$$R_t = R + \theta_\Pi (\Pi_t - 1) + \theta_Y (Y_t - Y) + mp_t, \quad (66)$$

$$D = \frac{R_{t-1}}{\Pi_t} D - T_t, \quad (67)$$

$$N_t = \int \exp(s_t^i) H_t^i di, \quad (68)$$

$$D = \int B_t^i di, \quad (69)$$

$$Y_t = \int C_t^i di + 0.5\varphi [\log(\Pi_t)]^2 Y_t, \quad (70)$$

$$s_t = \rho_s s_{t-1} + \epsilon_t^s, \quad (71)$$

$$mp_t = \rho_m mp_{t-1} + \epsilon_t^m. \quad (72)$$

Calibration. We calibrate the model in line with Auclert et al. (2021): $\beta = 0.982$, $\sigma = 2.0$, $\eta = 2$, $\chi = 0.786$, $\underline{B} = 0$, $\epsilon = 6$, $\varphi = 60$, $\theta_\Pi = 1.5$, $\theta_Y = 0$, $D = 5.6$, $\rho_s = 0.966$, $\rho_{mp} = 0.6$, $\sigma_s = 0.129$, $\rho_{mp} = 0.9$, $\sigma_{mp} = 0.0025$.

D.2 Mapping of the model to the general framework

We can map this HANK model in the general form of the outlined estimation procedure. The state variable, the structural shocks, the individual and aggregate control variables, and the needed deterministic steady state values are:

$$\mathbb{S}_t = \left\{ \{B_{t-1}^i\}_{i=1}^L, \{s_t^i\}_{i=1}^L, R_{t-1}^N, mp_t \right\}, \quad (73)$$

$$\nu_t = \left\{ \{\epsilon_t^{s,i}\}_{i=1}^L, \epsilon_t^{mp} \right\}, \quad (74)$$

$$\psi_t^I = \{H_t^i\}, \quad (75)$$

$$\psi_t^A = \{\Pi_t, W_t\}, \quad (76)$$

$$\psi^{DSS} = \{R\} \quad (77)$$

The parameters of the model are

$$\Theta = \{\beta, \sigma, \eta, \underline{B}, \chi, \theta_\Pi, \theta_Y, D, \epsilon, \varphi, \rho_s, \rho_m, \sigma_s, \sigma_m\} \quad (78)$$

When solving the model, we treat all parameters as calibrated.

We use two NNs to separate individual and aggregate policy functions. The

individual NN $\psi_{NN}^I(\cdot)$ solves for labor supply:

$$\left\{ \left(H_t^i \right) = \psi_{NN}^I \left(\mathbb{S}_t^i, \mathbb{S}_t, \tilde{\Theta} | \bar{\Theta} \right) \right\}_{i=1}^L, \quad (79)$$

where $\mathbb{S}_t^i = \{B_{t-1}^i, s_t^i\}$. The NN for the aggregate policy functions $\psi_{NN}^A(\cdot)$ determines the wage and inflation:

$$\begin{pmatrix} \Pi_t \\ W_t \end{pmatrix} = \psi_{NN}^A \left(\mathbb{S}_t, \tilde{\Theta} | \bar{\Theta} \right). \quad (80)$$

One challenge is that we must solve for the deterministic steady state values of the nominal rate, as the Taylor rule depends on them. This neural networks $\psi_{NN}^{SS}(\cdot)$ that maps the parameter values to the deterministic steady state values is:

$$\begin{pmatrix} R \\ Y \end{pmatrix} = \psi_{NN}^{SS} \left(\tilde{\Theta} | \bar{\Theta} \right). \quad (81)$$

Note that for $\theta_Y = 0$, we would not need to solve for Y .

D.3 Diagnostics: Loss function and approximation error

To investigate the accuracy of our solution, we report the average loss function over the training batches used to train the NN. We also provide a decomposition of the average loss function to illustrate more clearly that convergence is achieved for each individual loss component employed during training.⁴¹ The six loss functions are shown below:

$$L_1^b = \frac{1}{L} \sum_{i=1}^L \left(\left(\Psi^{FB} \left(1 - \beta R_t \mathbb{E}_t \left[\left(\frac{C_t^i}{C_{t+1}^i} \right)^\sigma \frac{1}{\Pi_{t+1}} \right], B_t^i - \underline{B} \right) \right)^2 \right), \quad (82)$$

$$L_2^b = \left(\varphi \log(\Pi_t) - \left((1 - \epsilon + \epsilon MC_t + \varphi \mathbb{E}_t \left[\frac{\Pi_{t+1}}{R_t} \log(\Pi_{t+1}) \frac{Y_{t+1}}{Y_t} \right]) \right) \right)^2, \quad (83)$$

⁴¹Solving the HANK model requires computing the DSS, as the monetary policy reaction function depends on DSS variables such as the interest rate and output. The DSS corresponds to a version of the economy without aggregate shocks; in our case, without monetary policy shocks.

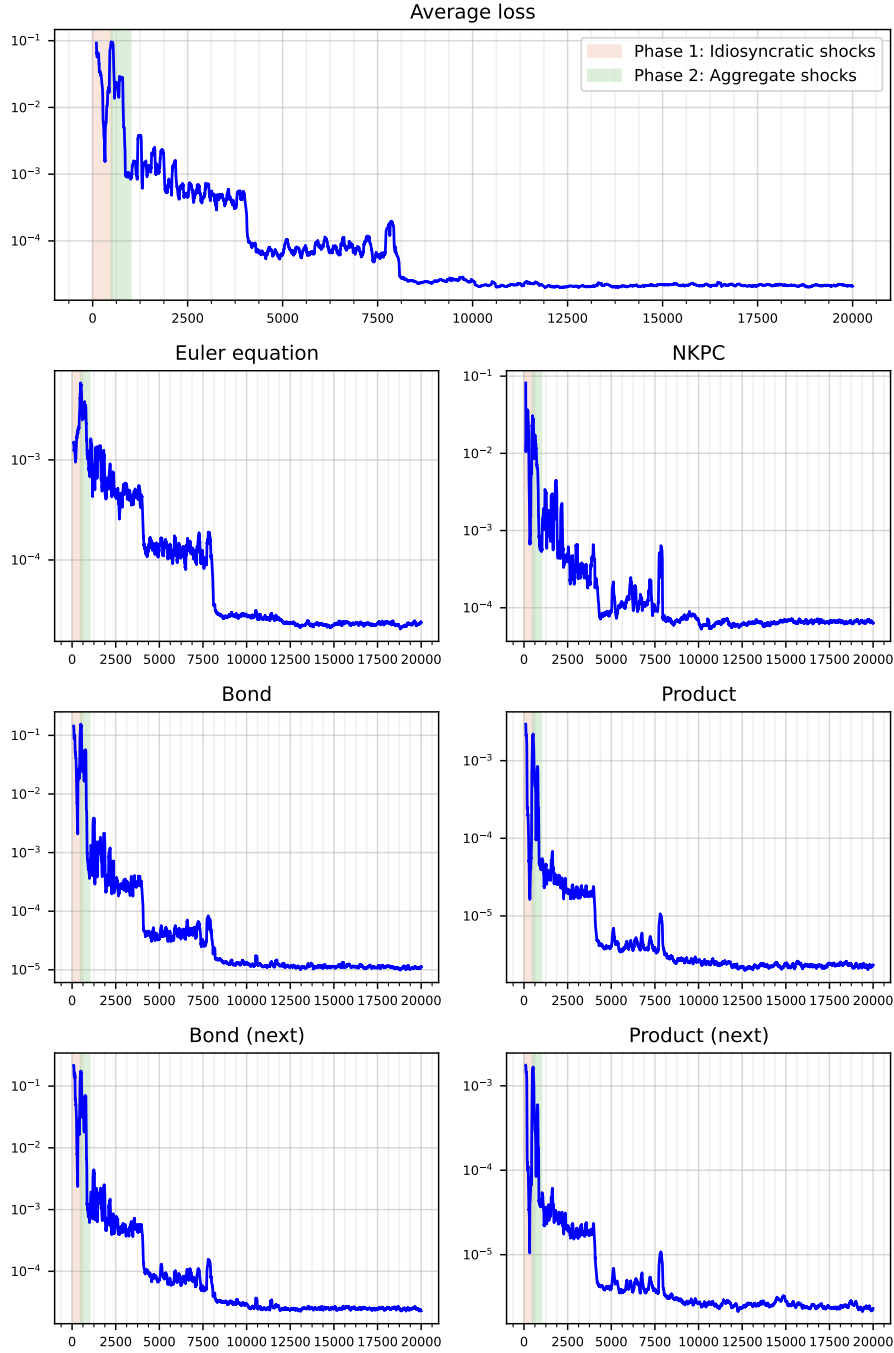


Figure 15: Convergence of the NN solution for the equilibrium. The figure shows the dynamics of the mean squared error during the training of the equilibrium. The upper plot shows the average loss, while the subcomponents (L_j^b , $j \in \{1, \dots, 6\}$) are defined in equations (82) - (87). The shaded areas indicate the periods in which we scale up idiosyncratic and aggregate risk. The vertical axis has a logarithmic scale.

$$L_3^b = \left(D - \frac{1}{L} \sum_{i=1}^L B_t^i \right)^2, \quad (84)$$

$$L_4^b = \left(Y_t - \frac{1}{L} \sum_{i=1}^L C_t^i - 0.5\varphi [\log(\Pi_t)]^2 Y_t \right)^2 \quad (85)$$

$$L_5^b = \frac{1}{M} \sum_{m=1}^M \left(\left(D - \frac{1}{L} \sum_{i=1}^L B_{t+1}^{i,m} \right)^2 \right), \quad (86)$$

$$L_6^b = \frac{1}{M} \sum_{m=1}^M \left(\left(Y_{t+1}^m - \frac{1}{L} \sum_{i=1}^L C_{t+1}^{i,m} - 0.5\varphi [\log(\Pi_{t+1}^m)]^2 Y_{t+1}^m \right)^2 \right), \quad (87)$$

where the first equation is the Euler equation for all households, the second equation is the NKPC, the third equation is current period bond market clearing, the fourth equation is the current period product market clearing, the fifth equation is next period bond market clearing, and the sixth equation is next period product market clearing. $\Psi^{FB}(\cdot, \cdot)$ is the Fischer-Burmeister function to account for the Karush-Kuhn-Tucker (KKT) conditions arising from the lower bound on assets $\underline{B}$. Note the superscript m indicates a Monte Carlo draw that we use to evaluate expectations. Note that we calculate each component for each batch, which we indicate with the superscript b .

The top panel of Figure 15 displays the evolution of the average loss function over optimization iterations (training), which we further average across batches:

$$\phi^L = \frac{1}{B} \sum_{b=1}^B \left[\frac{1}{6} \sum_{j=1}^6 L_j^b \right], \quad (88)$$

where we use an equal weight (1/6) for each loss function component, e.g. $\alpha_j = 1/6, \forall j$, which is the weights we use when training the NNs. In the lower panels of that figure, we plot the in-training evolution of the six loss functions defined above averaged across batches:

To facilitate convergence during training, we gradually introduce idiosyncratic shocks by increasing their variance from a small value to their target levels over the

course of the first 500 iterations. The red shaded area marks the stage at which this increase begins. We also introduce the aggregate shock (i.e., the monetary policy shock) gradually from period 500 to 1,000 by increasing its variance slowly from an arbitrarily low value to the level used in the model as indicated by the green area.

The training is successful, with clear convergence of the average loss. The breakdown of the loss into the individual residual errors (L_j^b , $j \in \{1, \dots, 6\}$ defined earlier) confirms that the NN learns all policy functions of interest effectively. After gradually introducing the idiosyncratic shocks, we introduce the aggregate shock (i.e., the monetary policy shock) by increasing its variance slowly from an arbitrarily low value to the level used in the model, as indicated by the red and green areas for the two types of shocks.

The next step is to define a diagnostic to assess how accurate the model solution is *after* the training of the NNs has been completed. We use the Euler equation errors, following Judd (1992) and Aruoba et al. (2006), but instead of focusing on one condition, we use the same set of equilibrium conditions used during training, which are the Euler equation, the New Keynesian Phillips Curve (NKPC), bond market clearing, and goods market clearing. The absolute error for these equations can be written as:

$$err_1 = \left| \Psi^{FB} \left(1 - \beta R_t \mathbb{E}_t \left[\left(\frac{C_t^i}{C_{t+1}} \right)^\sigma \frac{1}{\Pi_{t+1}} \right], B_t^i - \underline{B} \right) \right|, \quad (89)$$

$$err_2 = \left| \frac{\varphi \log(\Pi_t)}{1 - \epsilon} - \left(\left(1 + \frac{\epsilon}{1 - \epsilon} MC_t + \frac{\varphi}{1 - \epsilon} \mathbb{E}_t \left[\frac{\Pi_{t+1}}{R_t} \log(\Pi_{t+1}) \frac{Y_{t+1}}{Y_t} \right] \right) \right) \right|, \quad (90)$$

$$err_3 = \left| 1 - \frac{\frac{1}{L} \sum_{i=1}^L B_t^i}{D} \right|, \quad (91)$$

$$err_4 = \left| 1 - \frac{\frac{1}{L} \sum_{i=1}^L C_t^i - 0.5\varphi [\log(\Pi_t)]^2 Y_t}{Y_t} \right| \quad (92)$$

where the err_1 is for the Euler equation, err_2 for the NKPC, err_3 for bond market clearing, and err_4 for product market clearing. Note that we do not include the bond and product market-clearing next period, as they are already covered by the current

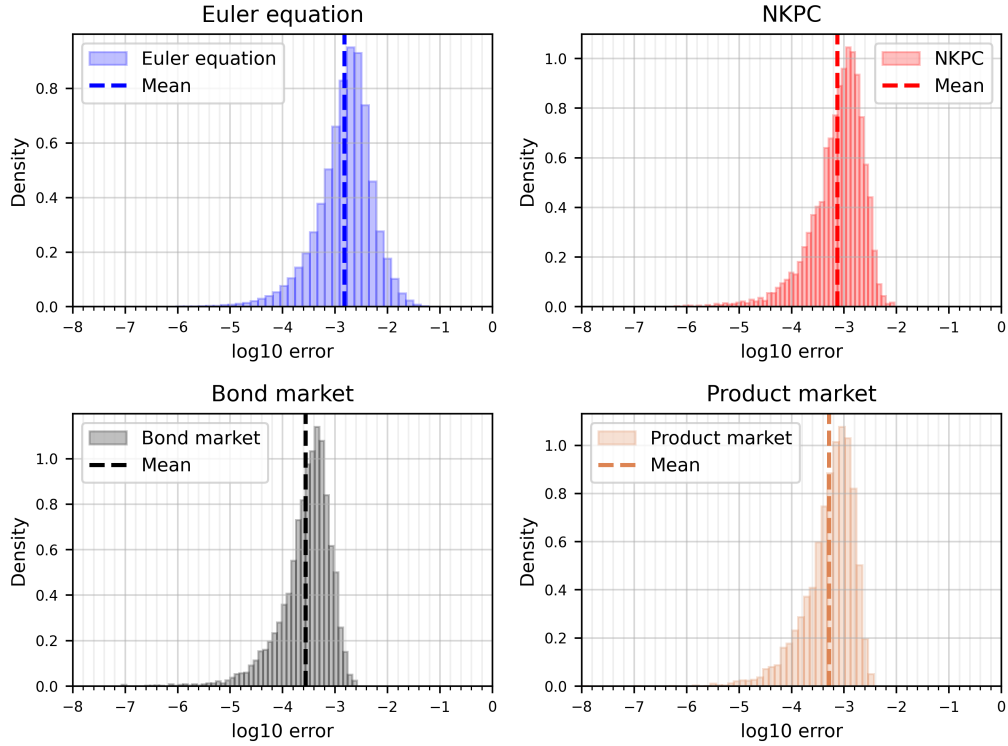


Figure 16: Distribution of post-training approximation errors. The figure shows the absolute error for the modified Euler equation (to account for the occasionally binding borrowing limit), the New Keynesian Phillips curve, bond market clearing and product market clearing. Errors are evaluated after the training of the NNs is complete. The distributions are based on a simulation of 10,000 periods. We evaluate the expectations, relevant to the modified Euler equation and the NKPC using 10,000 Monte Carlo draws. The horizontal axis displays the $\log_{10}$ of the value.

period counterparts.

We simulate our model for 10,000 periods and calculate for each period the approximation error.⁴² The results of the validation exercise for the four equilibrium conditions are shown in Figure 16. The residual errors are well contained, even in the right tails of the distributions. Each distribution is approximately bell-shaped, with modes close to the corresponding mean absolute errors. This suggests that large approximation errors are rare and unlikely to systematically affect the model's dynamics.

Moreover, the symmetry and concentration of the distributions around small val-

⁴²We draw 10,000 shocks to evaluate expectations.

ues further reinforce the reliability of the NN solution, indicating that the approximation performs consistently well across a wide range of simulated states.

D.4 Number of Agents and Number of Draws for Monte Carlo Integration

We are now testing how changing the number of agents and the number of draws for the Monte Carlo integration affects our results.

D.4.1 Number of Agents

Figure 17 shows the distribution of asset holdings across different levels of discretization, with the left panel showing the histogram and the right panel the corresponding cumulative distribution function. We compare results for $L=100, 200, 500$, and $1,000$ agents. Each distribution is obtained by averaging across 100 simulated economies (batches). The two figures illustrate that the benefit of increasing the number of tracked agents to capture heterogeneity is minimal, since the asset distribution changes only slightly.

Even more striking is the comparison based on the policy functions for labor and consumption. Figure 18 shows the bar graph of the asset distribution conditional on various levels of productivity (10th percentile, 50th percentile, and 90th percentile) along with the policy functions of labor and consumption for 100 agents (solid line) and 1,000 (dashed line).⁴³ The policy functions exhibit minimal change as the number of tracked agents increases from 100 to 1,000, supporting our claim that tracking more than 100 agents yields little additional benefit in approximating the asset distribution in this one-asset HANK model.

Nevertheless, computing time increases rapidly as the number of agents tracked L grows, as we will now show. The runtime ranges from 4h03min for $L = 100$ to

⁴³The other individual policy function determines the agent's asset demand, which is implied by the two individual policy functions and hence is not shown.

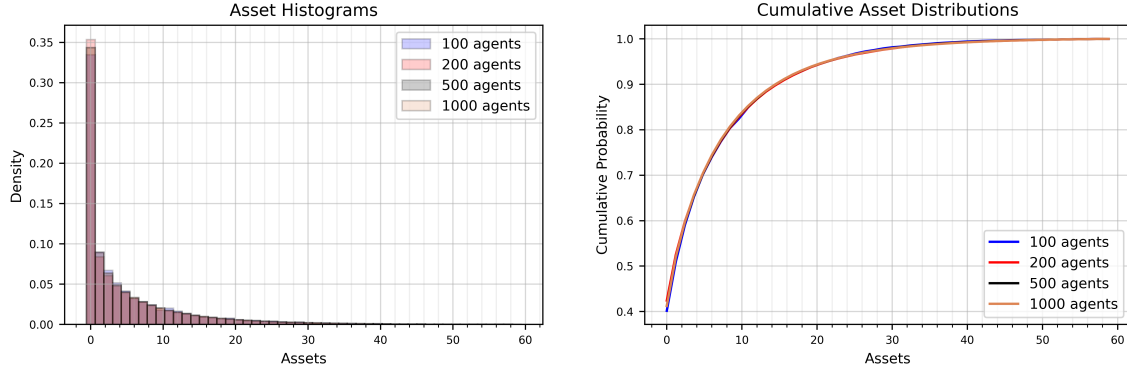


Figure 17: Asset distribution for variations in the number of agents L . The figure shows the histogram of asset holdings (left) and the corresponding cumulative distribution function (right) for different values of L used to solve the model with our NN methods. Specifically, we consider scenarios with 100, 200, 500, and 1,000 agents in equilibrium. The distribution is evaluated at the stochastic steady state, averaged over 100 simulated economies.

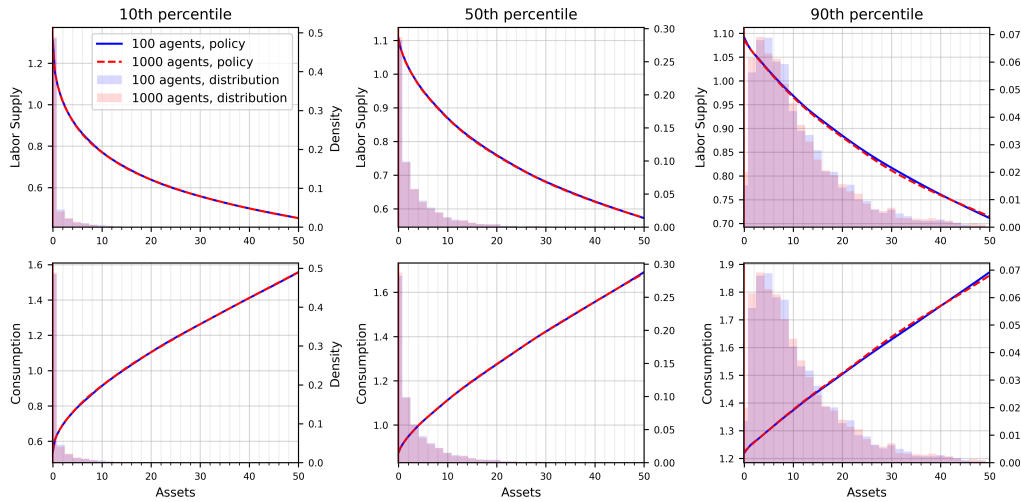


Figure 18: Policy functions for variations in number of agents L . Policy functions (labor supply - upper panels - and consumption - lower panels) and asset distribution conditional on individual productivity (10th percentile, 50th percentile, 90th percentile) for $L = 100$ and $L = 1,000$.

20h07min for $L = 1,000$, as reported in Table 6.⁴⁴ Notably, even the version of the model with 1,000 agents required only 6.4GB of memory (compared to just 0.9GB for $L = 100$). This reflects the new memory-efficient design of our solution code, which makes it feasible to run the nonlinear one-asset HANK model on a desktop machine. For instance, solving the model with $L = 100$ took 30h38min on a recent

⁴⁴These computations are performed on an Nvidia RTX 3090 GPU with 24GB of memory.

MacBook.⁴⁵

Number of Agents	GPU Computational Time	GPU Memory Usage (GB)
L=100	4h03min	0.9
L=200	5h45min	2.3
L=500	11h12min	3.7
L=1,000	20h07min	6.4

Table 6: Run time and memory usage for alternative number of agents. The calculation were computed with a Nvidia RTX 3090 with 24GB of memory as GPU.

D.4.2 Number of Monte Carlo Draws

We adopt a Monte Carlo approach to approximate expectations. In general, the accuracy of this approximation depends on using a sufficiently large number of Monte Carlo draws for next-period shock realizations. Here, we show that this number does not need to be large when using our neural network approach; in fact, a relatively modest number of draws is sufficient to achieve accurate approximations. The underlying reasons are similar to those that explain why only a limited number of agents need to be tracked in our framework.

In the baseline case with 100 agents, we vary the number of Monte Carlo draws used to approximate next-period variables—testing with 100, 250, and 500 draws. Importantly, the number of agents is fixed at L=100 across these exercises. These results, as shown in Figures 19 and 20, are reported for the one-asset version of the Auclert et al. (2021) model. The results show that 100 Monte Carlo draws approximate the expectations very good here. One reason that they perform so well is that we draw a new set of Monte Carlo draws during the training process for each iteration and batch, resulting in a very good approximation over the training process.

⁴⁵This computation used an Apple M4 Pro chip with a 14-core CPU and a 20-core GPU. We employed the Metal programming framework for macOS devices. When run on the CPU, the same task took approximately twice as long.

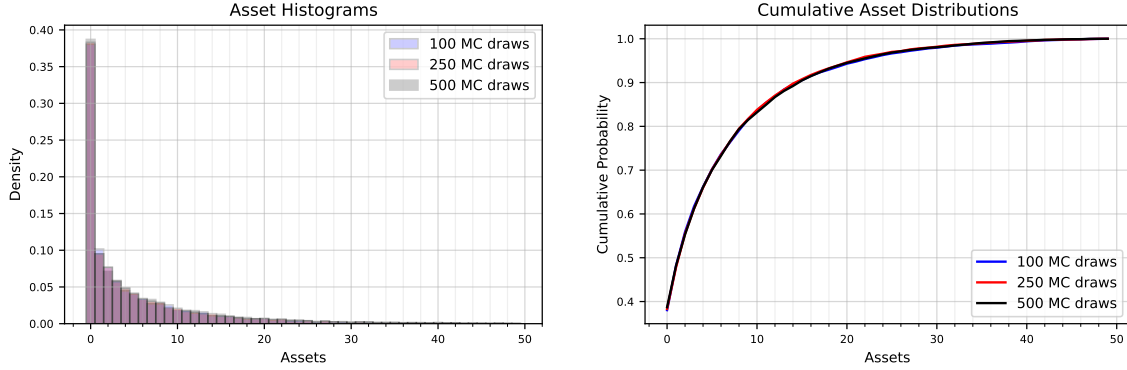


Figure 19: Asset distribution for variations in number of Monte Carlo draws MC . The figure shows the histogram of asset holdings (left) and the related cumulative density function (right) for an alternative number of Monte Carlo draws for the approximation of expectations. Specifically, we consider scenarios with 100, 250, and 500 Monte Carlo draws. We evaluate the distribution at its stochastic steady state, averaging over 100 economies.

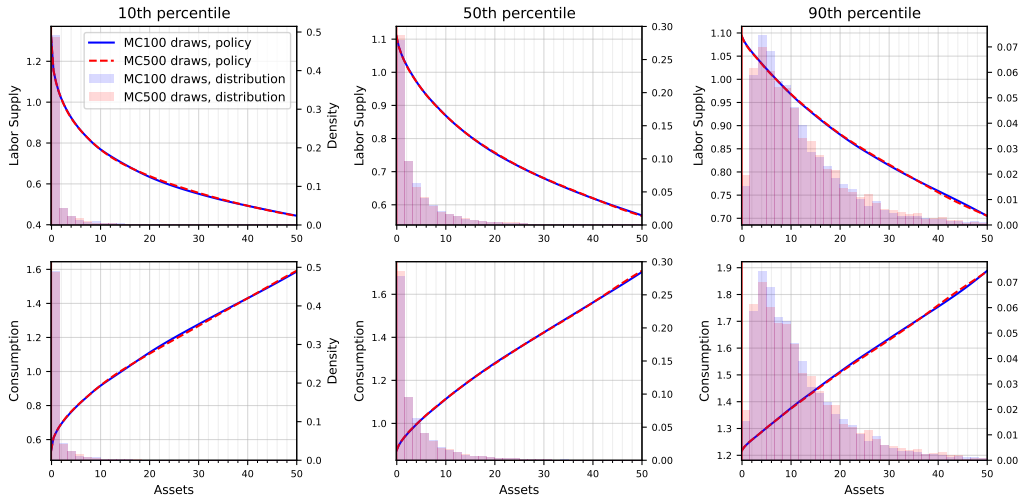


Figure 20: Policy functions for variations in number of Monte Carlo draws MC . Policy functions (labor supply - upper panels - and consumption - lower panels) and asset distribution conditional on individual productivity (10th percentile, 50th percentile, 90th percentile) for $MC = 100$ and $MC = 500$.

D.5 Comparison with Sequence Space Jacobian

We now compare again our neural network based solution method to a conventional (linear) approach, specifically the sequence space Jacobian (SSJ) approach of Auclert et al. (2021).

Figure 21 compares the asset distribution generated by our method and by the

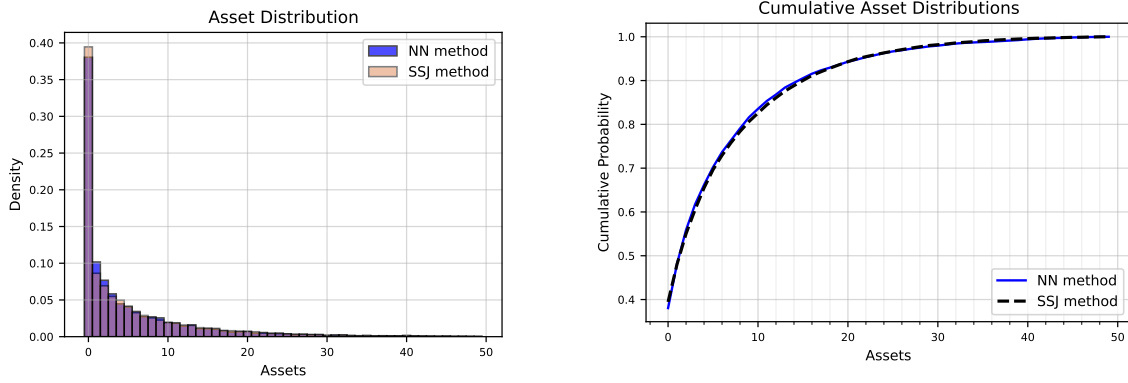


Figure 21: Asset distribution for our neural network (NN) method for comparison with sequence space Jacobian method. For the NN method, we evaluate the distribution at its stochastic steady state, averaging over 100 economies. For the SSJ, we use the deterministic steady state. Left panels: The histogram of asset holdings in equilibrium. Right panels: Cumulative density function for the asset holdings in equilibrium.

SSJ approach. For our method, we report the distribution at the stochastic steady state, which accounts for the presence of aggregate risk. In contrast, the SSJ method by construction assumes away aggregate uncertainty, so the distribution it produces corresponds to the deterministic steady state. Despite this key difference, the two distributions are remarkably similar. Although minor deviations are visible in the tails, the overall shape and key moments align closely, suggesting that our method provides a highly accurate approximation.

Notably, these deviations cannot be attributed solely to inaccuracies in either our method or the SSJ approach. Instead, they stem from a fundamental difference in the economic environments captured by the two methods. Incorporating aggregate uncertainty, as our method does, leads poorer households to work more in response to increased precautionary motives, resulting in fewer households clustered at the bottom of the asset distribution—reflected in the shorter blue bars in the lower asset bins. At the same time, this increased labor supply and precautionary saving among the poor depresses the equilibrium real interest rate, thereby reducing the incentive for wealthier households to save. As a result, rich households accumulate less wealth in equilibrium, leading to a thinner right tail in the asset distribution. This pattern

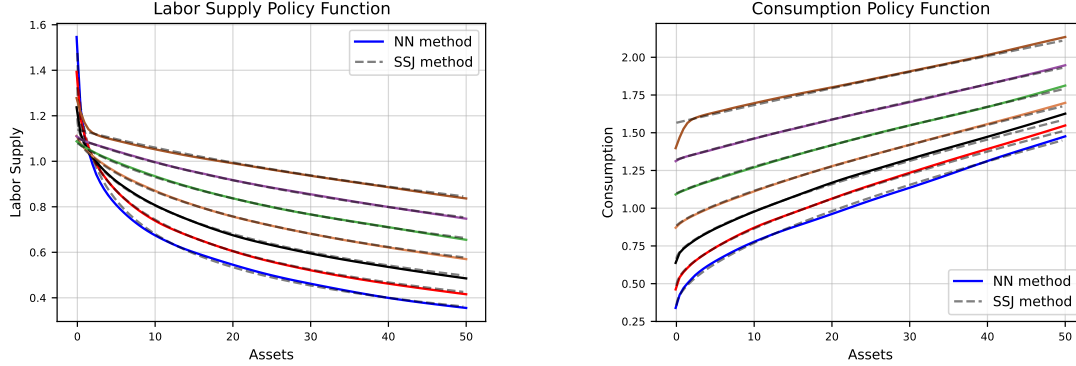


Figure 22: Individual policy functions comparison for the one-asset HANK model. The figure illustrates how the policy functions for labor and consumption vary for different assets levels and different productivity levels. We compare the neural network method with the SSJ method. We use the 7 productivity levels that the discretization approach of the SSJ provides for comparison. We evaluate the policy functions at the stochastic steady state (NN) and deterministic steady state (SSJ).

is clearly illustrated by the crossing of the blue solid and black dashed lines in the right panel of Figure 21, which plots the cumulative asset distribution.

Even in this relatively stylized model, the differences between the asset distributions implied by the nonlinear (our method) and linearized (SSJ) solutions are already notable. As we will show, these differences become even more pronounced if we increase aggregate risk.

The next step is to compare the two sets of policy functions that determine agents' labor supply and consumption decisions, as shown in Figure 22. The solid lines depict the policy functions approximated using our method, while the dashed lines correspond to the approximation obtained using the SSJ method from Auclert et al. (2021). Each color represents one of the seven idiosyncratic labor productivity levels considered in Auclert et al. (2021). The brown lines correspond to the most productive agents (top productivity bin), while the blue lines represent the least productive ones (bottom productivity bin.)

Note that these seven productivity levels are used solely for the purpose of comparing the results from our solution method with those from SSJ. It is worth recalling that our solution method does not rely on discretizing the productivity space. In-

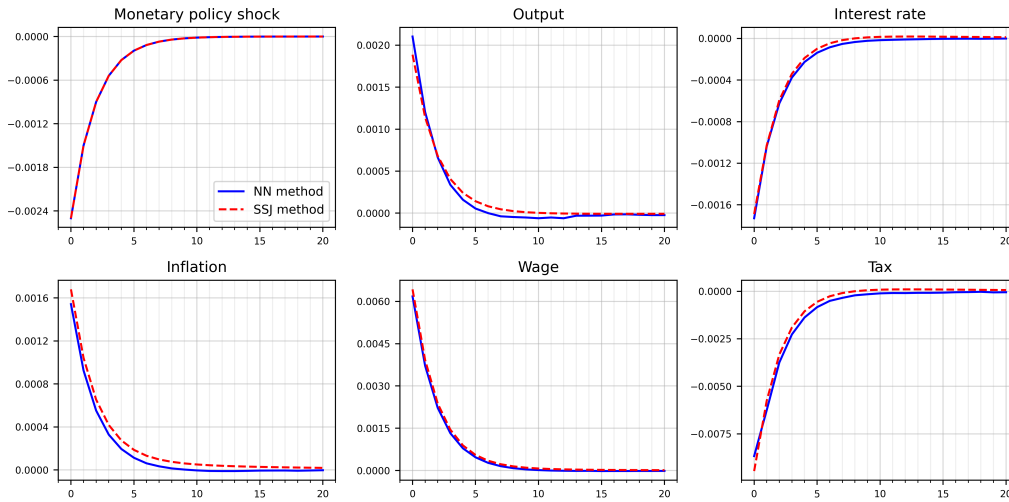


Figure 23: Impulse response functions comparison of neural network solution and sequence space Jacobian approach. The deviations from the stochastic steady state (for the NN method) and from the deterministic steady state (for the SSJ method) are reported for a one-standard-deviation negative monetary policy shock. $L=100$ and $MC = 100$.

stead, we treat idiosyncratic labor productivity as the outcome of a continuous AR(1) process, as specified in the model. For the purpose of this comparison, we evaluate the policy functions (solved for a continuous AR(1) process) at the values in line with the discretization in the code made available by Auclert and coauthors.

The upshot of the comparison is twofold. First, our method successfully replicates the policy functions approximated by the SSJ approach. Second, the main discrepancy arises in the behavior of the most productive agents with zero assets—and low productive agents with a large amount of assets. Both groups represents a tiny fraction of agents. Because so few agents follow such a path, the neural network has limited data to learn from, which results in a less accurate approximation of their behavior. Nonetheless, the inaccuracies in these cases are fairly limited, highlighting the strong extrapolative properties of neural networks.

Indeed, the effects of a monetary policy shock on the equilibrium dynamics of aggregate variables predicted by the two methods are quite similar. See Figure 23.

E Quantitative HANK model

We use a quantitative HANK model similar to those in Auclert et al. (2020), Bayer et al. (2024a), and Hagedorn et al. (2025). To begin with, we have two different sources of heterogeneity. In addition to idiosyncratic income risk, households have permanent discount factor heterogeneity as in Krueger et al. (2016). The model is a two-asset economy that features capital (subject to adjustment costs) next to government bonds. The agents hold a combined asset via a financial intermediary. The model features two sources of nominal rigidities by incorporating price wage stickiness, so that the model features a price as well as a wage Phillips curve. We incorporate inflation indexation for both price inflation and wage inflation, allowing a better fit with the data. Finally, the model features a total of 7 aggregate shocks: risk-premium, growth rate, monetary policy, wage mark-up, price mark-up, government spending, and investment.

Households. The economy consists of a continuum of households. The households are ex-ante heterogeneous with respect to their discount factors, that is the discount factor β^i varies across the households. The households choose consumption C_t^i , hold assets holdings at a financial intermediary Z_t^i . While the households also provide labor H_t^i , households are organized via labor unions. The utility function is:

$$E_0 \sum_{t=0}^{\infty} (\beta^i)^t \left[\left(\frac{1}{1-\sigma} \right) \left(\frac{C_t^i}{A_t} \right)^{1-\sigma} - \chi \left(\frac{1}{1+\eta} \right) (H_t^i)^{1+\eta} \right],$$

where A_t is the total factor productivity. The budget constraint in real terms can be written as:

$$C_t^i + \frac{Z_t^i}{R_t^Z} = \tau_t \left(\frac{W_t}{P_t A_t} \exp(s_t^i) H_t^i \right)^{1-\gamma\tau} + \frac{1}{\Pi_t} Z_{t-1}^i + Div_t^i, \quad (93)$$

where $Div_t^i = Div_t \exp(s_t^i)$ is the household specific real dividends, W_t is nominal wage, R_t^Z is the gross nominal return on the asset, Π_t is the gross inflation rate.⁴⁶ The taxation is progressive, which is governed by γ_τ , and τ_t determines the level of the post-tax income. The wage is divided by the productivity level to avoid a fiscal drag in taxation. The agents individual labor productivity s_t^i is stochastic and follows an AR(1) process: $s_t^i = \rho_s s_{t-1}^i + \epsilon_t^{s,i}$. The dividend payout is scaled with the level of the idiosyncratic shock. The agents face a borrowing limit $\underline{Z}$, which implies:

$$Z_t^i \geq \underline{Z}. \quad (94)$$

The first-order conditions with respect to consumption and asset holdings can be written as

$$1 = \beta R_t^Z \mathbb{E}_t \left[\left(\frac{\lambda_{t+1}^i}{\lambda_t^i} \right) \frac{1}{\Pi_{t+1}} \right] + \mu_t^i \quad (95)$$

$$\lambda_t^i = \left(\frac{C_t^i}{A_t} \right)^{-\sigma} \frac{1}{A_t}, \quad (96)$$

where $\mu_t^i \geq 0$ is the normalized Lagrange multiplier on the individual borrowing limit in equation (180).

Labor Union To incorporate wage stickiness in our model, we use a representative labour union that combines the differentiated labor services of each household. The effective labor input is

$$H_t = \left(\int_0^1 \exp(s_t^i) H_t^{i \frac{\epsilon_w - 1}{\epsilon_w}} di \right)^{\frac{\epsilon_w}{\epsilon_w - 1}} \quad (97)$$

The equilibrium nominal wage and the demand for the labor services of household i can be expressed as

$$W_t = \left(\int s_t^i W_t^{i 1 - \epsilon_w} di \right)^{\frac{1}{1 - \epsilon_w}} \quad \text{and} \quad h(W_t^i) = \left(\frac{W_t^i}{W_t} \right)^{-\epsilon_w} H_t \quad (98)$$

⁴⁶We abstract from conditioning the dividend payouts on the household specific discount factor.

The union sets the nominal wage W_t^i subject to wage adjustment costs a la Rotemberg. We also introduce inflation indexation so that the costs depend on the deviation from steady state inflation and previous period inflation, that is

$$\Gamma_t^W = (\Pi^W)^{1-\iota_w} (\Pi_{t-1}^W)^{\iota_w} \quad (99)$$

where wage inflation is given as $\Pi_t^W = W_t/W_{t-1}$. The parameter ι_w governs the degree of inflation indexation.

The maximization problem can be written as:

$$\max_{W_t^i} \int \left(\tau_t \left(\frac{W_t^i \exp(s_t^i)}{P_t A_t} h(W_t^i) \right)^{1-\gamma_\tau} - \frac{v(h(W_t^i))}{u'(C_t)} - \exp(s_t^i) \frac{\varphi^w}{2} \left(\frac{W_t^i}{\Gamma_t^W W_{t-1}^i} - 1 \right)^2 H_t A_t \right) di, \quad (100)$$

$$\text{where } u'(C_t) = \left(\frac{C_t}{A_t} \right)^{-\sigma} \frac{1}{A_t}, \text{ and } v'(H_t) = \chi \left(\frac{1}{1+\eta} \right) (H_t)^{1+\eta}. \quad (101)$$

To discount future profits, we are using the stochastic discount factor for the average agent, that is

$$\Lambda_{t,t+1} = \beta \frac{\lambda_{t+1}}{\lambda_t}, \quad (102)$$

$$\text{where } \lambda_t = \left(\frac{C_t}{A_t} \right)^{-\sigma} \frac{1}{A_t}, \text{ and } \beta = \int \beta^i di \quad (103)$$

Assuming a symmetric equilibrium, we can derive the wage Phillips curve as

$$(1 - \epsilon_w) \int [\exp(s_t^i)]^{1-\gamma_\tau} di \frac{\tau_t}{A_t} (1 - \gamma_\tau) \left(\frac{W_t}{P_t A_t} H_t \right)^{-\gamma_\tau} \frac{W_t}{P_t A_t} + \epsilon_w \frac{\chi(H_t)^\eta}{(C_t/A_t)^{-\sigma}} \quad (104)$$

$$- \varphi^w \left(\frac{\Pi_t^W}{\Gamma_t^W} - 1 \right) \frac{\Pi_t^W}{\Gamma_t^W} + \varphi^w E_t \Lambda_{t,t+1} \left(\frac{\Pi_{t+1}^W}{\Gamma_{t+1}^W} - 1 \right) \frac{\Pi_{t+1}^W}{\Gamma_{t+1}^W} \frac{H_{t+1}}{H_t} \frac{A_{t+1}}{A_t} + \mu_t^W = 0,$$

We also supplement the equation with a wage markup shock that follows an AR(1) process: $\mu_t^W = \rho_{\mu^W} \mu_{t-1}^W + \epsilon_t^{\mu^W}$. The Rotemberg costs for adjusting the wage costs are given back lump.

Firms. The firm sector consists of a continuum of final goods producers and intermediate goods firms. The final goods retailers buy the intermediate goods and transform them into a homogeneous final good using a CES production technology:

$$Y_t = \left(\int (Y_t^j)^{\frac{\epsilon-1}{\epsilon}} dj \right)^{\frac{\epsilon}{\epsilon-1}}, \quad (105)$$

where Y_t^j is the output of intermediate goods firm j . The equilibrium price of the final good and the demand for the intermediate goods of firm j can be expressed as:

$$P_t = \left(\int (P_t^j)^{1-\epsilon} dj \right)^{\frac{1}{1-\epsilon}}, \quad \text{and} \quad Y_t^j = \left(\frac{P_t^j}{P_t} \right)^{-\epsilon} Y_t. \quad (106)$$

Intermediate goods producers are monopolistically competitive.

The firm j uses labor N_t^j and capital K_{t-1} as inputs to produce output Y_t^j with the following production technology:

$$Y_t^j = (K_{t-1}^j)^\alpha (A_t H_t^j)^{1-\alpha}. \quad (107)$$

A_t is labor-augmenting productivity and follows a nonstationary process:

$$A_t = a_t A_{t-1}, \quad (108)$$

where the trend growth rate follows an AR(1) process:

$$a_t = \bar{a} \exp(z_t), \quad (109)$$

$$z_t = \rho_z z_{t-1} + \epsilon_t^z. \quad (110)$$

Labor and capital are hired in competitive markets, where the wage and rental rate are determined as follows:

$$W_t = (1 - \alpha) \psi_t \frac{Y_t}{H_t}, \quad (111)$$

$$r_t^k = \alpha \psi_t \frac{Y_t}{K_{t-1}}, \quad (112)$$

where ψ_t are the marginal costs. We discount future profits using the the stochastic discount factor for the average agent $\Lambda_{t,t+1}$.

The firm j sets the price of its goods to maximize its profit subject to the demand curve for intermediate goods and Rotemberg adjustment costs for changing prices:

$$\max_{P_t^j} P_t^j \left(\frac{P_t^j}{P_t} \right)^{-\epsilon_p} \frac{Y_t}{P_t} - \psi_t \left(\frac{P_t^j}{P_t} \right)^{-\epsilon_p} Y_t - \frac{\varphi^p}{2} \left(\frac{P_t^j}{\Gamma_t^P P_{t-1}^j} - 1 \right)^2 Y_t, \quad (113)$$

Inflation is indexed so that the costs are calculated relative to Γ_t^P , which is defined as

$$\Gamma_t^P = (\Pi^P)^{1-\iota_P} (\Pi_{t-1}^P)^{\iota_P} \quad (114)$$

where $\Pi_t = P_t/P_{t-1}$ and Π is the inflation target of the central bank.

Imposing a symmetric equilibrium and discounting future profits with the real interest rate, the New Keynesian Phillips curve can be written as:

$$\left[\varphi^p \left(\frac{\Pi_t}{\Gamma_t^P} - 1 \right) \frac{\Pi_t}{\Gamma_t^P} \right] = (1-\epsilon_p) + \epsilon_p \psi_t + \varphi^p \mathbb{E}_t \Lambda_{t,t+1} \left[\left(\frac{\Pi_{t+1}}{\Gamma_{t+1}^P} - 1 \right) \frac{\Pi_{t+1}}{\Gamma_{t+1}^P} \frac{Y_{t+1}}{Y_t} \right] + \mu_t^P, \quad (115)$$

We also supplement the equation with a price markup shock that follows an AR(1) process: $\mu_t^P = \rho_{\mu^P} \mu_{t-1}^P + \epsilon_t^{\mu^P}$ The Rotemberg adjustment costs are given back as a lump sum to the firm.

The real dividends paid out by the firm sector are:

$$Div_t = Y_t - W_t N_t - r_t^K K_{t-1}. \quad (116)$$

Financial Intermediary A representative competitive financial intermediary collects the financial assets from the households:

$$Z_t = \int Z_t^i di \quad (117)$$

The intermediary use the collected funds Z_t to invest them in capital K_t and government bonds D_t . Capital features adjustment costs, resulting in the following law of

motion:

$$K_t = (1 - \delta)K_{t-1} + \exp(\nu_t) \left(\gamma^A \left(\frac{I_t}{a\hat{K}_{t-1}} \right)^{1-\gamma^B} + \gamma^C \right) a\hat{K}_{t-1} \quad (118)$$

where the $\hat{\cdot}$ indicates that the previous capital stock is taken as given. ν_t is a marginal efficiency of investment shock that follows an AR(1) process: $\nu_t = \rho_\nu \nu_{t-1} + \epsilon_t^\nu$.

The return on the government bonds depends on the gross nominal interest rate R_t , which is set by the central bank. The return on the asset is also affected by risk-premium shocks ζ_t , which follows an AR(1) process: $\zeta_t = \rho_\zeta \zeta_{t-1} + \epsilon_t^\zeta$.

The profit maximization for the intermediary is then given as

$$\max_{K_t, D_t, I_t} r_t^K K_{t-1} + \frac{1}{\Pi_t} D_{t-1} - \frac{D_t}{\exp(\zeta_t) R_t} - I_t - \frac{1}{\Pi_t} Z_{t-1} + \frac{Z_t}{R_t^Z} \quad (119)$$

$$\text{s.t. } Z_t = K_t + D_t \quad (120)$$

$$K_t = (1 - \delta)K_{t-1} + \exp(\nu_t) \left(\gamma^A \left(\frac{I_t}{g\hat{K}_{t-1}} \right)^{1-\gamma^B} + \gamma^C \right) g\hat{K}_{t-1} \quad (121)$$

The first-order conditions with respect to government bonds, capital, and investment provide us with three equilibrium conditions:

$$\exp(\zeta_t) R_t = R_t^Z, \quad (122)$$

$$Q_t - \frac{1}{R_t^Z} = \beta E_t \left[\left(\frac{\tilde{C}_t}{\tilde{C}_{t+1}} \right)^\sigma \frac{1}{a_{t+1}} \right] \left(r_{t+1}^k + (1 - \delta) Q_{t+1} - \frac{1}{\Pi_{t+1}} \right), \quad (123)$$

$$\exp(\nu_t) Q_t = \left[\gamma^A (1 - \gamma^B) \left(\frac{I_t}{gK_{t-1}} \right)^{\gamma^B} \right]^{-1}, \quad (124)$$

where Q_t is the multiplier for the law of motion of capital.

Policy makers. The central bank sets the nominal interest R_t using a Taylor rule that responds to inflation and marginal cost (as a measure for the output gap) deviations from their targets Π and ψ . The ZLB restricts the nominal interest rate. The

rule is given as follows:

$$R_t = \left(R \left(\frac{\Pi_t}{\Pi} \right)^{\theta_\Pi} \left(\frac{\psi_t}{\psi} \right)^{\theta_Y} \right) \exp(mp_t), \quad (125)$$

where R and ψ denote the deterministic steady state (DSS) values of the nominal interest rate and output (in the detrended economy). While households face idiosyncratic shocks in the deterministic steady state, the economy does not face and the agents do not expect aggregate shocks. Additionally, there is also a persistent monetary policy shock mp_t to capture prolonged deviations from the rule:

$$mp_t = \rho_m mp_{t-1} + \epsilon_t^m. \quad (126)$$

The fiscal authority consumes g_t , adjusted for the level of productivity, which follows an AR(1) process: $g_t = \rho_g g_{t-1} + \epsilon_t^g$. It also sets τ_t to keep their debts D_t on a constant path adjusted for productivity gains:

$$D_t = \frac{R_{t-1}}{\Pi_t} D_{t-1} - T_t + A_t g_t, \quad (127)$$

$$T_t = \int \left(\frac{W_t}{P_t} \exp(s_t^i) H_t^i \right) di - \int \tau_t \left(\frac{W_t}{P_t A_t} \exp(s_t^i) H_t^i \right)^{1-\gamma_\tau} di. \quad (128)$$

Market clearing. Market clearing for the labor market and goods market requires

$$N_t = \int N_t^j dj = \int \exp(s_t^i) H_t^i di, \quad (129)$$

$$Y_t = \int C_t^i di + I_t + A_t g_t. \quad (130)$$

The bond market is already cleared via the financial intermediary.

E.1 Set of equations

$$\tilde{C}_t^i + \frac{\tilde{Z}_t^i}{R_t^Z} = \tilde{\tau}_t \left(\tilde{W}_t \exp(s_t^i) H_t^i \right)^{1-\gamma_\tau} + \frac{\tilde{Z}_{t-1}^i}{\Pi_t a_t} + \tilde{D}iv_t \exp s_t^i, \quad \forall i \quad (131)$$

$$1 = \beta^i R_t^Z \mathbb{E}_t \left[\left(\frac{\tilde{C}_t^i}{\tilde{C}_{t+1}^i} \right)^\sigma \frac{1}{\Pi_{t+1} a_{t+1}} \right] + \mu_t^i, \quad \forall i \quad (132)$$

$$Z_t^i \geq \underline{Z}, \forall i \quad (133)$$

$$(1 - \epsilon_w) \int [\exp(s_t^i)]^{1-\gamma_\tau} di \tilde{\tau}_t (1 - \gamma_\tau) \left(\tilde{W}_t H_t \right)^{-\gamma_\tau} \tilde{W}_t + \epsilon_w \frac{\chi(H_t)^\eta}{\tilde{C}_t^{-\sigma}} \quad (134)$$

$$- \varphi^w \left(\frac{\Pi_t^W}{\Gamma_t^W} - 1 \right) \frac{\Pi_t^W}{\Gamma_t^W} + \varphi^w E_t \Lambda_{t,t+1} \left[\left(\frac{\Pi_{t+1}^W}{\Gamma_{t+1}^W} - 1 \right) \frac{\Pi_{t+1}^W}{\Gamma_{t+1}^W} \frac{H_{t+1}}{H_t} a_t \right] + \mu_t^W = 0,$$

$$\Lambda_{t,t+1} = \beta \frac{\lambda_{t+1}}{\lambda_t} = \int \beta^i di \left(\frac{\tilde{C}_{t+1}}{\tilde{C}_t} \right)^{-\sigma} \frac{1}{a_t}, \quad (135)$$

$$\Gamma_t^W = (\Pi^W)^{1-\iota_w} (\Pi_{t-1}^W)^{\iota_w}, \quad (136)$$

$$\Pi_t^W = \Pi_t \frac{\tilde{W}_t}{\tilde{W}_{t-1}}, \quad (137)$$

$$\tilde{Y}_t = \left(\frac{\tilde{K}_{t-1}}{a_t} \right)^\alpha H_t^{1-\alpha} \quad (138)$$

$$a_t = \bar{a} \exp(z_t), \quad (139)$$

$$r_t^k = \alpha \psi_t a_t \frac{\tilde{Y}_t}{\tilde{K}_{t-1}} \quad (140)$$

$$\tilde{W}_t = (1 - \alpha) \psi_t \frac{\tilde{Y}_t}{H_t} \quad (141)$$

$$\tilde{Div}_t = \tilde{Y}_t - \tilde{W}_t N_t - r_t^K \tilde{K}_{t-1} / a_t, \quad (142)$$

$$\left[\varphi^p \left(\frac{\Pi_t}{\Gamma_t^P} - 1 \right) \frac{\Pi_t}{\Gamma_t^P} \right] = (1 - \epsilon^p) + \epsilon^p M C_t + \varphi^p \mathbb{E}_t \Lambda_{t,t+1} \left[\left(\frac{\Pi_{t+1}}{\Gamma_{t+1}^P} - 1 \right) \frac{\Pi_{t+1}}{\Gamma_{t+1}^P} \frac{\tilde{Y}_{t+1}}{\tilde{Y}_t} a_t \right] + \mu_t^P, \quad (143)$$

$$\Gamma_t^P = (\Pi)^{1-\iota_p} (\Pi_{t-1})^{\iota_p}, \quad (144)$$

$$\tilde{K}_t = (1 - \delta) \frac{\tilde{K}_{t-1}}{a_t} + \exp(\nu_t) \left(\gamma^A \left(\frac{a_t I_t}{a \hat{K}_{t-1}} \right)^{1-\gamma^B} + \gamma^C \right) \frac{a}{a_t} \tilde{K}_{t-1}, \quad (145)$$

$$\tilde{Z}_t = \tilde{K}_t + \tilde{D}_t, \quad (146)$$

$$\exp(\zeta_t) R_t = R_t^Z, \quad (147)$$

$$Q_t - \frac{1}{R_t^Z} = \beta E_t \left[\left(\frac{\tilde{C}_t}{\tilde{C}_{t+1}} \right)^\sigma \frac{1}{a_{t+1}} \right] \left(r_{t+1}^k + (1 - \delta) Q_{t+1} - \frac{1}{\Pi_{t+1}} \right), \quad (148)$$

$$\exp(\nu_t)Q_t = \left[\gamma^A(1 - \gamma^B) \left(\frac{I_t}{gK_{t-1}} \right)^{\gamma^B} \right]^{-1}, \quad (149)$$

$$R_t = \left(R \left(\frac{\Pi_t}{\Pi} \right)^{\theta_\Pi} \left(\frac{\psi_t}{\psi} \right)^{\theta_Y} \right) \exp(\epsilon_t^{mp_t}), \quad (150)$$

$$\tilde{D} = \frac{R_{t-1}}{\Pi_t a_t} \tilde{D} - \tilde{T}_t + g_t, \quad (151)$$

$$\tilde{T}_t = \int \left(\tilde{W}_t \exp(s_t^i) H_t^i \right) di - \int \tilde{\tau}_t \left(\tilde{W}_t \exp(s_t^i) H_t^i \right)^{1-\gamma_\tau} di \quad (152)$$

$$N_t = \int \exp(s_t^i) H_t^i di, \quad (153)$$

$$\tilde{Y}_t = \int \tilde{C}_t^i di + \tilde{X}_t + g_t. \quad (154)$$

Shocks

$$s_t^i = \rho_s s_{t-1}^i + \epsilon_t^{s,i}, \quad \forall i, \quad (155)$$

$$z_t = \rho_z z_{t-1} + \epsilon_t^z, \quad (156)$$

$$\mu_t^W = \rho_{\mu^W} \mu_{t-1}^W + \epsilon_t^{\mu^W}, \quad (157)$$

$$\mu_t^P = \rho_{\mu^P} \mu_{t-1}^P + \epsilon_t^{\mu^P}, \quad (158)$$

$$\nu_t = \rho_\nu \nu_{t-1} + \epsilon_t^\nu, \quad (159)$$

$$\zeta_t = \rho_\zeta \zeta_{t-1} + \epsilon_t^\zeta, \quad (160)$$

$$mp_t = \rho_m mp_{t-1} + \epsilon_t^m, \quad (161)$$

$$g_t = \rho_g g_{t-1} + \epsilon_t^g. \quad (162)$$

E.2 Mapping to general framework

We can map this HANK model in the general form of the outlined estimation procedure. The state variable, the structural shocks, the individual and aggregate control variables, and the needed deterministic steady state values are:

$$\mathbb{S}_t = \left\{ \left\{ \tilde{Z}_{t-1}^i \right\}_{i=1}^L, \left\{ s_t^i \right\}_{i=1}^L, \left\{ \beta^i \right\}_{i=1}^L, K_{t-1}, W_{t-1}, \Pi_{t-1}, \Pi_{t-1}^W, \zeta_t, z_t, \mu_t^W, \mu_t^P, \nu_t, mp_t, g_t \right\}, \quad (163)$$

$$\nu_t = \left\{ \left\{ \epsilon_t^{s,i} \right\}_{i=1}^L, \epsilon_t^\zeta, \epsilon_t^z, \epsilon_t^{\mu^W}, \epsilon_t^{\mu^P}, \epsilon_t^\nu, \epsilon_t^{mp}, \epsilon_t^g \right\}, \quad (164)$$

$$\psi_t^i = \{C_t^i\}, \quad (165)$$

$$\psi_t^A = \{\Pi_t, Q_t, \psi_t, H_t\}, \quad (166)$$

$$\psi_t^{DSS} = \{R\} \quad (167)$$

The parameters of the model are

$$\Theta = \{\eta, \sigma, \chi, \epsilon_w, \varphi^w, \iota_w, \alpha, \bar{a}, \gamma^\tau, \Pi, D, \rho_A, \sigma_s, \underline{Z}, \epsilon_p, \varphi^p, \iota_p, \delta, \gamma^A, \gamma^B, \gamma^C, \quad (168)$$

$$\Pi, \theta_\Pi, \theta_Y, \rho_s, \rho_\zeta, \rho_z, \rho_{\mu^W}, \rho_{\mu^P}, \rho_\nu, \rho_m, \rho_g, \sigma^s, \sigma^\zeta, \sigma^z, \sigma^{\mu^W}, \sigma^{\mu^P}, \sigma^\nu, \sigma^m, \sigma^g, \}$$

We use two NNs to separate between individual and aggregate policy functions.

The individual NN $\psi_{NN}^I(\cdot)$ solves for consumption:

$$\left\{ \left(C_t^i \right) = \psi_{NN}^I \left(\mathbb{S}_t^i, \mathbb{S}_t, \tilde{\Theta} | \bar{\Theta} \right) \right\}_{i=1}^L, \quad (169)$$

where $\mathbb{S}_t^i = \left\{ \tilde{B}_{t-1}^i, s_t^i \right\}$. The NN for the aggregate policy functions $\psi_{NN}^A(\cdot)$ determines inflation, multiplier for the law of motion of capital, marginal costs and labor supply:

$$\begin{pmatrix} \Pi_t \\ Q_t \\ \psi_t \\ H_t \end{pmatrix} = \psi_{NN}^A \left(\mathbb{S}_t, \tilde{\Theta} | \bar{\Theta} \right). \quad (170)$$

One challenge is that we must solve for the deterministic steady state values of the nominal rate, as the Taylor rule depends on them. This NNs $\psi_{NN}^{SS}(\cdot)$ that maps the parameter values to the deterministic steady state values is:

$$(R) = \psi_{NN}^{SS} \left(\tilde{\Theta} | \bar{\Theta} \right). \quad (171)$$

E.3 Predictive Checks

We assess the empirical bite of our quantitative HANK model. To examine this issue for our proof of concept, we rely on the model’s predictive checks.

In Bayesian econometrics, *prior predictive checks* assess whether the combination of a model and its prior yields plausible data before confronting the model with actual data. The prior predictive distribution is defined as follows:

$$p(y) = \int p(y \mid \theta) p(\theta) d\theta,$$

It shows how well the model’s likelihood function, $p(y \mid \theta)$, and prior, $p(\theta)$, combined together predict potential data, where y denotes such data and θ represents the model parameters. The prior predictive distribution is exactly the marginal likelihood of the model when evaluated at the actual observed data.

In Bayesian econometrics, the marginal likelihood plays a central role in assessing the posterior probability of a model relative to alternative specifications. Using terminology slightly outside strict Bayesian jargon, a larger posterior probability indicates a better model fit; that is, one that balances the likelihood of the observed data with the prior specification. If the data provide information about the model’s parameters that conflicts with the information encoded in the prior, the resulting marginal likelihood will be low, and the model will be less favored relative to competing models that achieve greater consistency between prior beliefs and observed evidence.⁴⁷

For our purposes, assessing the alignment of prior predictive checks with key empirical moments offers a convenient and informative way to assess the empirical adequacy and overall credibility of the model.

In practice, to compute predictive checks for our model, we draw parameter values from the support of the hypercube and simulate the model for 262 (quarterly) periods at each parameter draw. This procedure is computationally feasible because

⁴⁷See Leeper et al. (2017b) for an application of prior predictive checks to dynamic general equilibrium models.

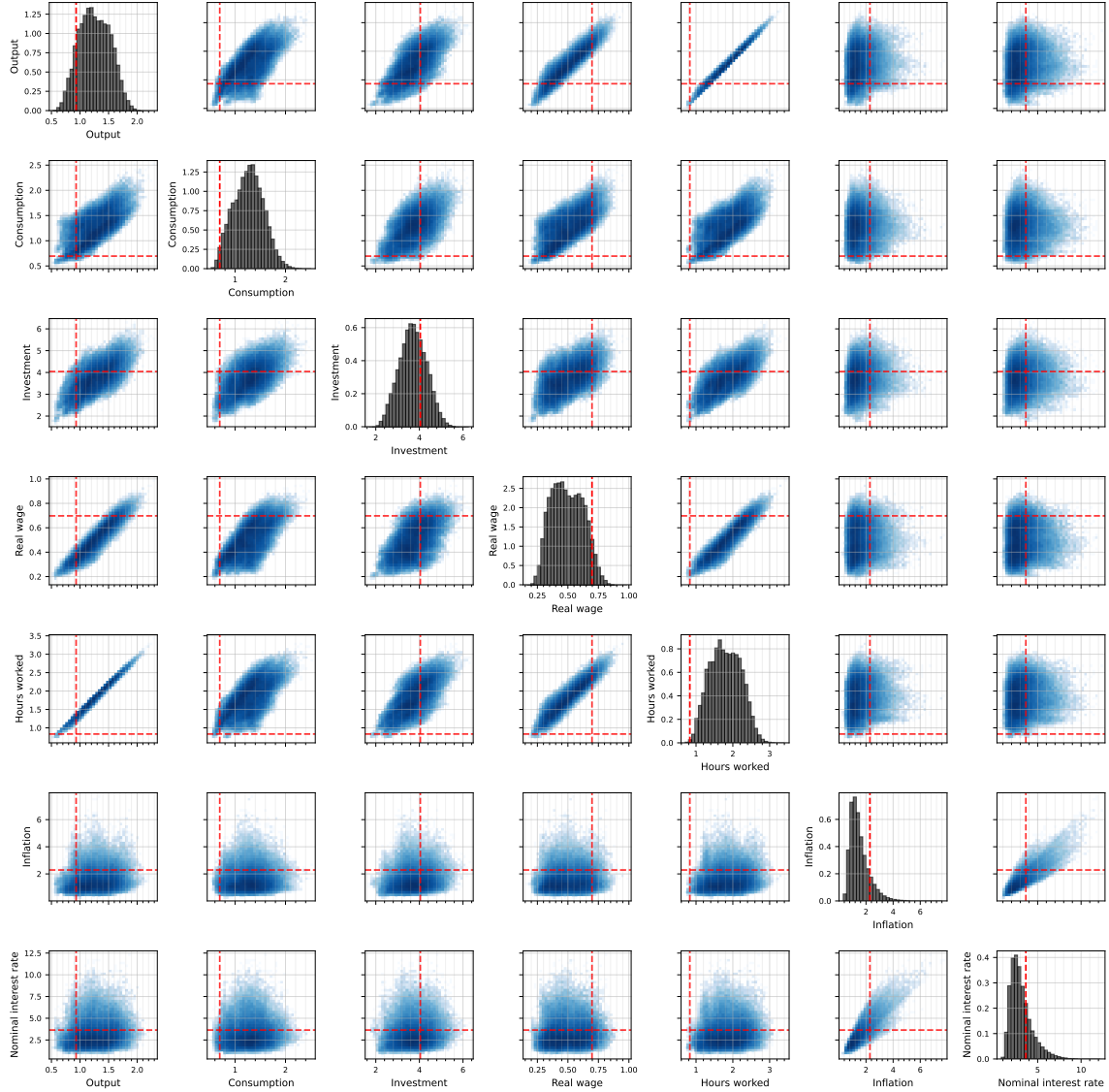


Figure 24: Predictive checks for the standard deviation of the observables. The figure shows the predictive checks for our model using 1,000 draws from the hypercube, for which the model is solved. The diagonal panels display the histogram for each observable: output, consumption, investment, real wage, hours worked, inflation, nominal interest rate. The off-diagonal panels show two-dimensional histograms that report the joint distribution among pairs of the observables. The red dashed lines indicate the corresponding values from the data.

our approach does not require re-solving the model or retraining the NNs at each parameter draw, as parameters are treated as pseudo-state variables. For each simulation, we compute the empirical statistic of interest. If the empirical moments lie within regions where the predictive distributions have probability mass, the model

can be considered capable of explaining the corresponding moments; otherwise, it is misspecified along that dimension.

The main results for the predictive checks is shown in Figure 24, where the standard deviation of the different observables is shown. Overall, the model's predictive checks capture the standard deviations of the data and most of the observed autocorrelations quite well. The main challenges lie in accounting for the negative autocorrelation of real wage growth and the persistence in hours worked, which are typical limitations of DSGE models.

E.4 Approximation Error

To evaluate the accuracy of our solution post-training, we use the set of equilibrium conditions employed during training, namely the consumption Euler equation, the first-order conditions for capital and investment, the Phillips curves for prices and wages, and the capital aggregation residuals:

$$err_1 = \left| \Psi^{FB} \left(1 - \beta R_t \mathbb{E}_t \left[\left(\frac{C_t^i}{C_{t+1}} \right)^\sigma \frac{1}{\Pi_{t+1} a_{t+1}} \right], B_t^i - \underline{B} \right) \right|, \quad (172)$$

$$err_2 = \left| Q_t - \frac{1}{R_t^Z} - \beta E_t \left[\left(\frac{\tilde{C}_t}{\tilde{C}_{t+1}} \right)^\sigma \frac{1}{a_{t+1}} \right] \left(r_{t+1}^k + (1 - \delta) Q_{t+1} - \frac{1}{\Pi_{t+1}} \right) \right|, \quad (173)$$

$$err_3 = \left| \frac{\varphi^p}{1 - \epsilon^p} \left(\frac{\Pi_t}{\Gamma_t^P} - 1 \right) \frac{\Pi_t}{\Gamma_t^P} - 1 - \frac{\epsilon^p}{1 - \epsilon^p} MC_t - \frac{\varphi^p}{1 - \epsilon^p} \mathbb{E}_t \Lambda_{t,t+1} \left[\left(\frac{\Pi_{t+1}}{\Gamma_{t+1}^P} - 1 \right) \frac{\Pi_{t+1}}{\Gamma_{t+1}^P} \frac{\tilde{Y}_{t+1}}{\tilde{Y}_t} a_t \right] - \frac{\mu_t^P}{1 - \epsilon^p} \right|, \quad (174)$$

$$err_4 = \left| Q_t - \left[\gamma^A (1 - \gamma^B) \left(\frac{I_t}{g K_{t-1}} \right)^{\gamma^B} \right]^{-1} \right|, \quad (175)$$

$$err_5 = \left| \int [\exp(s_t^i)]^{1-\gamma_\tau} di \tilde{\tau}_t (1 - \gamma_\tau) \left(\tilde{W}_t H_t \right)^{-\gamma_\tau} \tilde{W}_t + \frac{\epsilon_w}{1 - \epsilon_w} \frac{\chi(H_t)^\eta}{\tilde{C}_t^{-\sigma}} - \frac{\varphi^w}{1 - \epsilon_w} \left(\frac{\Pi_t^W}{\Gamma_t^W} - 1 \right) \frac{\Pi_t^W}{\Gamma_t^W} + \frac{\varphi^w}{1 - \epsilon_w} E_t \Lambda_{t,t+1} \left[\left(\frac{\Pi_{t+1}^W}{\Gamma_{t+1}^W} - 1 \right) \frac{\Pi_{t+1}^W}{\Gamma_{t+1}^W} \frac{H_{t+1}}{H_t} a_t \right] + \mu_t^W \right|, \quad (176)$$

$$err_6 = \left| \frac{K_t - \frac{1}{L} \sum_{i=1}^L K_t^i}{K_t} \right|, \quad (177)$$

E.5 Observation Equation

$$\begin{bmatrix} \text{Output growth} \\ \text{Consumption growth} \\ \text{Investment growth} \\ \text{Real wage growth} \\ \text{Hours growth} \\ \text{Inflation} \\ \text{Interest Rate} \end{bmatrix} = \begin{bmatrix} 100 \left(\frac{\tilde{Y}_t a_t}{\tilde{Y}_{t-1}} - 1 \right) \\ 100 \left(\frac{\tilde{C}_t a_t}{\tilde{C}_{t-1}} - 1 \right) \\ 100 \left(\frac{\tilde{I}_t a_t}{\tilde{I}_{t-1}} - 1 \right) \\ 100 \left(\frac{\tilde{W}_t a_t}{\tilde{W}_{t-1}} - 1 \right) \\ 100 \left(\frac{N_t}{N_{t-1}} - 1 \right) \\ 400 (\Pi_t - 1) \\ 400 (R_t - 1) \end{bmatrix} + u_t, \quad (178)$$

where $u + t$ is the measurement error. Note that we demean the growth series as in Bayer et al. (2024a).

E.6 Neural Network Particle Filter

The dynamics of total loss during the training of the NN are depicted in Figure 25. To assess potential concerns about overfitting of the NN particle filter, in the right chart of Figure 25, we report the total loss when the likelihood is evaluated at the parameters of the validation sample using the NN weights trained in every epoch (x-axis). The proximity of errors across epochs in both samples, depicted on the left and right charts, along with the consistent decrease in error over the validation sample, suggests that the NN is not overfitting.

F HANK model with the ZLB constraint

The model is a nonlinear HANK model that includes both idiosyncratic and aggregate risk. It incorporates two key features: households facing idiosyncratic income risk

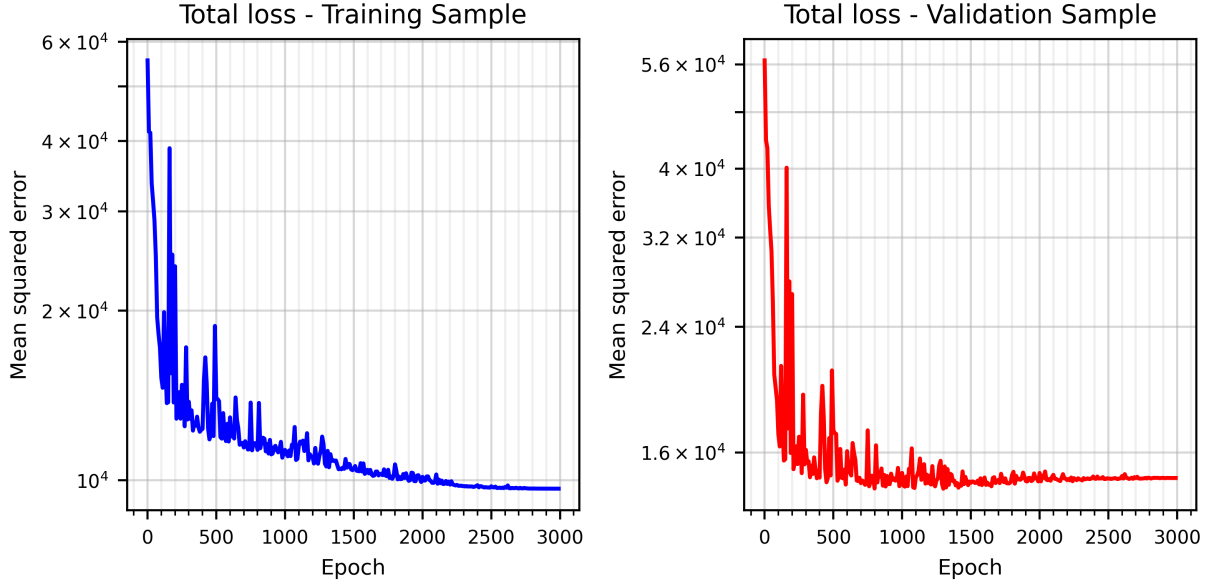


Figure 25: Training and validation convergence. The figure shows the total loss over the training sample (left) and over the validation sample (right). An epoch is completed when all the training or validation sample points are utilized. The vertical axis is expressed in a logarithmic scale.

and a borrowing limit; and the ZLB on the nominal interest rate, which constrains monetary policy. Additionally, the model includes demand, supply, and monetary policy shocks.

Households. The economy consists of a continuum of households. The households choose consumption C_t^i , labor H_t^i , and assets B_t^i to maximize their utility:

$$E_0 \sum_{t=0}^{\infty} \beta^t \exp(\zeta_t) \left[\left(\frac{1}{1-\sigma} \right) \left(\frac{C_t^i}{A_t} \right)^{1-\sigma} - \chi \left(\frac{1}{1+\eta} \right) (H_t^i)^{1+\eta} \right],$$

where ζ_t is an aggregate preference shock, which follows an AR(1) process $\zeta_t = \rho_{\zeta} \zeta_{t-1} + \epsilon_t^{\zeta}$. C_t is aggregate consumption and A_t is the total factor productivity. The budget constraint in real terms can be written as:

$$C_t^i + B_t^i = \tau_t \left(\frac{W_t}{P_t A_t} \exp(s_t^i) H_t^i \right)^{1-\gamma_{\tau}} + \frac{R_{t-1}}{\Pi_t} B_{t-1}^i + Div_t^i, \quad (179)$$

where $Div_t^i = Div_t \exp(s_t^i)$ is the household specific real dividends, W_t is nominal wage, R_t is the gross nominal interest rate, and Π_t is the gross inflation rate. The taxation is progressive, which is governed by γ_{τ} and τ_t determines the level of the

post-tax income. The wage is divided by the productivity level to avoid a fiscal drag in taxation. The agents individual labor productivity s_t^i is stochastic and follows an AR(1) process: $s_t^i = \rho_s s_{t-1}^i + \epsilon_t^{s,i}$. The dividend payout is scaled with the level of the idiosyncratic shock. The agents face a borrowing limit $\underline{B}$, which implies:

$$B_t \geq \underline{B}. \quad (180)$$

The first-order conditions can be written as

$$1 = \beta R_t \mathbb{E}_t \left[\left(\frac{\exp(\zeta_{t+1})}{\exp(\zeta_t)} \right) \left(\frac{\lambda_{t+1}^i}{\lambda_t^i} \right) \frac{1}{\Pi_{t+1}} \right] + \mu_t^i, \quad (181)$$

$$\lambda_t^i = \left(\frac{C_t^i}{A_t} \right)^{-\sigma} \frac{1}{A_t}, \quad (182)$$

$$\chi(H_t^i)^\eta = (\lambda_t^i) \left[\tau_t (1 - \gamma_\tau) \exp(s_t^i) \frac{W_t}{P_t A_t} \left(\exp(s_t^i) \frac{W_t}{P_t A_t} H_t^i \right)^{-\gamma_\tau} \right], \quad (183)$$

where $\mu_t^i \geq 0$ is the normalized Lagrange multiplier on the individual borrowing limit in equation (180).

Firms. The firm sector consists of a continuum of final goods producers and intermediate goods firms. The final goods retailers buy the intermediate goods and transform them into the homogeneous final good using a CES production technology:

$$Y_t = \left(\int_0^1 (Y_t^j)^{\frac{\epsilon-1}{\epsilon}} dj \right)^{\frac{\epsilon}{\epsilon-1}}, \quad (184)$$

where Y_t^j is the output of intermediate goods firm j . The equilibrium price of the final good and the demand for the intermediate goods of firm j can be expressed as:

$$P_t = \left(\int_0^1 (P_t^j)^{1-\epsilon} dj \right)^{\frac{1}{1-\epsilon}}, Y_t^j = \left(\frac{P_t^j}{P_t} \right)^{-\epsilon} Y_t. \quad (185)$$

Intermediate goods producers are monopolistically competitive. The firm j uses

labor N_t^j as input to produce output Y_t^j with the following production technology:

$$Y_t^j = A_t N_t^j. \quad (186)$$

Total factor productivity A_t follows a nonstationary process:

$$A_t = a_t A_{t-1}, \quad (187)$$

where the trend growth rate follows an AR(1) process:

$$a_t = \bar{a} \exp(z_t), \quad (188)$$

$$z_t = \rho_z z_{t-1} + \epsilon_t^z. \quad (189)$$

Labor is hired in competitive markets so that the wage is given as follows:

$$W_t/P_t = A_t MC_t. \quad (190)$$

The firm j sets the price of its goods to maximize its profit subject to the demand curve for intermediate goods and Rotemberg adjustment costs for changing prices:

$$\max_{P_t^j} P_t^j \left(\frac{P_t^j}{P_t} \right)^{-\epsilon} \frac{Y_t}{P_t} - MC_t \left(\frac{P_t^j}{P_t} \right)^{-\epsilon} Y_t - \frac{\varphi}{2} \left(\frac{P_t^j}{\Pi P_{t-1}^j} - 1 \right)^2 Y_t, \quad (191)$$

where Π is the inflation target of the central bank. Imposing a symmetric equilibrium and discounting future profits with the real interest rate, the New Keynesian Phillips curve can be written as:

$$\left[\varphi \left(\frac{\Pi_t}{\Pi} - 1 \right) \frac{\Pi_t}{\Pi} \right] = (1-\epsilon) + \epsilon MC_t + \beta \varphi \mathbb{E}_t \left(\frac{\exp(\zeta_{t+1})}{\exp(\zeta_t)} \right) \left(\frac{\lambda_{t+1}}{\lambda_t} \right) \left[\left(\frac{\Pi_{t+1}}{\Pi} - 1 \right) \frac{\Pi_{t+1}}{\Pi} \frac{Y_{t+1}}{Y_t} \right], \quad (192)$$

where $\Pi_t = P_t/P_{t-1}$ and $\lambda_t = (C_t/A_t)^{-\sigma} A_t^{-1}$. The Rotemberg adjustment costs are

ex-post given back. The real dividends of the firm sector can then be written as

$$Div_t = Y_t - W_t N_t. \quad (193)$$

Policy makers. The central bank sets the nominal interest R_t using a Taylor rule that responds to inflation and output deviations from their targets Π and Y . The ZLB restricts the nominal interest rate. The rule is given as follows:

$$R_t^N = \left(R \left(\frac{\Pi_t}{\Pi} \right)^{\theta_\Pi} \left(\frac{Y_t}{A_t Y} \right)^{\theta_Y} \right) \exp(mp_t), \quad (194)$$

$$R_t = \max [1, R_t^N], \quad (195)$$

where R and Y denote the deterministic steady state (DSS) values of the nominal interest rate and output (in the detrended economy). While households face idiosyncratic shocks in the deterministic steady state, the economy does not face and the agents do not expect aggregate shocks. Additionally, there is also a persistent monetary policy shock mp_t to capture prolonged deviations from the rule:

$$mp_t = \rho_m mp_{t-1} + \epsilon_t^m. \quad (196)$$

The fiscal authority sets τ_t to keep their debts D_t on a constant path adjusted for productivity gains:

$$D_t = \frac{R_{t-1}}{\Pi_t} D_{t-1} - T_t, \quad (197)$$

$$T_t = \int \left(\frac{W_t}{P_t} \exp(s_t^i) H_t^i \right) di - \int \tau_t \left(\frac{W_t}{P_t A_t} \exp(s_t^i) H_t^i \right)^{1-\gamma_\tau} di. \quad (198)$$

Market clearing. Market clearing for the labor market, bond market, and goods market requires

$$N_t = \int N_t^j dj = \int \exp(s_t^i) H_t^i di, \quad (199)$$

$$D_t = \int B_t^i di, \quad (200)$$

Calibrated Parameter Values			
Parameters		Value	Target/Source
β	Discount factor	0.9975	4% nominal interest rate
η	Inverse Frisch elasticity	0.72	Chetty et al. (2011)
σ	Relative risk aversion	1	Log utility
$\bar{a}$	Average growth rate	1.0033	Real GDP growth = 0.33% (quarterly)
χ	Disutility labor	0.74	Labor supply is approximately 1
γ^τ	Tax progressivity	0.18	Heathcote et al. (2017)
D	DSS government debt	1.0	Wealth share=25% GDP (Kaplan et al., 2018)
Π	Inflation target	1.00625	Inflation target = 2.5% (annualized)
ρ_s	Persistence labor prod.	0.9	Share of borrowers = 34%
ρ_ζ	Persistence pref. shock	0.7	Frequency of ZLB = 15%

Table 7: Calibration. The panel displays the calibrated parameters.

$$Y_t = \int C_t^i di. \quad (201)$$

Calibration Table 7 shows the calibrated parameters and its corresponding values.

F.1 Equilibrium conditions

To have stationarity, we need to define the variables as follows $\tilde{X}_t = \frac{X_t}{A_t}$.⁴⁸ The relevant conditions can then be written as:

$$\tilde{C}_t^i + \tilde{B}_t^i = \tilde{\tau}_t \left(\tilde{W}_t \exp(s_t^i) H_t^i \right)^{1-\gamma_\tau} + \frac{R_{t-1}}{\Pi_t a_t} \tilde{B}_{t-1}^i + \tilde{Div}_t^i, \quad (202)$$

$$1 = \beta R_t \mathbb{E}_t \left[\left(\frac{\exp(\zeta_{t+1})}{\exp(\zeta_t)} \right) \left(\frac{\tilde{C}_t^i}{\tilde{C}_{t+1}^i} \right)^\sigma \frac{1}{\Pi_{t+1} a_{t+1}} \right] + \mu_t^i, \quad (203)$$

$$\chi (H_t^i)^\eta = \left(\tilde{C}_t^i \right)^{-\sigma} \left[\tilde{\tau}_t (1 - \gamma_\tau) \exp(s_t^i) \tilde{W}_t \left(\exp(s_t^i) \tilde{W}_t H_t^i \right)^{-\gamma_\tau} \right], \quad (204)$$

$$\tilde{Y}_t = N_t, \quad (205)$$

$$\tilde{W}_t = MC_t, \quad (206)$$

$$\tilde{Div}_t = \tilde{Y}_t - \tilde{W}_t \tilde{Y}_t, \quad (207)$$

⁴⁸Note for simplicity, we denote detrended real wage also as $\tilde{X}_t = \frac{W_t}{P_t A_t}$.

$$\left[\varphi \left(\frac{\Pi_t}{\Pi} - 1 \right) \frac{\Pi_t}{\Pi} \right] = (1 - \epsilon) + \epsilon M C_t + \beta \varphi \mathbb{E}_t \left[\left(\frac{\exp(\zeta_{t+1})}{\exp(\zeta_t)} \right) \left(\frac{\tilde{C}_{t+1}}{\tilde{C}_t} \right)^{-\sigma} \left(\frac{\Pi_{t+1}}{\Pi} - 1 \right) \frac{\Pi_{t+1}}{\Pi} \frac{\tilde{Y}_{t+1}}{\tilde{Y}_t} \right], \quad (208)$$

$$R_t^N = \left(R \left(\frac{\Pi_t}{\Pi} \right)^{\theta_\Pi} \left(\frac{\tilde{Y}_t}{Y} \right)^{\theta_Y} \right) \exp(\epsilon_t^{mp_t}), \quad (209)$$

$$R_t = \max [1, R_t^N], \quad (210)$$

$$\tilde{D} = \frac{R_{t-1}}{\Pi_t a_t} \tilde{D} - \tilde{T}_t, \quad (211)$$

$$\tilde{T}_t = \int \left(\tilde{W}_t \exp(s_t^i) H_t^i \right) di - \int \tilde{\tau}_t \left(\tilde{W}_t \exp(s_t^i) H_t^i \right)^{1-\gamma_\tau} di \quad (212)$$

$$N_t = \int \exp(s_t^i) H_t^i di, \quad (213)$$

$$D = \int \tilde{B}_t^i di, \quad (214)$$

$$\tilde{Y}_t = \int \tilde{C}_t^i di. \quad (215)$$

F.2 Mapping to the general framework

The state variables and shocks of the HANK model are given as:

$$\mathbb{S}_t = \left\{ \left\{ \tilde{B}_{t-1}^i \right\}_{i=1}^L, \left\{ s_t^i \right\}_{i=1}^L, R_{t-1}^N, \zeta_t, a_t, mp_t \right\}, \quad (216)$$

$$\nu_t = \left\{ \left\{ \epsilon_t^{s,i} \right\}_{i=1}^L, \epsilon_t^\zeta, \epsilon_t^z, \epsilon_t^m \right\}. \quad (217)$$

The parameters of the model are divided into calibrated ($\bar{\theta}$) and estimated ones ($\tilde{\theta}$).

For instance, in the case of our application with US data, this yields:

$$\bar{\Theta} = \{\beta, \eta, \sigma, \bar{a}, \chi, \gamma^\tau, \Pi, D, \rho_s, \rho_\zeta\}, \quad (218)$$

$$\tilde{\Theta} = \{\sigma_s, \underline{B}, \varphi, \theta_\Pi, \theta_Y, \rho_z, \rho_m, \sigma_\zeta, \sigma_z, \sigma_m\}. \quad (219)$$

We use two NNs to separate between individual and aggregate policy functions. The individual NN $\psi_{NN}^I(\cdot)$ solves for labor supply:

$$\left\{ \left(H_t^i \right) = \psi_{NN}^I \left(\mathbb{S}_t^i, \mathbb{S}_t, \tilde{\Theta} | \bar{\Theta} \right) \right\}_{i=1}^L, \quad (220)$$

where $\mathbb{S}_t^i = \left\{ \tilde{B}_{t-1}^i, s_t^i \right\}$. The NN for the aggregate policy functions $\psi_{NN}^A(\cdot)$ determines the wage and inflation:

$$\begin{pmatrix} \Pi_t \\ \tilde{W}_t \end{pmatrix} = \psi_{NN}^A \left(\mathbb{S}_t, \tilde{\Theta} | \bar{\Theta} \right). \quad (221)$$

One challenge is that we must solve for the deterministic steady state values of the nominal rate and output, as the Taylor rule depends on them. To focus on the algorithm for the moment, let's assume that we have already trained a NN $\psi_{NN}^{SS}(\cdot)$ that maps the parameter values to the deterministic steady state values:

$$\begin{pmatrix} R \\ \tilde{Y} \end{pmatrix} = \psi_{NN}^{SS} \left(\tilde{\Theta} | \bar{\Theta} \right). \quad (222)$$

F.3 Training of the neural networks and mean squared error

We are training our NN based model solution over the extended parameter space in two steps. The details of the training process and the mean squared error are shown.

Training of the DSS NN. The NN to approximate output and nominal rate in the DSS is trained for 10,000 iterations. We conduct 15 optimization steps during each iteration step, including redrawing of the shocks, and use 100 Monte Carlo draws. After each iteration, the economy is simulated for 20 periods. The parameters are redrawn after every 40 iterations and are initialized with a longer simulation of 200 periods. Note that this step also requires training a temporary individual policy function, which provides the decision of households in the DSS. The convergence of the NN is shown in Figure 26, where the mean squared error averaged across the

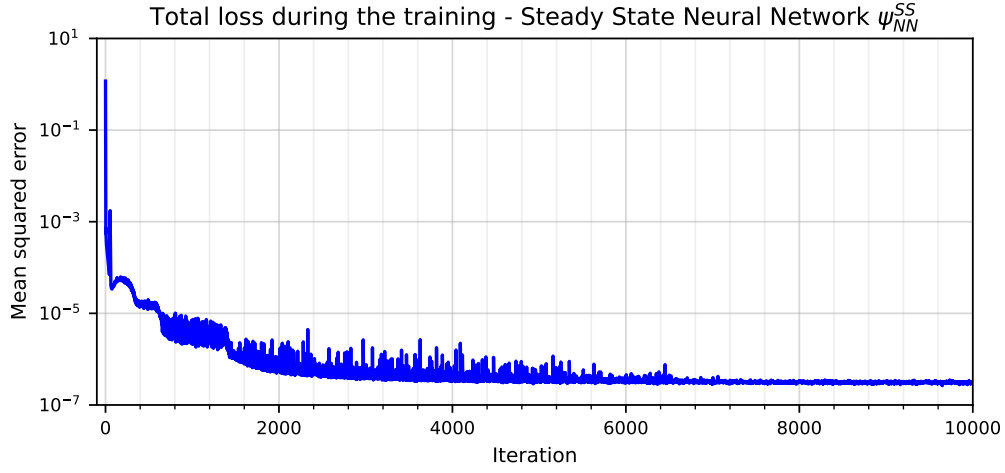


Figure 26: Convergence of the NN solution for the DSS of the HANK model. The figure shows the dynamics of the mean squared error, which is averaged across the equations that we minimize, associated with the deterministic steady state values of the nominal rate R and output Y . The vertical axis has a logarithmic scale.

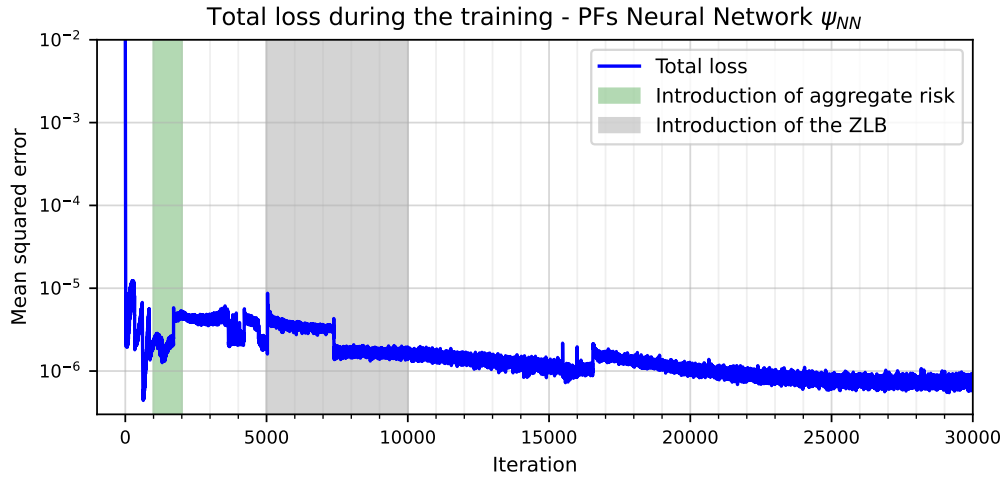


Figure 27: Convergence of the NN solution for the HANK model. The figure shows the dynamics of the mean squared error during the training of the extended individual and aggregate policy functions. The shaded areas indicate the periods in which we introduce aggregate risk and the ZLB. The vertical axis has a logarithmic scale.

equations that we minimize is shown. At the end of the training, the NN approximates the DSS with a very high accuracy.

Training of extended policy functions. We use 30,000 iterations to jointly train the NNs to approximate the extended individual and aggregate policy functions. We conduct 15 optimization steps during each iteration step, including redrawing of the shocks, and use 100 Monte Carlo draws. After each iteration, the economy is simulated for 20 periods. The parameters are redrawn after every 40 iterations and are initialized with a longer simulation of 200 period.⁴⁹ Figure 27 shows that the loss converges to an error in the magnitude of 10e-6. For better convergence, we slowly introduce aggregate risk between periods 1,000 and 2,000 by scaling the standard deviations of the aggregate shocks from a tiny value to the actual values. Similarly, we introduce the ZLB between periods 5,000 and 1,000 by slowly scaling the impact of the ZLB. In particular, we use the following functional form:

$$R_t = \max[R_t^N, 1] + a^{ZLB} \min[R_t^N - 1, 0]. \quad (223)$$

The parameter a^{ZLB} is stepwise lowered from 1 to 0 to move from the economy without ZLB to the ZLB economy.

F.4 How good is what we got?

The empirical fit of the model is quite promising given the stylized nature of the HANK model we have estimated. In Table 8, we compare the second moment of the observables in the estimated HANK model and in the data. The standard deviation of output growth, inflation, and the nominal interest rate in the model are quite close to the ones measured in the data.

The match is somewhat unsatisfactory, when it comes to the serial correlation of the three observables. This finding underscores the need for extensions of the HANK model aimed at enhancing its endogenous persistence mechanism. Introducing additional nominal and real frictions could be an effective strategy for improving the

⁴⁹During the first 1,000 iterations, we have a shorter simulation horizon of 10 periods and 100 periods if parameters are redrawn.

Standard deviations			Autocorrelations			Avg. Gini coef.		
	Model	Data		Model	Data		Model	Data
GDP	0.6947	0.5831	GDP	0.1355	0.4050	Wealth	0.8793	0.8410
Inflation	1.1511	0.9045	Inflation	0.8146	0.5456			
FFR	2.561	2.7537	FFR	0.7219	0.9707			

Table 8: Moments comparison: model vs. data. The table reports the standard deviations (left panel) and the autocorrelation coefficients (middle panel) in the model and in the U.S. data for the three observable variables: real GDP growth (GDP), GDP deflator growth (Inflation), and the federal funds rate (FFR). The right panel of the table shows the average Gini coefficient in the model and in the U.S. data (World Inequality Database). The model-implied moments are obtained by simulating the estimated model for 1,000,000 periods and with parameters set at the posterior median. The empirical moments are computed over the sample period used in estimation (1990:Q1-2019:Q4).

empirical fit of the stylized nonlinear HANK model we have estimated.

Acharya et al. (2023) estimate a HANK model using likelihood methods and find that its empirical performance is inferior to that of a state-of-the-art RANK model. One potential reason for this subpar performance is the challenge of accurately estimating certain parameters of the HANK model, particularly those affecting its DSS, due to the high computational cost of recomputing it. Our NN estimation method addresses this challenge by efficiently solving the steady state equilibrium of HANK models, allowing the estimation of these parameters. In contrast to Acharya et al. (2023), we estimate the fully nonlinear version of a HANK model, including the ZLB constraint.

In Table 8, we also present the average Gini index of the wealth distribution in our model, which we estimated without observing any cross-sectional data, alongside the corresponding index measured by the World Inequality Database for the US over the same sample period of our estimation. The two coefficients are quite close.

G Fine-tuning neural networks, and execution times.

In this appendix, we highlight how we fine-tune the NNs' weights for each model and application considered in the main paper.

Specifically, the following options of the networks could be used for fine-tuning: (1) activation function, (2) network width and depth, (3) optimizer, (4) loss function criterion, (5) policy function transformation, (6) input normalization, (7) number of training iterations, (8) learning rate, (9) batch size, (10) frequency of parameter draws, (11) simulation step size, (12) number of draws for Monte Carlo integration, and (13) number of internal optimization steps.

For the training of the neural-network particle filter, we test the adjustment of the following parameters: (1) activation function, (2) network width and depth, (3) optimizer, (4) batch size, (5) number of epochs, (6) number of draws, (7) size of the validation sample, and (8) learning rate.

In addition, the setup includes several options directly related to the particle filter itself: (1) measurement error, (2) number of particles, and (3) length of the data sample.

Below, we provide an overview of the NNs setup and execution time for the different models.

1. The linearized RANK model (Section 3.1) is specified as follows: The NN contains five hidden layers with 256 neurons each. The activation function for the hidden layers is continuously differentiable exponential linear units (CELU). The optimizer is AdamW. The learning rate follows a cosine annealing scheduler starting from an initial rate of 1×10^{-4} and terminating at a rate of 1×10^{-6} . The batch size is set to 1,000. The NN particle filter contains 4 hidden layers with 128 neurons each. The activation function for the hidden layers are CELU. The optimizer is AdamW.

The learning rate follows a cosine annealing scheduler from an initial level of 2×10^{-4} and terminating at 1×10^{-7} . The particle filter uses 100 particles.

2. The nonlinear RANK model with the ZLB constraint (Section 3.2) is specified as follows: The NN contains five hidden layers with 128 neurons each. The activation function for the four initial hidden layers is the Sigmoid Linear Unit (SiLU), while the fifth layer is Leaky Rectified Linear Unit (Leaky ReLU). The optimizer is AdamW. The learning rate follows a cosine annealing scheduler starting from an initial rate of $\times 10^{-3}$ and terminating at a rate of $\times 10^{-7}$. The batch size is set to 100. We generated a dataset comprising 1,000 likelihood samples using 5,000 particles, with a burn-in period of 50 and a total simulation length of 1,000 iterations. The NN particle filter consists of four hidden layers with 128 neurons each and employs SiLU activation functions. Training is performed using the AdamW optimizer with a cosine annealing learning rate schedule, starting from an initial rate of 1×10^{-3} and decaying to 1×10^{-7} over 10,000 epochs.

3. The one-asset HANK model of Auclert et al. (2021) (Section 3.3) is specified as follows: We specify three NNs for the model solution. Each NN contains 5 hidden layers with 128 neurons each. The activation function for the hidden layers are CELU. The optimizer is AdamW. The learning rate starts at 1×10^{-4} , and is then reduced at four fixed intervals by a factor of 0.1. The batch size is set to 50. The number of Monte Carlo integrations is set to 100. We use 100 households to compute the residuals for the individual optimization problems, while 100 households are used to keep track of the distribution and compute aggregates. The execution time is around 4 hours on an Nvidia RTX 3090 with 24GB of memory as GPU.

4. The quantitative HANK model (Section 3.4) is specified as follows: We specify three NNs for the model solution. Each NN contains 5 hidden layers with 128 neurons each. The activation function for the hidden layers is CELU. The optimizer

is AdamW. The learning rate starts at 1×10^{-4} , and is then reduced at four fixed intervals by a factor of 0.1. The batch size is set to 50. The number of Monte Carlo integrations is set to 100. We use 100 households to compute the residuals for the individual optimization problems, while 100 households are used to keep track of the distribution and compute aggregates. The NN for the particle filter features 2 hidden layers with 128 nodes. The activation function is CELU. The NN is trained over 3,000 epochs with a batch size of 100. The learning rate follows a cosine annealing schedule, starting at 1×10^{-4} and terminating at 1×10^{-8} . The validation share is 20%. The particle filter uses 10,000 particles. The total execution time is approximately 130 hours: 38 hours for the solution step, 90 hours for generating likelihood evaluations, 10 minutes for the NN particle filter, and 2 hours for the RWMH, using an Nvidia RTX 3090 GPU with 24 GB of memory.

5. The nonlinear HANK model with the ZLB constraint (Section 4) is specified as follows. We employ three NNs for the model solution. The NN trained to approximate the policy functions has five hidden layers with 128 neurons each and uses the CELU activation function. The optimizer is AdamW, with a batch size of 100 and 100 Monte Carlo integrations. We then train the NN particle filter. We draw from the parameter space and generate a data set comprising 10,000 likelihood values using 10,000 particles, with a burn-in period of 50 and a total simulation length of 144 iterations. We use 75% of these values for training and the remaining 25% for validation. The NN particle filter consists of four hidden layers with 64 neurons each and uses SiLU activation functions. Training is carried out over 1,000 epochs with a batch size of 50, using the AdamW optimizer with a cosine annealing learning rate schedule decreasing from 1×10^{-4} to 1×10^{-7} . The total execution time is approximately 27 hours. Model training takes around 6 hours, while the particle filter requires approximately 20 hours to generate 10,000 likelihood values. Training the NN particle filter and running the RWMH algorithm together take less than 1

hour.